AcreForge

Complete Owner's Manual and Rebuild Guide

U.S.-wide, local-first rural property planner. Live at https://jacobbaltins.github.io/acreforge/

This document contains everything needed to understand, operate, legally protect, monetize and **fully rebuild** the AcreForge website if you lose it: the full source code of the app is printed in Appendix A, and a complete backup zip is embedded inside this PDF as an attachment (see Part 2).

Part	Contents
1	What AcreForge is and how it works
2	Rebuild it from scratch (3 ways)
3	Using the app: every page
4	Technical reference (data, storage, code map)
5	Hosting and updating
6	Money, pricing and funding plan
7	Legal, privacy and safety
8	Troubleshooting and next steps
App. A	Full source of index.html and small files
App. B	Changelog and calculation notes

Part 1 - What AcreForge is

AcreForge is a single-page web app (one file, **index.html**) for planning rural land, farms, estates, homes and buildings anywhere in the U.S. It has no build step, no server and no database. Everything you create is saved in your own browser (localStorage). Maps come from free public services.

Main features

• About landing page and color themes (Settings > Appearance).

• Site Designer: infinite 2D canvas in feet, 71+ objects (barns, arenas, fences, roads, trees, utilities), snapping grid, rotate/resize/duplicate/delete, layers and labels, isometric 3D massing preview.

• Parcel import: KML or GeoJSON (county assessor/GIS or survey) sets true acreage and checks object placement. Acreage 5 to 1,000 (default 80).

• Building and Lot Builder: 12 library folders, presets (home, barn, shop, retail, full lot), editable rooms and items, 72/150/300/600 ft scales.

• U.S. state map (Census TIGER boundaries) and state terrain viewer (USGS aerial/topo imagery, AWS terrain, optional OpenStreetMap context and 3D buildings).

• Drawing modes for footprints, land areas, landmarks, proposed clearing, with perimeter and area measurements.

• Dashboard, notes, tasks, scenarios and finance sections; public export removes private data.

• Pricing preview (Free / Pro planned / Early Access) and a Pro waitlist button; payments are NOT active.

Technology

• Plain HTML, CSS and JavaScript in one file (about 285 KB).

• MapLibre GL JS 5.6.2 (bundled in /vendor): map rendering.

• us-states.js: U.S. Census 2024 cartographic state outlines (1:500,000) as a JS data file.

• Hosting: GitHub Pages from repo jacobbaltins/acreforge (free).

• Dormant: netlify/functions/stripe-webhook.js and netlify.toml (cannot run on GitHub Pages; Netlify account was credit-blocked).

Part 2 - Rebuild it from scratch

Way 1: Extract the embedded backup (fastest)

This PDF has the file **acreforge.zip** attached (complete site including the map library and state data). In Adobe Acrobat/Reader open the paperclip (Attachments) panel, save acreforge.zip, unzip it, and double-click index.html to run it locally. In Chrome/Edge's PDF viewer attachments may not show; use Acrobat Reader or run:

```
python -c "from pypdf import PdfReader;r=PdfReader('AcreForge-Complete-Manual.pdf');\
[open(k,'wb').write(v[0]) for k,v in r.attachments.items()]"
```

Way 2: Re-download from GitHub (if the repo still exists)

```
gh auth login
gh repo clone jacobbaltins/acreforge
cd acreforge
# open index.html, or serve it:
python -m http.server 8766   # then visit http://127.0.0.1:8766/index.html
```

Way 3: Rebuild by hand from this printout

1. Create a folder acreforge. Inside, create the files listed in Appendix A. Type or paste the code exactly (the full text is also in the embedded attachment, so use copy-paste from the attachment or PDF whenever possible).

2. Download the map library: run `npm pack maplibre-gl@5.6.2`, extract package/dist/maplibre-gl.js and maplibre-gl.css into a folder named **vendor**. (Or from https://unpkg.com/maplibre-gl@5.6.2/dist/.)

3. Recreate us-states.js: U.S. Census Bureau cartographic boundary file cb_2024_us_state_500k (https://www.census.gov/geographies/mapping-files/time-series/geo/cartographic-boundary.html), converted to the same JS data structure the app expects. If you cannot recreate it, the embedded zip contains it. Note: this data file is 528 KB and is NOT printed in this document.

4. Open index.html in a browser. Test: About page loads, Site Designer shows objects that can be dragged, state terrain map loads imagery.

5. Publish it (Part 5).

File list to have after a rebuild

File	Purpose
index.html	The entire app (printed in Appendix A)
us-states.js	State outline data (in embedded zip; regenerate from Census)
vendor/maplibre-gl.js / .css	Map library v5.6.2 (from npm/unpkg)
support.html, data-notice.html	Project status and privacy pages (Appendix A)
README.md, CHANGELOG.md, CALCULATIONS.md, CAMPAIGN.md	Documentation
netlify.toml, netlify/functions/stripe-webhook.js	Dormant Stripe webhook (Appendix A)

Part 3 - Using the app, page by page

• **About**: first screen; vision plus shortcuts and the pricing preview (Free, Pro planned, Early Access) and Join Pro waitlist.

• **Site Designer**: pick an object from the library, drag it onto the canvas, drag to move (snaps to grid, default 40 ft). Use the selected-object editor to rotate/resize. Mouse wheel zooms around the cursor; drag empty space to pan; Fit plan recenters. Objects always have an enlarged invisible grab area so they remain draggable at 1000 acres.

• **Parcel import**: Settings or Site Designer: import a single-polygon WGS84 KML/GeoJSON from a county assessor/GIS. Acreage is computed geodesically. A pasted Google Maps/Zillow URL is only a reference link.

• **Buildings and Lot Builder**: browse 12 library folders; load presets; edit rooms/items; choose drawing scale.

• **State map** and **State terrain**: view U.S. states; terrain uses USGS aerial by default, Topo toggle; enable Mapped features at neighborhood zoom for OSM homes, buildings (3D), roads, fields, woods and trees; Clear reference context resets them.

• **Overlays**: draw house/building footprints, land areas, landmarks and proposed clearing; app reports perimeter and area.

• **Settings**: acreage (5, 10, 20, 40, 80, 160, 320, 640, 1000), theme, grid size.

• **Public export** (local only): downloads published-project.json without boundary, address, notes or finances.

Where your data lives

Projects are stored in this browser under localStorage key `oregon-estate-planner-v1` (theme: oregon-estate-planner-theme). Clearing site data or using another browser/computer loses them. There is no cloud sync yet. To back up: in the browser console run `copy(localStorage.getItem('oregon-estate-planner-v1'))` and paste into a text file; to restore run `localStorage.setItem('oregon-estate-planner-v1', '...')`.

Part 4 - Technical reference

Code map (index.html)

• One `<style>` block with CSS variables and themes; the sidebar nav (13 items) and a content container.

• Last `<script>` block holds all app logic: state/defaultState (default acres 80), save/load to localStorage, page renderers, Site Designer (SVG with pointer events, data-object elements, mapHit for minimum hit size, dragGhost, grid snapping), Lot Builder IIFE, state map and MapLibre terrain viewer, parcel import, measurement math.

• End of file: the .pubbar IIFE (edit-mode switch and the waitlist button; PAY_LINK is blank. Put a Stripe Payment Link there to enable an Upgrade button).

• Editing is on for all visitors unless the address contains ?edit=0 (view-only legacy mode). The Public export button shows only on localhost.

Editing safely (Windows)

The file has very long lines. Edit with a code editor (VS Code) or script string replacement. After editing, syntax-check the last script: copy its contents to a .js file and run `node --check file.js`.

External services used (all free, no keys)

• USGS National Map imagery/topo tiles

• AWS Terrain Tiles (elevation)

• OpenStreetMap via the public Overpass API (only when Mapped features is enabled)

• U.S. Census TIGER boundaries (bundled in us-states.js)

Part 5 - Hosting and updating

Current host: GitHub Pages

Repo: github.com/jacobbaltins/acreforge (public), branch main, served at https://jacobbaltins.github.io/acreforge/. Pages rebuilds about 30 to 60 seconds after each push.

Update steps

```
# one-time setup (git needs an identity)
git config user.name  "jacobbaltins"
git config user.email "<your GitHub noreply email>"

# each update
gh repo clone jacobbaltins/acreforge   # first time only
# copy the edited index.html into the clone
git add -A
git commit -m "describe change"
git push
# then hard refresh the site (Ctrl+F5)
```

Recreate the hosting from nothing

```
gh repo create acreforge --public
# add all site files, commit, push to main
# GitHub: repo Settings > Pages > Source: Deploy from branch > main / (root)
# Site appears at https://<username>.github.io/acreforge/
```

Custom domain

• Buy a domain (about $20/yr). Repo Settings > Pages > Custom domain, then add DNS records at your registrar (GitHub shows the exact values) and enable Enforce HTTPS.

• Alternatives: Cloudflare Pages or Vercel (free tiers). Netlify is blocked by a credit limit on your account.

Part 6 - Money, pricing and funding plan

AcreForge can earn money, but payments are not active yet. Do these in order; spend little until the waitlist proves demand.

Step	Action	Approx. cost
1	Buy a domain and point it at Pages	$20/yr
2	Share the link in land, homestead, rural real estate and equestrian groups; track waitlist issues as demand	$0
3	Small marketing test (a post or ad)	$50-100
4	If real demand: LLC plus lawyer-reviewed Terms and Privacy Policy	varies by state/lawyer
5	Create Stripe account and a Payment Link; paste into PAY_LINK	Stripe fees per sale
6	Build accounts + cloud saving (e.g. Supabase) so Pro has real value	free tier first
7	Trademark search/clearance for the AcreForge name; patent counsel only if advised	attorney quote

Pricing ideas (not active)

• Free: full planner, browser-only saving.

• Pro (planned): cloud saving, PDF/DXF export, parcel import helper; try $8 to $15/month or about $79/year.

• Early Access: discounted founder pricing for the first users.

• Other revenue: one-time professional plan packages, builder/realtor team licenses, affiliate links for surveyors and contractors (disclose affiliates).

Reinvesting

Put revenue back into: domain/hosting, legal setup, accounts and cloud saving, then better parcel data features. Keep personal and business finances separate (LLC and a business bank account) before taking payments.

Crowdfunding

CAMPAIGN.md contains an unpublished, support-only GoFundMe draft with no invented goals, rewards or dates. A budget worksheet and organizer setup checklist are in the project files.

Part 7 - Legal, privacy and safety

• AcreForge is a working name and is NOT trademark-cleared. Run a USPTO search and get attorney advice before spending heavily on branding.

• Never claim 'patented' or 'patent pending' unless a patent application is actually filed.

• Keep the disclaimer everywhere: conceptual planning only; not a survey, engineering, permitting or legal advice; parcel lines must come from an assessor/GIS or survey.

• Attribution required: USGS The National Map, AWS Terrain Tiles, OpenStreetMap contributors (ODbL), U.S. Census boundaries, MapLibre GL (BSD license, included in vendor/MAPLIBRE-LICENSE.txt).

• Privacy: data stays in the user's browser; map requests go to USGS/AWS/Overpass. Details in data-notice.html (Appendix A).

• Do not take payments or collect personal data until Terms, a Privacy Policy and a payment processor are in place.

• The repository is public: never commit passwords, API keys, Stripe secrets or customer data.

• Waitlist posts are public GitHub issues; do not ask users for private info.

• Copyright: your own code and text are yours; keep the third-party licenses and credits.

• See your separate AcreForge-Ownership-and-Launch-Packet.pdf for the full ownership and launch guidance (consult a licensed attorney for advice).

Part 8 - Troubleshooting and next steps

Problem	Try this
Site shows an old version	Ctrl+F5; wait 1-2 minutes after a push; check the Pages build with gh api repos/jacobbaltins/acreforge/pages/builds/latest
Can't move objects	Reload; pick a smaller acreage in Settings; make sure the address has no ?edit=0; click on the object itself, not empty grid
Map blank or no imagery	Needs internet; USGS/AWS tiles may be temporarily down
Lost saved project	Saved in one browser only; use the same browser profile; export a backup before clearing data
git push fails 'author identity unknown'	Run git config user.name and user.email
Netlify deploy blocked	Account credit limit; use GitHub Pages
Stripe webhook not working	It cannot run on GitHub Pages; needs Netlify/Vercel/Cloudflare functions

Not done yet

- Payments, accounts and cloud saving
- Custom domain, LLC, reviewed Terms/Privacy, trademark clearance
- Testing DXF export and 3D buildings in a real city
- Mobile polish beyond the current layout

Appendix A - Source code

Files are printed in a small monospace font and wrapped at 154 characters; a trailing marker "~" at a line end means the line continued (it is a wrap, not part of the code). The attached zip has the exact originals, which is the safest way to restore. us-states.js and the MapLibre library are not printed.

A: index.html (entire app)

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<meta name="theme-color" content="#0a110f">
<title>AcreForge | U.S. Land & Estate Planner</title>
<link rel="stylesheet" href="./vendor/maplibre-gl.css">
<style>
:root{color-scheme:dark;--bg:#09110e;--panel:#111c18;--panel2:#17251f;--line:#283a31;--text:#ecf2ee;--muted:#9aa9a0;--green:#34d399;--amber:#fbbf24;--~
blue:#60a5fa;--rose:#fb7185;--purple:#e879f9;--radius:15px;font:15px/1.5 Inter,ui-sans-serif,system-ui,-apple-system,"Segoe UI",sans-serif}
*{box-sizing:border-box}body{margin:0;background:radial-gradient(ellipse at 60% -15%,#1a3025 0,transparent ~
42%),var(--bg);color:var(--text)}button,input,select,textarea{font:inherit;color:inherit}button{cursor:pointer}button:focus-visible,input:focus-~
visible,select:focus-visible,textarea:focus-visible,[tabindex]:focus-visible{outline:2px solid var(--green);outline-~
offset:2px}button:disabled{opacity:.45;cursor:not-allowed}
.shell{min-height:100vh;display:grid;grid-template-columns:240px minmax(0,1fr)}aside{position:sticky;top:0;height:100vh;background:#0c1511;border-~
right:1px solid var(--line);padding:20px 14px;display:flex;flex-direction:column;gap:18px;z-index:5}.brand{padding:5px 10px 14px}.brand h1{font-~
size:17px;line-height:1.25;margin:8px 0 4px}.brand p{font-size:11px;color:var(--muted);margin:0}.brandmark{font-~
size:22px;color:var(--green)}.nav{display:grid;gap:4px}.nav button{border:0;text-align:left;background:transparent;border-radius:9px;padding:9px ~
10px;color:#becac2;font-size:13px;display:flex;gap:10px;align-items:center}.nav button:hover{background:#17231d}.nav ~
button.active{background:#193428;color:#71edb9;font-weight:650}.nav .ico{width:20px;text-align:center;font-size:15px}@media(max-~
height:760px){.sidefoot{display:none}.nav button{padding:6px 10px}aside{overflow-y:auto}}.sidefoot{margin-top:auto;border-top:1px solid ~
var(--line);padding:12px 9px 0;color:var(--muted);font-size:11px}.main{min-width:0}.topbar{height:66px;display:flex;align-items:center;gap:12px;padding:0 ~
28px;border-bottom:1px solid #24352c;background:#0b1410cc;position:sticky;top:0;z-index:4;backdrop-filter:blur(12px)}.topbar strong{font-~
size:13px}.crumb{font-size:12px;color:var(--muted)}.push{margin-left:auto}.save-state{font-size:11px;color:var(--muted);display:flex;align-~
items:center;gap:7px}.dot{width:7px;height:7px;background:var(--green);border-radius:50%}.content{max-width:1500px;margin:auto;padding:26px 28px ~
60px}.page-head{display:flex;align-items:flex-start;justify-content:space-between;gap:20px;margin-bottom:21px}.eyebrow{text-transform:uppercase;letter-~
spacing:.12em;color:var(--green);font-size:10px;font-weight:750}.page-head h2{font-size:26px;line-height:1.2;margin:5px 0}.page-head ~
p{color:var(--muted);font-size:13px;margin:0;max-width:740px}.actions{display:flex;gap:8px;flex-wrap:wrap}.btn{border:1px solid ~
var(--line);background:#19251f;border-radius:9px;padding:8px 12px;font-size:12px;font-~
weight:600}.btn:hover{background:#22352a}.btn.primary{background:#0d8b5a;border-~
color:#0d8b5a;color:white}.btn.primary:hover{background:#0fa36a}.btn.danger{color:#ffc0c9;border-~
color:#64333b;background:#29171b}.btn.ghost{background:transparent}.grid{display:grid;gap:14px}.g2{grid-template-columns:repeat(2,minmax(0,1fr))}.g3{grid-~
template-columns:repeat(3,minmax(0,1fr))}.g4{grid-template-columns:repeat(4,minmax(0,1fr))}.card{background:linear-~
gradient(145deg,#14211b,#101a15);border:1px solid var(--line);border-radius:var(--radius);padding:17px;min-width:0}.card h3{font-size:14px;margin:0 0 ~
12px}.card h4{font-size:12px;margin:0 0 7px}.muted{color:var(--muted)}.small{font-size:11px}.tiny{font-size:10px}.kpi{min-height:112px}.kpi ~
.label{color:var(--muted);font-size:11px}.kpi .value{font-size:25px;font-weight:720;margin-top:8px;letter-spacing:-.03em}.kpi .hint{font-~
size:10px;color:var(--muted);margin-top:3px}.green{color:var(--green)}.amber{color:var(--amber)}.blue{color:var(--blue)}.purple{color:var(--~
purple)}.rose{color:var(--rose)}.row{display:flex;align-items:center;gap:10px}.between{display:flex;align-items:center;justify-content:space-~
between;gap:12px}.stack{display:grid;gap:12px}.pill{display:inline-flex;align-items:center;gap:5px;padding:3px 8px;border-~
radius:999px;background:#1c3026;color:#91e7bd;font-size:10px;font-~
weight:700}.pill.warn{background:#352b16;color:#f5d37a}.pill.rose{background:#351f24;color:#ffa3ae}.divider{border:0;border-top:1px solid ~
var(--line);margin:13px 0}.field{display:grid;gap:5px}.field label,.field .label{font-size:11px;color:#bac7be}.field input,.field select,.field ~
textarea,.input{width:100%;background:#0c1511;border:1px solid #304238;border-radius:8px;padding:8px 10px;font-size:12px}.field ~
textarea{resize:vertical;min-height:70px}.inline-fields{display:grid;grid-template-columns:repeat(auto-fit,minmax(135px,1fr));gap:10px}.notice{border:1px ~
solid #3c4831;background:#242616;border-radius:10px;padding:11px 13px;color:#ddd9ad;font-size:11px}.notice.greenish{border-~
color:#28513e;background:#10271d;color:#b5e9ce}.notice.blueish{border-color:#29445e;background:#101e2b;color:#c3d9f3}.empty{padding:25px;text-~
align:center;color:var(--muted);border:1px dashed var(--line);border-radius:12px}.progress{height:7px;background:#28352e;border-~
radius:99px;overflow:hidden}.progress i{display:block;height:100%;background:var(--green);border-radius:99px}.table-wrap{overflow:auto;border:1px solid ~
var(--line);border-radius:10px}table{width:100%;border-collapse:collapse;font-size:11px;min-width:560px}th,td{text-align:left;padding:9px 10px;border-~
bottom:1px solid #26372e;vertical-align:middle}th{background:#142019;color:#aab8af;font-weight:650;position:sticky;top:0}tr:last-child td{border-~
bottom:0}td input,td select{background:#0c1511;border:1px solid #304238;border-radius:6px;padding:5px 6px;font-size:11px;max-width:180px}
.designer-layout{display:grid;grid-template-columns:minmax(0,1fr) 335px;gap:16px;align-items:start}.map-~
card{padding:14px;overflow:hidden;background:linear-gradient(160deg,#18251d,#101912);box-shadow:0 20px 55px #0003}.map-toolbar{display:flex;gap:6px;align-~
items:center;flex-wrap:wrap;padding:3px 3px 12px}.toolrow{display:flex;gap:5px;flex-wrap:wrap}.toolbtn{border:1px solid ~
var(--line);background:#0c1511;border-radius:7px;padding:6px 8px;font-size:10px}.toolbtn.active{border-~
color:#35d399;color:#8de9bc;background:#163225}.map-viewport{height:clamp(390px,46vw,620px);background:#0a130f;border:1px solid #34493a;border-~
radius:12px;overflow:hidden;position:relative;box-shadow:inset 0 0 38px #0006;cursor:grab;touch-action:none}.map-viewport.is-panning{cursor:grabbing}.map-~
viewport svg{display:block;width:100%;height:100%;touch-action:none}.map-meta{display:flex;gap:14px;flex-wrap:wrap;color:var(--muted);font-~
size:10px;padding:10px 3px 0}.map-side{display:grid;gap:12px}.map-side .card{padding:14px}.object-list{max-~
height:300px;overflow:auto;display:grid;gap:4px}.object-row{display:flex;align-items:center;gap:7px;border:1px solid transparent;border-~
radius:7px;background:transparent;padding:7px;text-align:left;font-size:10px;width:100%}.object-row:hover,.object-row.selected{background:#1b2c22;border-~
color:#315442}.swatch{width:10px;height:10px;border-radius:3px;flex:none}.palette-filters{display:flex;gap:5px;overflow:auto;padding:9px 0;scrollbar-~
width:thin}.palette-filter{flex:none;border:1px solid #314437;background:#0b1510;color:#b4c4b8;border-radius:999px;padding:5px 8px;font-size:9px}.palette-~
filter.active{background:#18402e;border-color:#2d8a5d;color:#a6f1c6}.palette-grid{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:6px;max-~
height:330px;overflow:auto;padding:2px 3px 2px 0}.palette-item{min-width:0;display:grid;grid-template-columns:26px 1fr;grid-template-rows:auto ~
auto;column-gap:7px;align-items:center;text-align:left;border:1px solid #273a2e;background:#0c1511;border-radius:8px;padding:7px}.palette-~
item:hover{background:#17281e;border-color:#477652}.palette-item.active{background:#183526;border-color:#46cc88;box-shadow:inset 0 0 0 1px ~
#46cc8833}.palette-glyph{grid-row:1/3;width:25px;height:25px;display:grid;place-items:center;border-radius:7px;background:color-mix(in ~
srgb,var(--swatch),#0b1510 68%);color:var(--swatch);font-size:10px;font-weight:800;border:1px solid color-mix(in srgb,var(--swatch),transparent ~
58%)}.palette-name{font-size:9px;line-height:1.2;overflow:hidden;text-overflow:ellipsis;display:-webkit-box;-webkit-line-clamp:2;-webkit-box-~
orient:vertical}.palette-size{font-size:8px;color:#819287;margin-top:3px}.palette-head{grid-column:1/-1;font-size:9px;font-weight:700;letter-~
spacing:.08em;text-transform:uppercase;color:#8de9bc;padding:8px 2px 2px;position:sticky;top:0;background:#101912;z-index:1}.palette-head ~
span{color:#819287;font-weight:400}.palette-hint{margin-top:8px;padding:8px;border-radius:8px;background:#14241a;color:#a7c3ad;font-size:9px}.layer-~
grid{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:7px}.layer-chip{display:flex;align-items:center;gap:6px;font-~
size:9px;color:#b6c3ba}.layer-chip input{accent-color:#34d399;margin:0}.scene-card{grid-column:1/-1}.scene-~
canvas{width:100%;height:210px;display:block;background:radial-gradient(ellipse at 50% 68%,#183323,#0a130f 74%);border-radius:9px;touch-~
action:none;cursor:grab}.scene-canvas:active{cursor:grabbing}
.tabbar{display:flex;gap:5px;overflow:auto;border-bottom:1px solid var(--line);margin-bottom:14px}.tabbar button{white-~
space:nowrap;background:transparent;border:0;color:var(--muted);padding:9px 10px;font-size:11px;border-bottom:2px solid transparent}.tabbar ~
button.active{color:var(--green);border-color:var(--green)}.check-row{display:flex;align-items:flex-start;gap:9px;padding:8px 0;border-bottom:1px solid ~
#26372e;font-size:11px}.check-row:last-child{border:0}.check-row input[type=checkbox]{accent-color:#10b981;margin-top:2px}.check-row .task-~
text{flex:1}.check-row small{color:var(--muted);display:block;font-size:10px;margin-top:2px}.phase{display:grid;grid-template-columns:110px 1fr ~
auto;gap:12px;align-items:center;padding:12px 0;border-bottom:1px solid var(--line)}.phase:last-child{border:0}.phase h4{margin:0;font-size:12px}.phase ~
p{margin:3px 0 0;font-size:10px;color:var(--muted)}.barline{height:8px;background:#25352c;border-radius:99px;overflow:hidden}.barline ~
i{display:block;height:100%;background:linear-gradient(90deg,#10b981,#6ee7b7)}.banner{margin-top:18px;padding:12px;border:1px solid #35453b;border-~
radius:10px;background:#101a14;color:#b4c2b9;font-size:10px}.manual-search{max-width:390px}.hide{display:none!important}.mobile-nav{display:none}.range-~
row{display:grid;grid-template-columns:1fr 95px;align-items:center;gap:12px}.range-row input[type=range]{width:100%;accent-color:#10b981}.range-row ~
input[type=number]{width:100%}.key-value{display:flex;justify-content:space-between;gap:15px;padding:7px 0;border-bottom:1px solid #26372e;font-~
size:11px}.key-value:last-child{border:0}.report-copy{line-height:1.75;font-size:12px}.object-actions{display:flex;gap:6px;flex-wrap:wrap}.file-~
hidden{display:none}
@media(max-width:1050px){.g4{grid-template-columns:repeat(2,minmax(0,1fr))}.designer-layout{grid-template-columns:1fr}.map-side{display:grid;grid-~
template-columns:repeat(2,minmax(0,1fr));gap:12px}.palette-card{grid-column:1/-1}.map-viewport{height:510px}}
@media(max-width:720px){.shell{display:block}aside{position:fixed;inset:auto 0 0;height:auto;width:100%;padding:6px 7px calc(6px + env(safe-area-inset-~
bottom));border:0;border-top:1px solid var(--line);background:#0a130ff5}.brand,.sidefoot{display:none}.nav{display:flex;gap:2px;overflow-x:auto;scrollbar-~
width:none}.nav button{display:grid;justify-items:center;gap:0;padding:5px 3px;min-width:58px;font-size:10px;flex:0 0 auto;white-space:nowrap}.nav ~
.ico{font-size:16px}.topbar{height:55px;padding:0 14px}.content{padding:18px 13px 100px}.page-head{display:block}.page-head h2{font-size:23px}.page-head ~
.actions{margin-top:12px}.g2,.g3{grid-template-columns:1fr}.g4{grid-template-columns:repeat(2,minmax(0,1fr));gap:8px}.card{padding:14px}.kpi{min-~
height:96px}.kpi .value{font-size:21px}.map-side{grid-template-columns:1fr}.map-viewport{height:400px}.designer-layout{display:flex;flex-~
direction:column}.designer-layout>div{width:100%}.palette-grid{max-height:360px}.phase{grid-template-columns:85px 1fr}.phase>strong{grid-column:2}.save-~
state{display:none}.table-wrap table{min-width:540px}.crumb{display:none}}
```

```
@media(prefers-reduced-motion:reduce){*,*::before,*::after{scroll-behavior:auto!important;animation-duration:.01ms!important;transition-~
duration:.01ms!important}}
@media print{body{background:#fff;color:#17251f}.shell{display:block}aside,.topbar,.actions,.nav{display:none!important}.content{padding:0;max-~
width:none}.card{background:#fff;border:1px solid #bbb;break-~
inside:avoid}.banner,.notice{background:#fff;color:#333}.hide{display:block!important}.page{display:block!important}svg{max-width:100%}}
#siteMap text{font-family:inherit}
#usaMap{background:#0a130f;border-radius:12px}
.map-hud{position:absolute;left:12px;top:12px;display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:4px 10px;width:min(440px,calc(100% - ~
24px));box-sizing:border-box;padding:7px 10px;border:1px solid #405347;border-radius:9px;background:#0a130fe8;color:#d4e1d6;font-size:10px;line-~
height:1.25;backdrop-filter:blur(8px);pointer-events:none}.map-hud span{min-width:0;white-space:normal}.map-hud .toolbtn{pointer-events:auto}.map-~
card:has(#usaMap) .map-hud{display:none}
.terrain-map{position:absolute;inset:0;overflow:hidden;background:#dfe6dc;cursor:crosshair;touch-action:none}.terrain-~
tiles{position:absolute;inset:0;will-change:transform}.terrain-tiles img{position:absolute;width:256px;height:256px;max-width:none;user-~
select:none;pointer-events:none}.terrain-map-marker{position:absolute;left:50%;top:50%;width:32px;height:42px;transform:translate(-50%,-100%);filter:drop-~
shadow(0 2px 3px #07110d99);pointer-events:none}.terrain-map-status{position:absolute;left:12px;top:12px;max-width:min(430px,calc(100% - ~
24px));padding:8px 10px;border:1px solid #35483a;border-radius:9px;background:#0b1510e8;color:#e2ece2;font-size:10px;line-height:1.4;pointer-~
events:none}.terrain-map-credit{position:absolute;right:8px;bottom:7px;padding:3px 6px;border-radius:4px;background:#f8faf4e8;color:#243b2d;font-~
size:9px}.map-card:has(#usaMap) .map-hud,.map-card:has(#terrainMap) .map-hud{display:none}
.map-card:has(#usaMap) .map-toolbar .toolrow,.map-card:has(#usaMap) .map-toolbar [data-action^="zoom-"],.map-card:has(#usaMap) .map-toolbar ~
#gridSelect,.map-card:has(#usaMap) .map-toolbar [data-action="toggle-grid"],.map-card:has(#usaMap) #mapViewport~.map-meta>:nth-child(-n+4),.map-~
card:has(#terrainMap) .map-toolbar .toolrow,.map-card:has(#terrainMap) .map-toolbar [data-action^="zoom-"],.map-card:has(#terrainMap) .map-toolbar ~
#gridSelect,.map-card:has(#terrainMap) .map-toolbar [data-action="toggle-grid"],.map-card:has(#terrainMap) #mapViewport~.map-meta>:nth-~
child(-n+4){display:none}
.state-hit{fill:transparent;stroke:transparent;cursor:pointer;pointer-events:all}.state-hit:hover,.state-hit:focus{fill:#c9e7bc33;stroke:#efffe2;stroke-~
width:2;outline:none}
.terrain-map{cursor:grab;background:#dfe6dc}.terrain-map:active{cursor:grabbing}.terrain-canvas{position:absolute;inset:0}.terrain-map-~
status{z-index:2;max-width:min(480px,calc(100% - 24px))}.terrain-map-credit{z-index:2;left:8px;right:auto;max-width:calc(100% - 16px)}.maplibregl-ctrl-~
group{background:#f9faf8!important}.maplibregl-ctrl-group button{color:#17251f!important}.terrain-feature-panel{max-height:280px;overflow:auto}.terrain-~
feature-row{display:grid;grid-template-columns:1fr auto;align-items:center;gap:5px;padding:7px;border-bottom:1px solid #26372e;font-size:10px}.terrain-~
feature-row button{border:0;background:transparent;text-align:left;padding:2px;color:#d6e5da}.terrain-feature-row.selected{background:#1b2c22}.terrain-~
feature-measure{color:#9baea1;font-size:9px}.terrain-hint{font-size:10px;color:#b9c9bd}
</style>
<style>
.terrain-osm-status{position:absolute;left:8px;bottom:30px;z-index:2;max-width:min(420px,calc(100% - 16px));padding:5px 8px;border:1px solid ~
#34483b;border-radius:6px;background:#0b1510e8;color:#e2ece2;font-size:9px;line-height:1.35;pointer-events:none}
.terrain-reference-options{position:relative}.terrain-reference-options summary{list-style:none;cursor:pointer;border:1px solid var(--line);border-~
radius:7px;padding:6px 9px;background:#0c1511;font-size:10px}.terrain-reference-options summary::-webkit-details-marker{display:none}.terrain-reference-~
options[open] .terrain-reference-checks{position:absolute;z-index:3;top:calc(100% + 5px);left:0;min-~
width:190px;display:grid;gap:8px;padding:10px;border:1px solid var(--line);border-radius:8px;background:#101a15;box-shadow:0 8px 24px #0008}.terrain-~
reference-checks{display:none}.terrain-reference-checks label{display:flex;align-items:center;gap:8px;font-size:11px}.terrain-reference-checks ~
input{accent-color:var(--green);margin:0}
.terrain-guidance{align-self:center;max-width:420px;color:var(--muted);font-size:10px;line-height:1.35}
.terrain-map-credit a{color:inherit}
.scene-canvas{height:clamp(270px,34vw,400px)}
.scene-canvas.is-panning{cursor:grabbing}
.designer-details{padding:0!important;overflow:hidden}
.designer-details>summary{display:flex;align-items:center;justify-content:space-between;gap:10px;padding:13px 14px;cursor:pointer;list-style:none;font-~
size:12px;font-weight:650}
.designer-details>summary::-webkit-details-marker{display:none}
.designer-details>summary::after{content:"+";color:var(--muted);font-size:16px;font-weight:400}
.designer-details[open]>summary::after{content:"–"}
.designer-details>.details-body{padding:0 14px 14px}
.scene-card{grid-column:1/-1}
.scene-options{position:relative}
.scene-options summary{cursor:pointer;list-style:none;border:1px solid var(--line);border-radius:7px;padding:5px 8px;background:#0c1511;font-size:10px}
.scene-options summary::-webkit-details-marker{display:none}
.scene-options[open] .row{position:absolute;right:0;bottom:calc(100% + 6px);padding:7px;border:1px solid var(--line);border-~
radius:8px;background:#101a15;box-shadow:0 8px 24px #0008}
</style>
<link rel="icon" href="data:image/svg+xml,%3Csvg xmlns=%27http://www.w3.org/2000/svg%27 viewBox=%270 0 32 32%27%3E%3Crect width=%2732%27 height=%2732%27 ~
rx=%276%27 fill=%27%23234d2f%27/%3E%3Cpath d=%27M4 24l8-12 5 7 3-4 8 9z%27 fill=%27%23e8d9a8%27/%3E%3C/svg%3E"><meta name="description" content="Plan U.S.~
 rural land, farms, estates, homes, and buildings with site layouts, terrain, budgets, and reports."><meta property="og:title" content="AcreForge | U.S. ~
Land &amp;amp; Estate Planner"><meta property="og:description" content="Plan U.S. rural land, farms, estates, homes, and buildings with site layouts, terrain,~
 budgets, and reports."><meta property="og:type" content="website"><meta name="theme-color" content="#234d2f"></head>
<body>
<div class="shell">
<aside>
 <div class="brand"><div class="brandmark">❋</div><h1>AcreForge</h1><p>Nationwide land &amp;amp; estate planner</p></div>
 <nav class="nav" id="nav" aria-label="Main navigation"></nav>
 <div class="sidefoot"><div class="row"><span class="dot"></span><span>Saved in this browser</span></div><div style="margin-top:7px">No account or server ~
required</div></div>
</aside>
<main class="main">
 <header class="topbar"><strong id="topTitle">AcreForge</strong><span class="crumb" id="topCrumb"></span><span class="push"></span><span class="save-~
state" id="saveState" role="status"><i class="dot"></i> Saved locally</span></header>
 <div class="content" id="content"></div>
</main>
</div>
<input class="file-hidden" id="importFile" type="file" accept=".json,application/json">
<input class="file-hidden" id="parcelFile" type="file" accept=".geojson,.json,.kml,.xml,application/geo+json,application/vnd.google-earth.kml+xml">
<script src="./vendor/maplibre-gl.js"></script>
<script src="./us-states.js"></script>
<script>
(()=>{"use strict";
const STORAGE="oregon-estate-planner-v1",RECOVERY="oregon-estate-planner-recovery-v1",CELL=40;
const navItems=[
 ["about","■","About"],["dashboard","■","Dashboard"],["designer","■","Site Designer"],["buildings","■","Buildings"],["homes","■","Lot Builder"],["infrastr~
ucture","■","Infrastructure"],["financials","$","Financials"],["operations","■","Operations"],["compliance","✔","Compliance"],["scenarios","■","Scenarios"~
],["reports","■","Reports"],["manual","■","Manual"],["settings","■","Settings"]
];
const templates={
 barn:{name:"20-Stall Main Barn",cat:"Building",w:144,h:48,height:24,color:"#b45309",label:"B",cost:1200000,method:"fixed",stalls:20,phase:"1"},
 arena:{name:"Indoor Arena",cat:"Building",w:200,h:100,height:38,color:"#64748b",label:"A",cost:900000,method:"fixed",phase:"1"},
 shop:{name:"Workshop",cat:"Building",w:100,h:60,height:22,color:"#0891b2",label:"W",cost:450000,method:"fixed",phase:"2"},
 feedstore:{name:"Feed & Tack Store",cat:"Building",w:80,h:50,height:18,color:"#b7791f",label:"FT",cost:180000,method:"fixed",phase:"2"},
 farmstore:{name:"Farm & General Store",cat:"Building",w:72,h:48,height:16,color:"#c2410c",label:"GS",cost:210000,method:"fixed",phase:"2"},
 tackshop:{name:"Tack Shop",cat:"Building",w:48,h:36,height:14,color:"#a16207",label:"TS",cost:125000,method:"fixed",phase:"2"},
 event:{name:"Event Barn & Food Hall",cat:"Building",w:120,h:80,height:30,color:"#c026d3",label:"E",cost:1600000,method:"fixed",phase:"3"},
 eventcenter:{name:"Event ~
Center",cat:"Building",w:180,h:120,height:36,color:"#d946ef",label:"EC",cost:2800000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 expo:{name:"Exposition Hall",cat:"Building",w:200,h:100,height:32,color:"#a855f7",label:"EX",cost:1900000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 fairbarn:{name:"Livestock Exhibition ~
Barn",cat:"Building",w:160,h:80,height:24,color:"#c084fc",label:"LB",cost:1250000,method:"fixed",phase:"2",layer:"Fairgrounds"},
 grandstand:{name:"Grandstand",cat:"Venue",w:140,h:45,height:28,color:"#f97316",label:"GS",cost:620000,method:"fixed",phase:"3",layer:"Fairgrounds"},
```

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 rodeoring:{name:"Rodeo / Show ~
Ring",cat:"Venue",w:180,h:120,height:2,color:"#d97706",label:"R",cost:240000,method:"area",phase:"2",layer:"Fairgrounds",shape:"circle"},
 amphitheater:{name:"Amphitheater",cat:"Venue",w:150,h:110,height:20,color:"#e879f9",label:"AM",cost:480000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 pavilion:{name:"Open-Air Pavilion",cat:"Venue",w:100,h:70,height:20,color:"#fb923c",label:"PV",cost:210000,method:"fixed",phase:"2",layer:"Fairgrounds"},
 eventlawn:{name:"Event Lawn",cat:"Venue",w:240,h:160,height:1,color:"#65a30d",label:"EL",cost:3,method:"area",phase:"3",layer:"Fairgrounds"},
 plaza:{name:"Festival Plaza",cat:"Venue",w:140,h:100,height:1,color:"#d6a76c",label:"PL",cost:14,method:"area",phase:"3",layer:"Fairgrounds"},
 vendor:{name:"Vendor Booth Row",cat:"Venue",w:160,h:24,height:10,color:"#fb7185",label:"VB",cost:160000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 foodcourt:{name:"Food Truck Court",cat:"Venue",w:120,h:90,height:1,color:"#f59e0b",label:"FC",cost:12,method:"area",phase:"3",layer:"Fairgrounds"},
 stage:{name:"Performance Stage",cat:"Venue",w:48,h:32,height:18,color:"#ec4899",label:"ST",cost:95000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 ticket:{name:"Ticket & Welcome Gate",cat:"Venue",w:36,h:24,height:12,color:"#38bdf8",label:"G",cost:45000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 restroom:{name:"Public Restroom ~
Pavilion",cat:"Building",w:48,h:32,height:12,color:"#60a5fa",label:"WC",cost:165000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 bleachers:{name:"Portable Bleacher ~
Zone",cat:"Venue",w:100,h:24,height:12,color:"#fb923c",label:"BL",cost:85000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 camping:{name:"Tent Camping Field",cat:"Camping",w:240,h:160,height:1,color:"#14b8a6",label:"TC",cost:4,method:"area",phase:"3",layer:"Fairgrounds"},
 rv:{name:"RV / Camper Area",cat:"Camping",w:220,h:120,height:1,color:"#0d9488",label:"RV",cost:16,method:"area",phase:"3",layer:"Fairgrounds"},
 checkin:{name:"Event Check-in & First ~
Aid",cat:"Venue",w:40,h:30,height:12,color:"#f43f5e",label:"CI",cost:72000,method:"fixed",phase:"3",layer:"Fairgrounds"},
 stand:{name:"Farm Stand Market",cat:"Building",w:50,h:30,height:16,color:"#e11d48",label:"F",cost:150000,method:"fixed",phase:"3"},
 yurt:{name:"Caretaker Yurts",cat:"Building",w:70,h:30,height:12,color:"#eab308",label:"Y",cost:90000,method:"fixed",phase:"2"},
 pasture:{name:"Rotational Pasture",cat:"Pasture",w:320,h:320,height:1,color:"#22a652",label:"P",cost:6000,method:"acre",phase:"1"},
 road:{name:"Gravel Road",cat:"Road",w:120,h:20,height:1,color:"#8b8478",label:"R",cost:40,method:"linear",phase:"1"},
 mainroad:{name:"Primary Gravel Access Road",cat:"Road",w:1600,h:24,height:1,color:"#8b8478",label:"AR",cost:45,method:"linear",phase:"1"},
 farmroad:{name:"Farm Service Road",cat:"Road",w:600,h:18,height:1,color:"#a0957d",label:"SR",cost:28,method:"linear",phase:"1"},
 curvedroad:{name:"Curved Gravel Road",cat:"Road",w:180,h:60,height:1,color:"#8b8478",label:"CR",cost:40,method:"linear",phase:"1",layer:"Roads"},
 path:{name:"Walking Path",cat:"Path",w:80,h:8,height:1,color:"#d6c7a1",label:"",cost:10,method:"linear",phase:"2"},
 fence:{name:"Fencing",cat:"Fence",w:80,h:8,height:4,color:"#a16207",label:"",cost:12,method:"linear",phase:"1"},
 railfence:{name:"Post-and-Rail Fence",cat:"Fence",w:120,h:12,height:4,color:"#9a6b3e",label:"PR",cost:24,method:"linear",phase:"1"},
 boardfence:{name:"Board Fence",cat:"Fence",w:120,h:12,height:5,color:"#b0895b",label:"BF",cost:32,method:"linear",phase:"1"},
 wirefence:{name:"Woven-Wire Horse Fence",cat:"Fence",w:120,h:10,height:5,color:"#7c6f55",label:"WW",cost:18,method:"linear",phase:"1"},
 electricfence:{name:"Electric Tape Fence",cat:"Fence",w:120,h:8,height:4,color:"#eab308",label:"EF",cost:8,method:"linear",phase:"1"},
 noclimbfence:{name:"No-Climb Mesh Fence",cat:"Fence",w:120,h:10,height:5,color:"#64748b",label:"NC",cost:22,method:"linear",phase:"1"},
 parking:{name:"Parking Area",cat:"Parking",w:180,h:100,height:1,color:"#475569",label:"PK",cost:8,method:"area",phase:"3"},
 trailer:{name:"Trailer Parking",cat:"Parking",w:180,h:60,height:1,color:"#334155",label:"TP",cost:10,method:"area",phase:"2"},
 busparking:{name:"Bus & Shuttle Drop-~
off",cat:"Parking",w:140,h:60,height:1,color:"#475569",label:"BUS",cost:12,method:"area",phase:"3",layer:"Fairgrounds"},
 overflow:{name:"Overflow Grass ~
Parking",cat:"Parking",w:240,h:180,height:1,color:"#647a50",label:"OP",cost:2,method:"area",phase:"3",layer:"Fairgrounds"},
 manure:{name:"Manure Storage",cat:"Operations",w:80,h:60,height:8,color:"#854d0e",label:"M",cost:45000,method:"fixed",phase:"1"},
 hay:{name:"Hay Storage",cat:"Operations",w:100,h:60,height:20,color:"#ca8a04",label:"H",cost:85000,method:"fixed",phase:"1"},
 yard:{name:"Equipment Yard",cat:"Operations",w:100,h:80,height:1,color:"#78716c",label:"EY",cost:25000,method:"area",phase:"2"},
 outdoor:{name:"Outdoor Arena",cat:"Arena",w:200,h:100,height:1,color:"#a16207",label:"OA",cost:120000,method:"area",phase:"2"},
 roundpen:{name:"Small Circular Round Pen",cat:"Arena",w:60,h:60,height:1,color:"#65a30d",label:"RP",cost:15000,method:"fixed",phase:"2",shape:"circle"},
 largaroundpen:{name:"Large Circular Round ~
Pen",cat:"Arena",w:100,h:100,height:1,color:"#4d7c0f",label:"LRP",cost:24000,method:"fixed",phase:"2",shape:"circle"},
 circlepaddock:{name:"Circular Turnout ~
Pen",cat:"Pasture",w:140,h:140,height:1,color:"#16a34a",label:"CP",cost:1500,method:"acre",phase:"1",shape:"circle"},
 paddock:{name:"Turnout Paddock",cat:"Pasture",w:160,h:120,height:1,color:"#16a34a",label:"T",cost:1500,method:"acre",phase:"1"},
 pond:{name:"Pond / Water Feature",cat:"Water",w:120,h:80,height:1,color:"#0ea5e9",label:"W",cost:4,method:"area",phase:"2"},
 well:{name:"Well Location",cat:"Utility",w:20,h:20,height:8,color:"#38bdf8",label:"W",cost:25000,method:"fixed",phase:"1"},
 tank:{name:"Water Tank",cat:"Utility",w:40,h:40,height:24,color:"#0284c7",label:"WT",cost:35000,method:"fixed",phase:"1"},
 septic:{name:"Septic Field",cat:"Utility",w:160,h:100,height:1,color:"#a3a38d",label:"S",cost:45000,method:"area",phase:"2"},
 reserve:{name:"Septic Reserve Area",cat:"Utility",w:160,h:100,height:1,color:"#d4d4aa",label:"SR",cost:0,method:"area",phase:"2"},
 utility:{name:"Utility Corridor",cat:"Utility",w:200,h:16,height:1,color:"#facc15",label:"U",cost:90,method:"linear",phase:"1"},
 firewater:{name:"Fire-Water Storage",cat:"Utility",w:60,h:60,height:20,color:"#ef4444",label:"FW",cost:60000,method:"fixed",phase:"1"},
 drainage:{name:"Stormwater / Drainage",cat:"Water",w:100,h:60,height:1,color:"#14b8a6",label:"D",cost:2,method:"area",phase:"1"},
 tree:{name:"Tree / Shade ~
Tree",cat:"Landscape",w:24,h:24,height:35,color:"#16a34a",label:"T",cost:450,method:"fixed",phase:"1",layer:"Landscape",shape:"circle"},
 bush:{name:"Native Bush",cat:"Landscape",w:14,h:14,height:6,color:"#4d7c0f",label:"",cost:85,method:"fixed",phase:"1",layer:"Landscape",shape:"circle"},
 shrubcluster:{name:"Native Shrub ~
Cluster",cat:"Landscape",w:48,h:40,height:7,color:"#3f6b31",label:"SC",cost:260,method:"fixed",phase:"1",layer:"Landscape",shape:"circle"},
 grove:{name:"Tree Grove",cat:"Landscape",w:120,h:100,height:28,color:"#15803d",label:"TG",cost:450,method:"fixed",phase:"1",layer:"Landscape"},
 orchard:{name:"Orchard Block",cat:"Landscape",w:180,h:120,height:22,color:"#65a30d",label:"OR",cost:2200,method:"acre",phase:"2",layer:"Landscape"},
 oak:{name:"Oak (broadleaf ~
tree)",cat:"Landscape",w:40,h:40,height:50,color:"#2f8f3a",label:"OK",cost:600,method:"fixed",phase:"1",layer:"Landscape",shape:"circle"},
 conifer:{name:"Douglas-fir / ~
Conifer",cat:"Landscape",w:28,h:28,height:80,color:"#0f5132",label:"DF",cost:350,method:"fixed",phase:"1",layer:"Landscape",shape:"circle"},
 fruittree:{name:"Fruit / Nut ~
Tree",cat:"Landscape",w:20,h:20,height:18,color:"#84cc16",label:"FT",cost:120,method:"fixed",phase:"1",layer:"Landscape",shape:"circle"},
 forest:{name:"Forest Stand",cat:"Landscape",w:300,h:240,height:70,color:"#14532d",label:"FOR",cost:300,method:"acre",phase:"1",layer:"Landscape"},
 riparian:{name:"Riparian / Wetland ~
Buffer",cat:"Landscape",w:200,h:60,height:8,color:"#0f766e",label:"RB",cost:2500,method:"acre",phase:"2",layer:"Landscape"},
 meadow:{name:"Native Meadow / ~
Wildflowers",cat:"Landscape",w:200,h:150,height:2,color:"#c084fc",label:"MW",cost:1500,method:"acre",phase:"2",layer:"Landscape"},
 lawn:{name:"Lawn / Turf",cat:"Landscape",w:120,h:80,height:1,color:"#4ade80",label:"LN",cost:0.6,method:"area",phase:"2",layer:"Landscape"},
 flowerbed:{name:"Flower / Ornamental ~
Bed",cat:"Landscape",w:40,h:20,height:2,color:"#f472b6",label:"FB",cost:12,method:"area",phase:"2",layer:"Landscape"},
 croprows:{name:"Row Crop Field",cat:"Farm",w:300,h:200,height:3,color:"#ca8a04",label:"RC",cost:1200,method:"acre",phase:"2",layer:"Farmland"},
 hayfield:{name:"Hay Field",cat:"Farm",w:300,h:250,height:2,color:"#a3b83c",label:"HF",cost:900,method:"acre",phase:"1",layer:"Farmland"},
 vineyard:{name:"Vineyard",cat:"Farm",w:240,h:160,height:7,color:"#7c3aed",label:"VY",cost:18000,method:"acre",phase:"3",layer:"Farmland"},
 berries:{name:"Berry / Hazelnut Rows",cat:"Farm",w:200,h:120,height:6,color:"#be185d",label:"BR",cost:9000,method:"acre",phase:"3",layer:"Farmland"},
 vegplot:{name:"Vegetable Garden / Market ~
Plot",cat:"Farm",w:100,h:60,height:2,color:"#65a30d",label:"VG",cost:3,method:"area",phase:"2",layer:"Farmland"},
 greenhouse:{name:"Greenhouse / Hoop House",cat:"Farm",w:30,h:96,height:14,color:"#99f6e4",label:"GH",cost:30,method:"sqft",phase:"3",layer:"Farmland"},
 irrpond:{name:"Irrigation ~
Pond",cat:"Farm",w:120,h:90,height:1,color:"#38bdf8",label:"IP",cost:80000,method:"fixed",phase:"3",layer:"Farmland",shape:"circle"},
 silo:{name:"Silo / Grain ~
Bin",cat:"Farm",w:30,h:30,height:50,color:"#94a3b8",label:"SI",cost:40000,method:"fixed",phase:"3",layer:"Farmland",shape:"circle"},
 equipshed:{name:"Equipment / Tractor Shed",cat:"Farm",w:60,h:40,height:16,color:"#78716c",label:"ES",cost:45,method:"sqft",phase:"2",layer:"Farmland"},
 home:{name:"Custom Home (Home Builder)",cat:"Building",w:40,h:30,height:20,color:"#c9a27a",label:"H",cost:275,method:"sqft",phase:"1"},
 homelot:{name:"House Lot (single family)",cat:"City",w:60,h:100,height:24,color:"#fb923c",label:"HL",cost:210,method:"sqft",phase:"3",layer:"Town"},
 townhomes:{name:"Townhome Row",cat:"City",w:120,h:50,height:30,color:"#f59e0b",label:"TH",cost:190,method:"sqft",phase:"3",layer:"Town"},
 apartments:{name:"Apartment Building",cat:"City",w:140,h:90,height:45,color:"#e879f9",label:"APT",cost:230,method:"sqft",phase:"3",layer:"Town"},
 shopfront:{name:"Shop / Main Street Store",cat:"City",w:40,h:70,height:22,color:"#f43f5e",label:"SH",cost:220,method:"sqft",phase:"3",layer:"Town"},
 office:{name:"Office / Civic Building",cat:"City",w:100,h:70,height:40,color:"#60a5fa",label:"OF",cost:260,method:"sqft",phase:"3",layer:"Town"},
 school:{name:"School",cat:"City",w:200,h:120,height:22,color:"#facc15",label:"SCH",cost:300,method:"sqft",phase:"3",layer:"Town"},
 firestation:{name:"Fire / Emergency Station",cat:"City",w:90,h:60,height:24,color:"#ef4444",label:"FS",cost:320,method:"sqft",phase:"3",layer:"Town"},
 citypark:{name:"Town Park / Green",cat:"City",w:200,h:150,height:1,color:"#22c55e",label:"PK",cost:6,method:"area",phase:"3",layer:"Town"},
 plaza:{name:"Plaza / Town Square",cat:"City",w:100,h:100,height:1,color:"#d6d3d1",label:"PZ",cost:25,method:"area",phase:"3",layer:"Town"},
 citystreet:{name:"City Street",cat:"City",w:300,h:36,height:1,color:"#64748b",label:"ST",cost:420,method:"linear",phase:"3",layer:"Town"},
 citylot:{name:"Parking Lot (town)",cat:"City",w:120,h:80,height:1,color:"#475569",label:"PL",cost:9,method:"area",phase:"3",layer:"Town"},
 hedge:{name:"Hedge / Windbreak",cat:"Landscape",w:200,h:16,height:14,color:"#4d7c0f",label:"H",cost:18,method:"linear",phase:"1",layer:"Landscape"},
 garden:{name:"Native / Pollinator ~
Garden",cat:"Landscape",w:80,h:60,height:1,color:"#d946ef",label:"NG",cost:8,method:"area",phase:"2",layer:"Landscape"},
```

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 playground:{name:"Family Play Area",cat:"Venue",w:80,h:60,height:1,color:"#f472b6",label:"PLAY",cost:18,method:"area",phase:"3",layer:"Fairgrounds"},
 light:{name:"Area Lighting Zone",cat:"Utility",w:120,h:80,height:22,color:"#fde047",label:"L",cost:42000,method:"fixed",phase:"3",layer:"Utilities"},
 firelane:{name:"Emergency Access Lane",cat:"Road",w:240,h:24,height:1,color:"#ef4444",label:"EA",cost:55,method:"linear",phase:"1",layer:"Access"},
 trail:{name:"Nature / Walking Trail",cat:"Path",w:240,h:10,height:1,color:"#b7a888",label:"TR",cost:12,method:"linear",phase:"2",layer:"Access"},
 custom:{name:"Custom Building",cat:"Building",w:60,h:60,height:18,color:"#38bdf8",label:"C",cost:150,method:"sqft",phase:"2"}
};
const paletteGroups=["All","Equestrian","Fairgrounds","Landscape","Farmland","Town & city","Buildings","Access & parking","Infrastructure"];
const taskSeed=[
 ["EFU zoning & farm use","Confirm parcel zoning and review uses permitted outright versus conditional.","High","Planning"],
 ["Farm qualification","Verify farm-use qualification and any current farm tax-deferral status.","High","Planning"],
 ["Farm dwelling","Confirm eligibility and siting rules for farm-help or accessory farm dwelling.","High","Planning"],
 ["Agritourism events","Confirm event limits, conditions, parking, and farm-use relationship with county planning.","High","Planning"],
 ["Farm stand rules","Verify current farm-stand siting, size, sales mix, and promotional-activity limits.","Medium","Planning"],
 ["Access & sight distance","Review driveway permits, road access, event traffic, and sight distance.","High","Site"],
 ["Water source","Confirm well log, water rights or exempt use, capacity, and water-quality requirements.","High","Utilities"],
 ["Septic feasibility","Obtain site evaluation sized for proposed occupancy, kitchen, and events.","High","Utilities"],
 ["Fire access & flow","Review access roads, turnarounds, fire flow, and storage with fire authority.","High","Safety"],
 ["Stormwater & erosion","Determine permit obligations, disturbance area, drainage, and erosion controls.","High","Site"],
 ["Building design","Obtain licensed structural design and confirm snow, wind, accessibility, and permit needs.","High","Buildings"],
 ["Agricultural labor","Confirm current wage, overtime, workers' compensation, and recordkeeping requirements.","Medium","Operations"],
 ["Worker housing","Verify applicable housing, sanitation, and safety standards before providing housing.","Medium","Operations"],
 ["Food service","Confirm state and county licensing for proposed food handling and service.","Medium","Operations"],
 ["Professional review","Review land-use, tax, insurance, environmental, and financing assumptions with qualified advisers.","High","Other"]
];
const manualSeed=[
 ["Site Design","Zoning & siting","Place daily horse and worker circulation near the barn while separating public event traffic. Verify all siting ~
assumptions against parcel constraints and county direction."],
 ["Site Design","Circulation & parking","Plan all-weather emergency access, turnarounds, event parking, and a public-facing farm stand approach."],
 ["Site Design","Pasture layout","Sketch rotational paddocks with shared water access and convenient connections to the barn. Confirm stocking and acreage~
 needs with an agricultural professional."],
 ["Site Design","Utilities corridor","Coordinate power, water, data, septic, and drainage corridors early; reserve room for utility access and future ~
maintenance."],
 ["Schematics","Main barn","Planning concept: center aisle, 12 × 12 ft stalls, wash area, tack/feed rooms, and horse-safe circulation. Obtain licensed ~
plans before construction."],
 ["Schematics","Indoor arena","Planning concept: 100 × 200 ft clear-span riding area, footing, ventilation, viewing, and accessible entry."],
 ["Schematics","Event barn & food hall","Plan kitchen, restrooms, accessible routes, life safety, occupancy-driven septic, and event traffic together."],
 ["Schematics","Workshop & housing","Plan equipment service bays, secure storage, and any caretaker housing only after applicable review."],
 ["Capital Budget","Phasing","A common planning sequence is site infrastructure and core equestrian facilities first, then support facilities, then ~
discretionary public-facing uses."],
 ["Capital Budget","Contingency","Carry an explicit contingency and separate allowances for site work, permits, fees, utility extensions, and ~
escalation."],
 ["Capital Budget","Funding","Model financing terms and draw timing with a lender; do not treat illustrative payback as a lending decision."],
 ["Staffing Ratios","Barn staff","Staffing depends on service level, horse care, hours, and geography; use local operators and advisers to validate ~
ratios."],
 ["Staffing Ratios","Farm & grounds","Account for seasonal workloads, equipment operation, maintenance, safety, and relief coverage."],
 ["Side Businesses","Boarding & training","Model board tiers, occupancy, lessons, clinics, and arena rentals separately to expose their costs and ~
seasonality."],
 ["Side Businesses","Agritourism events","Test a conservative event calendar and verify county approvals, traffic, noise, fire, and food-service ~
requirements."],
 ["Side Businesses","Farm stand & hay","Separate farm-produced and incidental goods; keep records and verify current applicable limits."]
];
const $=id=>document.getElementById(id), esc=s=>String(s??"").replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;","'":"&#39;"}[c]));
const fmt=n=>"$"+Math.round(Number(n)||0).toLocaleString("en-US"), num=(x,d=0)=>Number.isFinite(+x)?+x:d, uid=()=>Math.random().toString(36).slice(2,10);
function starterItems(){return [
 makeItem("barn",320,220),makeItem("arena",560,220),makeItem("shop",320,340),makeItem("feedstore",470,340),makeItem("farmstore",620,340),makeItem("tacksho~
p",770,340),makeItem("event",1120,220),makeItem("stand",1390,370),makeItem("yurt",320,500),
 makeItem("pasture",520,650),makeItem("pasture",920,650),makeItem("roundpen",460,430),makeItem("largaroundpen",560,420),makeItem("circlepaddock",720,540),
 makeItem("mainroad",160,580),makeItem("curvedroad",1780,560),makeItem("farmroad",160,630),makeItem("railfence",460,620),makeItem("wirefence",600,620),mak~
eItem("electricfence",740,690),makeItem("noclimbfence",880,620),makeItem("boardfence",1020,620),
 makeItem("tree",1700,300),makeItem("tree",1740,340),makeItem("bush",1680,360),makeItem("bush",1710,380),makeItem("shrubcluster",1780,350),makeItem("grove~
",1940,280)
 ];}
function makeItem(type,x,y){const t=templates[type];return{id:uid(),type,name:t.name,x,y,w:t.w,h:t.h,height:t.height,color:t.color,label:t.label,rotation:~
0,phase:t.phase,status:"Preliminary",costMethod:t.method,unitCost:t.cost,shape:t.shape||"rect",visible:true,locked:false};}
function defaultState(){return{schemaVersion:1,appVersion:"1.17.0",updatedAt:new ~
Date().toISOString(),project:{name:"AcreForge",county:"",community:"",acres:80,aspect:1.5,goal:"Equestrian estate & ~
farm"},scenarios:[{id:"base",name:"Starter concept",items:starterItems().map(i=>({...i,x:Math.round(i.x*.8)})),finance:{rate:850,occupied:16,events:12,eve~
ntFee:5500,fixed:215000,stallCost:3000,eventCost:1800,other:120000,contingency:12,payrollOffset:false,linked:true,financing:0,interest:7,term:20}}],active~
Scenario:"base",tasks:{},operations:{barnRatio:9,landRatio:30,eventRatio:2},layers:Object.fromEntries(["Buildings","Pastures","Roads","Paths","Fences","Pa~
rking","Utilities","Water","Operations","Fairgrounds","Landscape","Farmland","Town","Access","Labels","Grid"].map(x=>[x,true])),notes:[]};}
function normalizeState(raw){if(!raw||typeof raw!=="object"||!Array.isArray(raw.scenarios)||!raw.project)throw Error("The selected file is not a valid ~
estate project.");if(num(raw.schemaVersion,1)>1)throw Error("This project was saved by a newer app version; update the app before importing it.");const ~
base=defaultState();const scenarios=raw.scenarios.map((sc,i)=>{if(!sc||typeof sc!=="object"||!Array.isArray(sc.items))throw Error("A scenario is missing ~
its site-object ~
list.");if(sc.items.some(o=>!o||!templates[o.type]||!Number.isFinite(+o.x)||!Number.isFinite(+o.y)||!Number.isFinite(+o.w)||!Number.isFinite(+o.h)))throw ~
Error("A scenario contains an unsupported or invalid site object; the current project was not ~
changed.");return{...base.scenarios[0],...sc,id:String(sc.id||"scenario-"+i),name:String(sc.name||"Scenario ~
"+(i+1)),items:sc.items,finance:{...base.scenarios[0].finance,...(sc.finance||{})}}});if(!scenarios.length)throw Error("The project must contain at least ~
one scenario.");const layers=raw.layers&&typeof raw.layers==="object"?{...base.layers,...raw.layers}:base.layers;const project={...base.project,...raw.pro~
ject};project.parcelBoundary=project.parcelBoundary?normalizeParcelBoundary(project.parcelBoundary):null;project.mapFeatures=normalizeMapFeatures(project.~
mapFeatures);const s={...base,...raw,project,operations:{...base.operations,...(raw.operations||{})},layers,tasks:raw.tasks&&typeof raw.tasks==="object"?r~
aw.tasks:{},notes:Array.isArray(raw.notes)?raw.notes:[],scenarios};s.project.acres=Math.max(1,Math.min(1000,num(s.project.acres,base.project.acres)));s.pr~
oject.aspect=Math.max(.25,Math.min(4,num(s.project.aspect,1.5)));if(project.parcelBoundary)s.project.acres=project.parcelBoundary.acres;s.activeScenario=s~
.scenarios.some(x=>x.id===raw.activeScenario)?raw.activeScenario:s.scenarios[0].id;s.scenarios.forEach(sc=>sc.items=sc.items.map(i=>({...makeItem(i.type,+~
i.x,+i.y),...i,id:String(i.id||uid()),name:String(i.name||templates[i.type].name).slice(0,100),color:/^#[0-9a-~
f]{6}$/i.test(i.color||"")?i.color:templates[i.type].color,w:Math.max(1,Math.min(10000,num(i.w,templates[i.type].w))),h:Math.max(1,Math.min(10000,num(i.h,~
templates[i.type].h))),height:Math.max(1,Math.min(300,num(i.height,templates[i.type].height))),unitCost:Math.max(0,num(i.unitCost,templates[i.type].cost))~
,rotation:[0,90,180,270].includes(+i.rotation)?+i.rotation:0,shape:["rect","circle","triangle"].includes(i.shape)?i.shape:"rect",phase:["1","2","3"].inclu~
des(String(i.phase))?String(i.phase):"1",costMethod:["fixed","acre","linear","sqft","area"].includes(i.costMethod)?i.costMethod:templates[i.type].method})~
));return s;}
function migrateLegacy(raw){const s=defaultState();s.project.acres=Math.max(1,num(raw.ac,s.project.acres));const v=raw.v||[],c=raw.c||[];s.scenarios[0].it~
ems=(raw.items||[]).filter(i=>templates[i.type]).map(i=>({...makeItem(i.type,num(i.x)*CELL,num(i.y)*CELL),w:num(i.fw,num(i.w)*CELL),h:num(i.fh,num(i.h)*CE~
LL),height:num(i.ht,templates[i.type].height/CELL)*CELL,shape:i.shape||"rect",unitCost:num(raw.cx?.[i.type],templates[i.type].cost)}));Object.assign(s.sce~
narios[0].finance,{rate:num(v[0],850),occupied:num(v[1],16),events:num(v[2],12),eventFee:num(v[3],5500),other:num(v[4],120000),fixed:num(v[5],215000),stal~
lCost:num(v[6],3000),eventCost:num(v[7],1800),payrollOffset:!!c[0],linked:c[1]!==false});return s;}
const normalizeStateBase=normalizeState;normalizeState=function(raw){const next=normalizeStateBase(raw);if(Array.isArray(raw?.scenarios))raw.scenarios.for~
Each((source,scenarioIndex)=>source?.items?.forEach((item,itemIndex)=>{const target=next.scenarios[scenarioIndex]?.items[itemIndex];if(!target)return;targ~
et.rotation=Number.isFinite(+item.rotation)?((+item.rotation%360)+360)%360:0;target.shape=["rect","roundrect","circle","triangle","poly"].includes(item.sh~
ape)?item.shape:"rect"}));return next};
let state;try{const raw=localStorage.getItem(STORAGE);if(raw)state=normalizeState(JSON.parse(raw));else{const legacy=localStorage.getItem("st3");state=leg~
acy?migrateLegacy(JSON.parse(legacy)):defaultState();if(legacy)localStorage.setItem(STORAGE,JSON.stringify(state));}}catch(e){state=defaultState();state._~
recoveryWarning="Saved data could not be read; a clean starter project was loaded. If a recovery snapshot exists, export it before continuing.";}
if(state.project.name==="Oregon Estate Planner")state.project.name="AcreForge";
state.project.mapFeatures=normalizeMapFeatures(state.project.mapFeatures);
```

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let activePage="about",tool="select",selectedId=null,undoStack=[],redoStack=[],manualFilter="All",manualQuery="",complianceFilter="All",paletteQuery="",paletteGroup="All",dragStart=null,saveTimer=null,sceneAngle=.65,sceneZoom=1,sceneAuto=!matchMedia("(prefers-reduced-motion: reduce)").matches,sceneCut=false,sceneTimer=null,sceneDrag=null,mapZoom=1,mapPan={x:0,y:0},mapAspect=1.5,gridSize=40,snapEnabled=true,mapView="flat",selectedStateMap=null,terrainMapCenter=null,terrainZoom=6,terrainLayer="imagery",terrainMapInstance=null,terrainDrawMode="navigate",terrainDraft=[],terrainSelectedFeatureId=null,terrain3d=true;
let scenePan={x:0,y:0},detailsOpen={library:false,objects:false,layers:false};
const STATE_MAP_CENTERS={AL:[32.8,-86.8],AK:[64,-152],AZ:[34.2,-111.7],AR:[34.8,-92.2],CA:[37.2,-119.5],CO:[39,-105.5],CT:[41.6,-72.7],DE:[39,-75.4],DC:[38.91,-77.04],FL:[28,-82],GA:[32.7,-83.3],HI:[20.8,-156.3],ID:[44.2,-114.5],IL:[40,-89],IN:[39.9,-86.2],IA:[42.1,-93.5],KS:[38.5,-98.3],KY:[37.5,-85],LA:[31.1,-92],ME:[45.3,-69],MD:[39,-76.7],MA:[42.3,-71.8],MI:[44.3,-85.5],MN:[46,-94],MS:[32.7,-89.7],MO:[38.5,-92.5],MT:[46.8,-110.5],NE:[41.5,-99.8],NV:[39.3,-116.6],NH:[43.8,-71.6],NJ:[40.1,-74.5],NM:[34.5,-106],NY:[42.9,-75.5],NC:[35.5,-79.5],ND:[47.5,-100.5],OH:[40.2,-82.8],OK:[35.6,-97.5],OR:[44,-120.5],PA:[41,-77.5],RI:[41.6,-71.5],SC:[33.8,-80.9],SD:[44.5,-100],TN:[35.8,-86],TX:[31,-99],UT:[39.3,-111.7],VT:[44,-72.7],VA:[37.5,-79],WA:[47.4,-120.7],WV:[38.6,-80.6],WI:[44.5,-89.7],WY:[43,-107.5]};
const USGS_TILES={topo:"https://basemap.nationalmap.gov/arcgis/rest/services/USGSTopo/MapServer/tile",imagery:"https://basemap.nationalmap.gov/arcgis/rest/services/USGSImageryTopo/MapServer/tile"};
const scenario=()=>state.scenarios.find(s=>s.id===state.activeScenario)||state.scenarios[0], items=()=>scenario().items;
function save(){clearTimeout(saveTimer);$("saveState").innerHTML='<i class="dot" style="background:#fbbf24"></i> Saving…';saveTimer=setTimeout(()=>{try{const prev=localStorage.getItem(STORAGE);if(prev)localStorage.setItem(RECOVERY,prev);state.updatedAt=new Date().toISOString();localStorage.setItem(STORAGE,JSON.stringify(state));$("saveState").innerHTML='<i class="dot"></i> Saved locally'}catch(e){$("saveState").textContent="Storage unavailable — export backup"}},300);}
function snapshot(){undoStack.push(JSON.stringify(items()));if(undoStack.length>50)undoStack.shift();redoStack=[];}
function undo(){if(!undoStack.length)return;redoStack.push(JSON.stringify(items()));scenario().items=JSON.parse(undoStack.pop());selectedId=null;save();render();}
function redo(){if(!redoStack.length)return;undoStack.push(JSON.stringify(items()));scenario().items=JSON.parse(redoStack.pop());selectedId=null;save();render();}
function shapeAreaSqft(w,h,shape="rect"){if(shape==="circle")return Math.PI*w*h/4;if(shape==="triangle")return w*h/2;if(shape==="roundrect"){const radius=Math.min(w,h)*.2;return Math.max(0,w*h-(4-Math.PI)*radius*radius)}if(shape==="poly"){const cut=Math.min(w,h)*.18;return Math.max(0,w*h-2*cut*cut)}return w*h;}
function acresFor(i){return shapeAreaSqft(i.w,i.h,i.shape)/43560;}
function footprintPoints(item){const{w,h}=item,x=item.x,y=item.y,cut=Math.min(w,h)*.18;let points=item.shape==="triangle"?[[w/2,0],[w,h],[0,h]]:item.shape==="poly"?[[cut,0],[w-cut,0],[w,cut],[w,h-cut],[w-cut,h],[cut,h],[0,h-cut],[0,cut]]:item.shape==="circle"?Array.from({length:24},(_,n)=>[w/2+Math.cos(n*Math.PI/12)*w/2,h/2+Math.sin(n*Math.PI/12)*h/2]):[[0,0],[w,0],[w,h],[0,h]];const angle=num(item.rotation)*Math.PI/180,c=Math.cos(angle),s=Math.sin(angle),cx=w/2,cy=h/2;return points.map(([px,py])=>[x+cx+(px-cx)*c-(py-cy)*s,y+cy+(px-cx)*s+(py-cy)*c]);}
function parcelAreaAcres(ring){const radius=6371007.1809;let sum=0;for(let i=0;i<ring.length;i++){const a=ring[i],b=ring[(i+1)%ring.length],lon1=a[0]*Math.PI/180,lon2=b[0]*Math.PI/180,lat1=a[1]*Math.PI/180,lat2=b[1]*Math.PI/180;let delta=lon2-lon1;if(delta>Math.PI)delta-=2*Math.PI;if(delta< -Math.PI)delta+=2*Math.PI;sum+=delta*(2+Math.sin(lat1)+Math.sin(lat2));}return Math.abs(sum*radius*radius/2)/4046.8564224;}
function normalizeParcelBoundary(raw){if(!raw||!Array.isArray(raw.coordinates))return null;const coordinates=raw.coordinates.map(poly=>{if(!Array.isArray(poly)||!poly.length)throw Error("Parcel boundary contains an empty polygon.");return poly.map(ring=>{if(!Array.isArray(ring)||ring.length<4)throw Error("Each parcel ring must contain at least four coordinate pairs.");const points=ring.map(p=>{const lon=Number(p?.[0]),lat=Number(p?.[1]);if(!Number.isFinite(lon)||!Number.isFinite(lat)||Math.abs(lon)>180||Math.abs(lat)>90)throw Error("Parcel coordinates must be longitude/latitude pairs in WGS84.");return[lon,lat]});const first=points[0],last=points.at(-1);if(first[0]!==last[0]||first[1]!==last[1])points.push([...first]);return points;});});const outer=coordinates.flatMap(p=>p[0]),lat0=outer.reduce((s,p)=>s+p[1],0)/outer.length,lon0=outer.reduce((s,p)=>s+p[0],0)/outer.length,rad=6371007.1809,ft=3.280839895,cos=Math.max(.01,Math.cos(lat0*Math.PI/180));const points=outer.map(([lon,lat])=>[(lon-lon0)*Math.PI/180*rad*cos*ft,(lat-lat0)*Math.PI/180*rad*ft]),minX=Math.min(...points.map(p=>p[0])),maxX=Math.max(...points.map(p=>p[0])),minY=Math.min(...points.map(p=>p[1])),maxY=Math.max(...points.map(p=>p[1]));const acres=coordinates.reduce((sum,poly)=>sum+Math.max(0,parcelAreaAcres(poly[0])-poly.slice(1).reduce((v,r)=>v+parcelAreaAcres(r),0)),0);if(!Number.isFinite(acres)||acres<=0||maxX-minX<1||maxY-minY<1)throw Error("Parcel boundary has no measurable area.");const polygons=coordinates.map(poly=>poly.map(ring=>ring.map(([lon,lat])=>{let dx=lon-lon0;if(dx>180)dx-=360;if(dx< -180)dx+=360;return[dx*Math.PI/180*rad*cos*ft-minX,((lat-lat0)*Math.PI/180*rad*ft)-minY]})));return{coordinates,polygons,acres,width:maxX-minX,height:maxY-minY,source:String(raw.source||"Imported boundary").slice(0,120),importedAt:raw.importedAt||new Date().toISOString(),center:[lon0,lat0]};}
function parseGeoCoordinates(coords){if(!Array.isArray(coords)||coords.length<4)throw Error("Parcel rings must contain at least four longitude/latitude points.");return coords.map(p=>{if(!Array.isArray(p)||p.length<2)throw Error("Invalid GeoJSON coordinate; expected [longitude, latitude].");return[Number(p[0]),Number(p[1])];});}
function parseGeoJSONParcel(json){let geometry=json;if(json.type==="FeatureCollection"){if(json.features?.length!==1)throw Error("Export one parcel boundary at a time (the GeoJSON must contain exactly one feature).");geometry=json.features[0]?.geometry;}else if(json.type==="Feature")geometry=json.geometry;if(geometry?.type==="Polygon")return[geometry.coordinates.map(parseGeoCoordinates)];if(geometry?.type==="MultiPolygon")return geometry.coordinates.map(poly=>poly.map(parseGeoCoordinates));throw Error("GeoJSON must contain a Polygon or MultiPolygon feature.");}
function parseKmlParcel(text){const doc=new DOMParser().parseFromString(text,"application/xml");if(doc.querySelector("parsererror"))throw Error("This KML file is not valid XML.");const polygons=[...doc.getElementsByTagNameNS("*","Polygon")].map(poly=>[...poly.getElementsByTagNameNS("*","LinearRing")].map(ring=>{const raw=ring.getElementsByTagNameNS("*","coordinates")[0]?.textContent?.trim();if(!raw)throw Error("A KML parcel ring has no coordinates.");return raw.split(/\s+/).map(pair=>{const values=pair.split(",");return[Number(values[0]),Number(values[1])];});}));if(!polygons.length)throw Error("No KML Polygon was found. Export the parcel outline as KML or GeoJSON.");if(polygons.length>1)throw Error("This KML contains multiple parcel polygons. Export a single parcel boundary at a time.");return polygons;}
async function importParcelFile(file){try{const text=await file.text(),coords=/\.kml$|\.xml$/i.test(file.name)?parseKmlParcel(text):parseGeoJSONParcel(JSON.parse(text)),boundary=normalizeParcelBoundary({coordinates:coords,source:file.name});if(state.project.parcelBoundary&&!confirmAction("Replace the imported parcel boundary? Your placed objects will stay at their current map coordinates."))return;state.project.parcelBoundary=boundary;state.project.acres=boundary.acres;if(!state.project.locationUrl)state.project.locationUrl=`https://www.google.com/maps/@${boundary.center[1]},${boundary.center[0]},16z`;save();render();alert(`Parcel boundary imported: ${boundary.acres.toFixed(2)} acres calculated from the boundary. Source: ${boundary.source}. Confirm the source is an authoritative parcel or survey file.`);}catch(e){alert("Parcel import failed: "+e.message);}}
function pointInParcel(x,y){const b=state.project.parcelBoundary;if(!b)return true;const insideRing=ring=>{let inside=false;for(let i=0,j=ring.length-1;i<ring.length;j=i++){const a=ring[i],c=ring[j];if((a[1]>y)!==(c[1]>y)&&x<(c[0]-a[0])*(y-a[1])/(c[1]-a[1])+a[0])inside=!inside;}return inside;};return b.polygons.some(poly=>insideRing(poly[0])&&!poly.slice(1).some(insideRing));}
function mapHit(w,h){const u=propertyDims().w/mapZoom/560*18;if(w>=u&&h>=u)return"";const W=Math.max(w,u),H=Math.max(h,u);return`<rect x="${w/2-W/2}" y="${h/2-H/2}" width="${W}" height="${H}" fill="#fff" fill-opacity="0.001"/>`}
function objectOutsideParcel(item){if(!state.project.parcelBoundary)return false;const points=footprintPoints(item),center=[item.x+item.w/2,item.y+item.h/2];if(![...points,center].every(p=>pointInParcel(...p)))return true;const cross=(a,b,c)=>{const v=(b[0]-a[0])*(c[1]-a[1])-(b[1]-a[1])*(c[0]-a[0]);return Math.abs(v)<1e-7?0:v>0?1:2;},intersects=(a,b,c,d)=>cross(a,b,c)!==cross(a,b,d)&&cross(c,d,a)!==cross(c,d,b),edges=points.map((p,i)=>[p,points[(i+1)%points.length]]);return state.project.parcelBoundary.polygons.some(poly=>poly.some(ring=>ring.some((a,i)=>edges.some(([c,d])=>intersects(a,ring[(i+1)%ring.length],c,d)))));}
function safeMapReference(value){try{const u=new URL(value);return u.protocol==="https:"&&(/(^|\.)google\.com$/.test(u.hostname)||u.hostname==="maps.app.goo.gl"||/(^|\.)zillow\.com$/.test(u.hostname))?u.href:"";}catch{return"";}}
function propertyDims(){const boundary=state.project.parcelBoundary;if(boundary)return{w:boundary.width,h:boundary.height,acres:boundary.acres};const acres=Math.max(1,num(state.project.acres,80)),a=Math.max(.25,Math.min(4,num(state.project.aspect,1.5)));const w=Math.sqrt(acres*43560*a),h=acres*43560/w;return{w,h,acres};}
function siteMapBounds(){const{w,h}=propertyDims(),baseW=w+200,baseH=h+200,aspect=Math.max(.35,Math.min(3,mapAspect));let viewW=baseW,viewH=baseH;if(viewW/viewH>aspect)viewH=viewW/aspect;else viewW=viewH*aspect;viewW/=mapZoom;viewH/=mapZoom;return{x:w/2-viewW/2+mapPan.x,y:h/2-viewH/2+mapPan.y,w:viewW,h:viewH};}
function areaSqft(i){return shapeAreaSqft(i.w,i.h,i.shape);}
function compute(){const its=items(),f=scenario().finance,dims=propertyDims();const by=t=>its.filter(i=>i.type===t);let used=0,past=0;for(const i of its){const ar=acresFor(i);if(templates[i.type].cat==="Pasture")past+=ar;else if(!["road","path","fence","utility"].includes(i.type))used+=ar;}
const open=Math.max(0,dims.acres-used-past), buildings=its.filter(i=>templates[i.type].cat==="Building").length;
const costs=its.reduce((sum,i)=>{const t=templates[i.type],area=areaSqft(i),method=i.costMethod||t.method,u=num(i.unitCost,t.cost);if(method==="acre")return sum+area/43560*u;if(method==="linear")return sum+linearLength(i)*u;if(method==="sqft")return sum+area*u;if(method==="area")return sum+area*u;return sum+u;},0);
const capex=costs*(1+Math.max(0,num(f.contingency))/100), barnCount=by("barn").length, eventCount=by("event").length, stand=by("stand").length, arena=by("arena").length, linked=!!f.linked;
const maxStalls=barnCount*20, occupied=Math.max(0,Math.min(200,linked?Math.min(num(f.occupied),maxStalls):num(f.occupied))),events=Math.max(0,Math.min(100,linked?Math.min(num(f.events),eventCount*18):num(f.events)));
const other=linked?(stand?42000:0)+(arena?48000:0)+(past>=10?30000:0):num(f.other);
const gross=occupied*num(f.rate)*12+events*num(f.eventFee)+other, ~
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payroll=f.payrollOffset?90000:0,opex=Math.max(0,num(f.fixed)-payroll)+occupied*num(f.stallCost)+events*num(f.eventCost),noi=gross-opex;
const margin=num(f.rate)*12-num(f.stallCost),needed=num(f.fixed)-payroll+events*num(f.eventCost)-events*num(f.eventFee)-other;
const breakeven=needed<=0?0:margin<=0?Infinity:Math.ceil(needed/margin);
const debt=Math.max(0,num(f.financing)),rate=Math.max(0,num(f.interest))/100/12,months=Math.max(0,num(f.term))*12;let ~
pmt=0;if(debt>0&&months>0){pmt=rate===0?debt/months:debt*(rate*Math.pow(1+rate,months))/(Math.pow(1+rate,months)-1)}const ~
outOfBounds=state.project.parcelBoundary?its.filter(objectOutsideParcel).length:0;
return{dims,used,past,open,buildings,costs,capex,gross,opex,noi,occupied,events,other,breakeven,pmt,annualDebt:pmt*12,afterDebt:noi-pmt*12,outOfBounds};}
function pageHead(kicker,title,sub,actions=""){return`<div class="page-head"><div><div ~
class="eyebrow">${kicker}</div><h2>${title}</h2><p>${sub}</p></div><div class="actions">${actions}</div></div>`}
function btn(text,action,cls=""){return`<button class="btn ${cls}" data-action="${action}">${text}</button>`}
function kpi(label,value,hint="",color=""){return`<div class="card kpi"><div class="label">${label}</div><div class="value ${color}">${value}</div><div ~
class="hint">${hint}</div></div>`}
function selectOpts(items,current){return items.map(x=>`<option value="${esc(x[0])}" ${x[0]===current?"selected":""}>${esc(x[1])}</option>`).join("");}
function navRender(){$("nav").innerHTML=navItems.map(([id,icon,label])=>`<button data-page="${id}" class="${activePage===id?"active":""}" aria-~
current="${activePage===id?"page":"false"}"><span class="ico" aria-hidden="true">${icon}</span><span>${label}</span></button>`).join("");}
function render(){navRender();$("topTitle").textContent=state.project.name||"AcreForge";$("topCrumb").textContent=" / ~
"+(navItems.find(x=>x[0]===activePage)?.[2]||"");
const fn={about:aboutPage,dashboard:dashboardPage,designer:designerPage,buildings:buildingsPage,infrastructure:infrastructurePage,financials:financialsPag~
e,operations:operationsPage,compliance:compliancePage,scenarios:scenariosPage,reports:reportsPage,manual:manualPage,settings:settingsPage};$("content").in~
nerHTML=(fn[activePage]||dashboardPage)();clearInterval(sceneTimer);if(activePage==="settings"&&state.project.parcelBoundary){const ~
acres=document.querySelector('[data-project="acres"]');if(acres){acres.readOnly=true;acres.title="Calculated from imported parcel ~
boundary";acres.closest(".field")?.insertAdjacentHTML("afterend",`<div class="notice">${state.project.parcelBoundary.acres.toFixed(2)} acres calculated ~
from imported boundary (${esc(state.project.parcelBoundary.source)}). <button class="toolbtn" data-action="clear-parcel">Remove parcel ~
outline</button></div>`);}}if(activePage==="designer"){const bar=document.querySelector(".map-toolbar");bar?.insertAdjacentHTML("beforeend",`<button ~
class="toolbtn" data-action="toggle-map-view">${mapView==="flat"?"USA map":"Flat plan"}</button><button class="toolbtn" data-action="import-parcel">Import~
 parcel boundary</button><input class="input" style="width:220px;padding:6px 8px" data-project="locationUrl" aria-label="Google Maps or Zillow reference ~
link" placeholder="Google Maps / Zillow link" value="${esc(state.project.locationUrl||"")}">`);const meta=document.querySelector(".map-~
meta");if(meta){const ~
b=state.project.parcelBoundary,href=safeMapReference(state.project.locationUrl||"");meta.insertAdjacentHTML("beforeend",b?`<span>${esc(b.source)} · ~
${b.acres.toFixed(2)} geodesic acres</span><button class="toolbtn" data-action="clear-parcel">Remove parcel</button>`:"<span>Import a survey or assessor ~
parcel file for real acreage.</span>");if(href)meta.insertAdjacentHTML("beforeend",`<a class="toolbtn" href="${esc(href)}" target="_blank" rel="noopener ~
noreferrer">Open property reference ■</a>`);}if(mapView==="globe"){const plan=$("siteMap");if(plan)plan.style.display="none";$("mapViewport")?.insertAdjac~
entHTML("beforeend",drawGlobe());}draw3d();sceneTimer=setInterval(()=>{if(sceneAuto&&!sceneDrag){sceneAngle+=.006;draw3d()}},100);}}
function aboutPage(){return`<section class="about-hero"><div class="eyebrow">ACREFORGE · U.S. PROJECT PLANNER</div><h2>Plan the land.<br>Build what ~
belongs on it.</h2><p class="about-lead">I’m building AcreForge to bring rural property planning, site layout, and home and building design together in ~
one approachable workspace for people across the United States.</p><p class="muted">The idea is to explore a property like a planner and build on it like ~
a game: map the land, place barns and infrastructure, sketch homes and floor plans, and see how the pieces fit before taking concepts to qualified ~
professionals.</p><div class="actions"><button class="btn primary" data-page="designer">Start with the site plan</button><button class="btn" data-~
page="homes">Design a home</button><button class="btn ghost" data-page="dashboard">Open project dashboard</button></div></section><div class="grid g3" ~
style="margin:14px 0"><article class="card"><div class="eyebrow">FREE</div><h3>$0</h3><p class="small muted">Map, site plan, building and lot builder, ~
local saving. Available now.</p></article><article class="card"><div class="eyebrow">PRO · PLANNED</div><h3>Pricing TBD</h3><p class="small ~
muted">Planned: cloud-saved projects, PDF/DXF export, and premium map data. Not yet available; no payments are taken.</p></article><article ~
class="card"><div class="eyebrow">EARLY ACCESS</div><h3>Join the waitlist</h3><p class="small muted">Use the Join Pro waitlist button in the top bar to ~
tell me what you would pay for.</p></article></div><div class="grid g3 about-features"><article class="card"><div class="eyebrow">01 · LAND</div><h3>Plan ~
a property</h3><p class="small muted">Arrange buildings, arenas, pasture, roads, utilities, and other site elements on a measured planning grid. Explore ~
terrain and map context where available.</p></article><article class="card"><div class="eyebrow">02 · BUILD</div><h3>Design homes & buildings</h3><p ~
class="small muted">Move from a site concept into building layouts, room plans, furniture placement, dimensions, and a 3D view.</p></article><article ~
class="card"><div class="eyebrow">03 · DECIDE</div><h3>Keep the plan connected</h3><p class="small muted">Track planning assumptions, costs, scenarios, ~
due-diligence questions, and project notes alongside the design.</p></article></div><div class="grid g2" style="margin-top:14px"><article ~
class="card"><h3>What I’m trying to build</h3><p class="small muted">A practical, visual planning tool for families and rural landowners who want to think~
 through an estate, farm, or equestrian property in one place—from the overall site down to the rooms in a home.</p><p class="small muted">It’s being ~
developed iteratively, with a local-first approach: your project is saved in this browser, and you can export a backup.</p><a class="btn ghost" ~
href="support.html">Project status & data use ■</a></article><article class="card"><h3>Important boundary</h3><p class="small muted">This is for early-~
stage concepts and organization, not construction documents or professional approvals. Maps, dimensions, cost assumptions, and layouts must be checked ~
against the actual property and applicable requirements by the right professionals. Planning and regulatory information is not tailored to any particular ~
city, county, or state.</p><button class="btn" data-page="manual">Read the planning notes</button></article></div>`}
function dashboardPage(){const c=compute(),sc=scenario(),pct=taskSeed.filter(t=>state.tasks[t[0]]?.done).length/taskSeed.length*100, ~
high=taskSeed.filter(t=>t[2]==="High"&&!state.tasks[t[0]]?.done).length;
return pageHead("PROJECT OVERVIEW",esc(state.project.name),`${esc(state.project.county||"U.S. property planning")} · ${sc.name} · planning assumptions ~
only`,btn("Print summary","print","ghost"))+
`<div class="grid g4">${kpi("Property area",`${num(state.project.acres)} ac`,`${Math.round(c.dims.w).toLocaleString()} × ~
${Math.round(c.dims.h).toLocaleString()} ft`,"green")}${kpi("Open / unassigned",`${c.open.toFixed(1)} ac`,`${c.past.toFixed(1)} ac pasture · ~
${c.used.toFixed(1)} ac built`,"blue")}${kpi("Illustrative capital",fmt(c.capex),`Includes ${num(sc.finance.contingency)}% ~
contingency`,"amber")}${kpi("Annual net operating income",fmt(c.noi),`Before financing · ${c.gross<0?"":"Illustrative"}` ,c.noi>=0?"green":"rose")}</div>
<div class="grid g2" style="margin-top:14px"><div class="card"><div class="between"><h3>Planning progress</h3><span class="pill">${Math.round(pct)}% ~
reviewed</span></div><div class="progress"><i style="width:${pct}%"></i></div><p class="small muted">${high} high-priority due-diligence items remain ~
incomplete. A checked item records your progress; it is not evidence of approval.</p><div class="row">${btn("Open due ~
diligence","page:compliance","ghost")}${btn("Review site layout","page:designer","ghost")}</div></div>
<div class="card"><div class="between"><h3>Current project snapshot</h3><span class="pill warn">Illustrative</span></div><div class="key-~
value"><span>Scenario</span><strong>${esc(sc.name)}</strong></div><div class="key-value"><span>Buildings on ~
plan</span><strong>${c.buildings}</strong></div><div class="key-value"><span>Estimated gross revenue</span><strong>${fmt(c.gross)}/yr</strong></div><div ~
class="key-value"><span>Estimated operating expense</span><strong>${fmt(c.opex)}/yr</strong></div><div class="key-value"><span>Stall ~
breakeven</span><strong>${Number.isFinite(c.breakeven)?c.breakeven+" stalls":"Not reached"}</strong></div></div></div>
<div class="grid g3" style="margin-top:14px"><div class="card"><h3>Next planning steps</h3><div class="check-row"><span class="amber">01</span><div>Verify~
 parcel acreage, county, and permitted uses<small>County planning · parcel-specific review</small></div></div><div class="check-row"><span ~
class="amber">02</span><div>Confirm water, septic, and emergency access feasibility<small>Site feasibility · qualified ~
professionals</small></div></div><div class="check-row"><span class="amber">03</span><div>Stress-test costs, occupancy, and project phases<small>Financial~
 model · lender / contractor inputs</small></div></div></div>
<div class="card"><h3>Areas assigned</h3><div class="key-value"><span>Structures & hardscape</span><strong>${c.used.toFixed(2)} ac</strong></div><div ~
class="key-value"><span>Pasture</span><strong>${c.past.toFixed(2)} ac</strong></div><div class="key-value"><span>Open / ~
unassigned</span><strong>${c.open.toFixed(2)} ac</strong></div><div class="notice" style="margin-top:10px">Footprints are conceptual and do not account ~
for slopes, easements, setbacks, wetlands, or parcel geometry.</div></div>
<div class="card"><h3>Quick actions</h3><div class="stack">${btn("Add site object","page:designer","primary")}${btn("Adjust financial ~
assumptions","page:financials")}${btn("Compare scenarios","page:scenarios")}${btn("Export project ~
backup","export","ghost")}</div></div></div>${disclaimerShort()}`;}
function layerFor(type){const t=templates[type],c=t?.cat||"";if(t?.layer)return t.layer;if(c==="Building"||c==="Arena")return"Buildings";if(c==="Pasture")~
return"Pastures";if(c==="Road")return"Roads";if(c==="Path")return"Paths";if(c==="Fence")return"Fences";if(c==="Parking")return"Parking";if(c==="Utility")r~
eturn"Utilities";if(c==="Water")return"Water";if(c==="Landscape")return"Landscape";return"Operations";}
function paletteCategory(type){const t=templates[type];if(t.cat==="Farm")return"Farmland";if(t.cat==="City")return"Town & city";if(t.layer==="Fairgrounds"~
||t.cat==="Venue"||t.cat==="Camping")return"Fairgrounds";if(t.cat==="Landscape")return"Landscape";if(t.cat==="Utility"||t.cat==="Water")return"Infrastruct~
ure";if(t.cat==="Road"||t.cat==="Path"||t.cat==="Fence"||t.cat==="Parking")return"Access & parking";if(["barn","arena","pasture","paddock","roundpen","lar~
garoundpen","circlepaddock","outdoor","manure","hay"].includes(type))return"Equestrian";return"Buildings";}
const palItem=([id,t])=>`<button class="palette-item ${tool===id?"active":""}" data-palette-item="${id}" title="${esc(t.name)} · ${t.w} × ${t.h} ft"><span~
 class="palette-glyph" style="--swatch:${esc(t.color)}">${esc(t.label||"✦")}</span><span class="palette-name">${esc(t.name)}</span><span class="palette-~
size">${t.w}×${t.h} ft</span></button>`;
function paletteCount(g){return Object.keys(templates).filter(id=>g==="All"||paletteCategory(id)===g).length;}
function paletteHtml(){const list=filteredPalette();if(!list.length)return`<div class="empty">No objects match this ~
search.</div>`;if(paletteGroup!=="All")return list.map(palItem).join("");return paletteGroups.slice(1).map(g=>{const ~
items=list.filter(([id])=>paletteCategory(id)===g);return items.length?`<div class="palette-head">${esc(g)} ~
<span>${items.length}</span></div>`+items.map(palItem).join(""):"";}).join("");}
function filteredPalette(){return Object.entries(templates).filter(([id,t])=>(paletteGroup==="All"||paletteCategory(id)===paletteGroup)&&(`${t.name} ~
${t.cat} ${id}`.toLowerCase().includes(paletteQuery.toLowerCase())));}
function mapObjectDetail(i){const w=i.w,h=i.h,mid=w/2,cx=i.color||templates[i.type].color;let s="";
if(i.type==="tree")s=`<path d="M${mid} ${h*.68}v${h*.2}" stroke="#713f12" stroke-width="${Math.max(2,w*.09)}"/><circle cx="${mid}" cy="${h*.43}" ~
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r="${Math.min(w,h)*.43}" fill="#22a447"/><circle cx="${mid-w*.12}" cy="${h*.36}" r="${Math.min(w,h)*.22}" fill="#4ade80" opacity=".9"/>`;
else if(i.type==="bush"||i.type==="shrubcluster"){const count=i.type==="bush"?1:7;for(let n=0;n<count;n++){const ~
x=i.type==="bush"?mid:w*(.2+(n%3)*.24),y=i.type==="bush"?h/2:h*(.25+Math.floor(n/3)*.24),r=Math.min(w,h)*(i.type==="bush"?.4:.16);s+=`<circle cx="${x}" ~
cy="${y}" r="${r}" fill="${n%2?"#3f6b31":"#568b43"}" stroke="#91b77b" stroke-width="1"/><circle cx="${x-r*.25}" cy="${y-r*.25}" r="${r*.28}" ~
fill="#b7d39d" opacity=".64"/>`}}
else if(i.type==="grove"||i.type==="orchard"){const rows=i.type==="orchard"?3:3,cols=i.type==="orchard"?5:4;for(let r=0;r<rows;r++)for(let ~
q=0;q<cols;q++){const x=8+(q+.5)*Math.max(0,w-16)/cols,y=8+(r+.5)*Math.max(0,h-16)/rows,radius=Math.max(5,Math.min(13,w/cols*.36,h/rows*.36));s+=`<circle ~
cx="${x}" cy="${y}" r="${radius}" fill="${(q+r)%2?"#166534":"#22a447"}" stroke="#86efac" stroke-width="1"/><circle cx="${x-radius*.27}" ~
cy="${y-radius*.27}" r="${radius*.3}" fill="#86efac" opacity=".72"/>`}}
else if(["hedge","fence","railfence","boardfence","wirefence","electricfence","noclimbfence"].includes(i.type)){if(i.type==="hedge"){s=`<path d="M0 ~
${h/2}H${w}" stroke="#365f24" stroke-width="${Math.max(3,h*.62)}" stroke-dasharray="3 3"/>`;for(let n=0;n<=Math.min(24,Math.floor(w/18));n++){const ~
x=n*w/Math.min(24,Math.floor(w/18));s+=`<circle cx="${x}" cy="${h/2}" r="${Math.max(2,h*.24)}" fill="${n%2?"#568039":"#365f24"}"/>`}}else{const count=Math~
.max(3,Math.min(24,Math.floor(w/24))),mesh=i.type==="wirefence"||i.type==="noclimbfence",electric=i.type==="electricfence",rails=i.type==="railfence"?3:i.~
type==="boardfence"?3:2;for(let r=0;r<rails;r++){const y=h*(.2+r*.3);s+=electric?`<path d="M0 ${y}H${w}" stroke="#fef08a" stroke-~
width="${Math.max(1.5,h*.12)}" stroke-dasharray="8 4"/>`:i.type==="boardfence"?`<rect x="0" y="${y-h*.07}" width="${w}" height="${Math.max(2,h*.14)}" ~
fill="#e4c89f" stroke="#8b6342" stroke-width="1"/>`:`<path d="M0 ${y}H${w}" stroke="${i.type==="railfence"?"#e4c89f":"#d8d1bf"}" stroke-~
width="${Math.max(1.5,h*.1)}"/>`}if(mesh)for(let n=0;n<=count;n++){const x=n*w/count;s+=`<path d="M${x} ${h*.13}v${h*.74}" stroke="#e2e8e0" stroke-~
opacity=".8" stroke-width="${Math.max(1,h*.045)}"/>`}for(let n=0;n<=Math.floor(count/2);n++){const x=n*w/Math.max(1,Math.floor(count/2));s+=`<rect ~
x="${x-Math.max(1,h*.08)}" y="${h*.05}" width="${Math.max(2,h*.16)}" height="${h*.9}" rx="1" fill="#8b6342"/>`}}}
else if(["road","mainroad","farmroad"].includes(i.type)){s=`<path d="M0 ${h*.16}V${h*.84}M0 ${h/2}H${w}M${w} ${h*.16}V${h*.84}" stroke="#d9d0bd" stroke-~
opacity=".8" stroke-width="${Math.max(1,h*.06)}" ${i.type==="mainroad"?'stroke-dasharray="12 8"':""}/>`}
else if(i.type==="parking"||i.type==="trailer"||i.type==="busparking"||i.type==="overflow"){s=`<path d="M${w*.08} ${h*.16}v${h*.68}M${w*.92} ~
${h*.16}v${h*.68}" stroke="#e2e8f0" stroke-opacity=".75" stroke-width="2"/>`;const n=Math.min(12,Math.max(2,Math.floor(w/22)));for(let q=1;q<n;q++){const ~
x=w*.08+q*w*.84/n;s+=`<path d="M${x} ${h*.16}v${h*.27}m0 ${h*.14}v${h*.27}" stroke="#e2e8f0" stroke-opacity=".65" stroke-width="1.4"/>`}}
else if(i.type==="eventcenter"||i.type==="expo"||i.type==="fairbarn"||i.type==="event"||i.type==="barn"||i.type==="shop"||i.type==="feedstore"||i.type==="~
farmstore"||i.type==="tackshop"||i.type==="stand"||i.type==="restroom"){s=`<path d="M${w*.08} ${h*.18}L${w/2} ${h*.04}L${w*.92} ${h*.18}" fill="none" ~
stroke="#fff" stroke-opacity=".75" stroke-width="${Math.max(2,Math.min(6,w*.018))}"/><path d="M${w*.08} ${h*.2}V${h*.88}M${w*.92} ${h*.2}V${h*.88}" ~
stroke="#fff" stroke-opacity=".42" stroke-width="2"/>`;const count=Math.max(2,Math.min(8,Math.floor(w/36)));for(let q=0;q<count;q++){const ~
x=(q+1)*w/(count+1);s+=`<rect x="${x-w*.022}" y="${h*.3}" width="${w*.044}" height="${h*.16}" rx="2" fill="#dcfce7" fill-opacity=".78"/>`}}
else if(i.type==="grandstand"||i.type==="bleachers"){for(let n=0;n<6;n++){const y=h*.18+n*h*.095;s+=`<path d="M${w*.12} ${y}H${w*.9-n*w*.035}" ~
stroke="#fed7aa" stroke-width="${Math.max(2,h*.025)}"/>`}s+=`<path d="M${w*.12} ${h*.14}L${w*.24} ${h*.82}" stroke="#fff" stroke-opacity=".45" stroke-~
width="2"/>`}
else if(i.type==="rodeoring"||i.type==="outdoor"||i.type==="arena"||["roundpen","largaroundpen","circlepaddock"].includes(i.type)){if(["roundpen","largaro~
undpen","circlepaddock"].includes(i.type)){s=`<ellipse cx="${w/2}" cy="${h/2}" rx="${w*.4}" ry="${h*.4}" fill="none" stroke="#f8fafc" stroke-opacity=".78"~
 stroke-width="2"/>`;for(let r=0;r<3;r++)s+=`<ellipse cx="${w/2}" cy="${h/2}" rx="${w*(.18+r*.09)}" ry="${h*(.18+r*.09)}" fill="none" stroke="#fff" ~
stroke-opacity=".55" stroke-width="1"/>`;if(i.type!=="circlepaddock")s+=`<path d="M${w*.12} ${h*.5}H${w*.27}M${w*.73} ${h*.5}H${w*.88}" stroke="#f8fafc" ~
stroke-width="3"/>`}else{const pad=Math.min(w,h)*.12;s=`<rect x="${pad}" y="${pad}" width="${w-2*pad}" height="${h-2*pad}" ~
rx="${i.type==="rodeoring"?Math.min(w,h)/2:2}" fill="none" stroke="#f8fafc" stroke-opacity=".76" stroke-width="2"/>`;if(i.type==="rodeoring")s+=`<ellipse ~
cx="${w/2}" cy="${h/2}" rx="${w*.32}" ry="${h*.32}" fill="none" stroke="#fff" stroke-opacity=".35" stroke-width="2"/>`;if(i.type==="arena")for(let ~
r=0;r<4;r++)s+=`<path d="M${pad} ${h*.35+r*h*.09}H${w-pad}" stroke="#fff" stroke-opacity=".12" stroke-width="1"/>`}}
else if(i.type==="vendor"||i.type==="foodcourt"){const n=Math.min(12,Math.max(2,Math.floor(w/22)));for(let q=0;q<n;q++){const x=q*w/n+2;s+=`<rect x="${x}"~
 y="${h*.2}" width="${w/n-4}" height="${h*.6}" rx="2" fill="#fff" fill-opacity=".13" stroke="#fff" stroke-opacity=".72" stroke-width="1.5"/>`}}
else if(i.type==="camping"||i.type==="rv"){const n=8;for(let q=0;q<n;q++){const ~
x=10+(q%4)*Math.max(16,w*.2),y=10+Math.floor(q/4)*Math.max(14,h*.42);if(i.type==="camping")s+=`<path d="M${x} ${y+12}l8-12 8 12z" fill="#5eead4" ~
stroke="#ecfeff" stroke-width="1"/>`;else s+=`<rect x="${x}" y="${y}" width="24" height="14" rx="3" fill="#e2e8f0" stroke="#fff"/><rect x="${x+3}" ~
y="${y+2}" width="8" height="5" fill="#38bdf8"/>`}}
else if(i.type==="eventlawn"||i.type==="plaza"||i.type==="garden"||i.type==="playground"){const count=16;for(let q=0;q<count;q++){const ~
x=8+(q*37%Math.max(12,w-12)),y=8+(q*23%Math.max(12,h-12));if(i.type==="playground")s+=`<circle cx="${x}" cy="${y}" r="4" ~
fill="${q%2?"#fb7185":"#60a5fa"}"/>`;else s+=`<circle cx="${x}" cy="${y}" r="${i.type==="garden"?3:2}" ~
fill="${i.type==="garden"?(q%2?"#f0abfc":"#fda4af"):"#d9f99d"}" opacity=".8"/>`}}
else if(i.type==="amphitheater"){for(let n=0;n<5;n++)s+=`<path d="M${w*.12+n*w*.075} ${h*.18+n*h*.115}Q${w*.5} ${h*.02+n*h*.12} ${w*.88-n*w*.075} ~
${h*.18+n*h*.115}" fill="none" stroke="#f5d0fe" stroke-opacity=".8" stroke-width="2"/>`}
else if(i.type==="stage")s=`<path d="M${w*.15} ${h*.22}H${w*.85}V${h*.78}H${w*.15}z" fill="#fff" fill-opacity=".18" stroke="#fff" stroke-~
width="2"/><circle cx="${w/2}" cy="${h*.42}" r="${Math.min(w,h)*.08}" fill="#fde68a"/>`;
else if(i.type==="ticket"||i.type==="checkin")s=`<path d="M${w*.08} ${h*.86}V${h*.35}Q${w/2} ${-h*.12} ${w*.92} ${h*.35}V${h*.86}" fill="none" ~
stroke="#fff" stroke-width="3"/><path d="M${w*.35} ${h*.45}V${h*.85}m${w*.3} ${-h*.4}v${h*.4}" stroke="#fff" stroke-width="2"/>`;
else if(i.type==="path"||i.type==="trail"||i.type==="firelane")s=`<path d="M0 ${h/2}H${w}" stroke="#fff" stroke-opacity=".64" stroke-~
width="${Math.max(2,h*.18)}" stroke-dasharray="${i.type==="firelane"?"8 5":"none"}"/>`;
const detail=mapObjectDetail(i),badge=String(i.label||t.name.slice(0,2)).slice(0,4),badgeWidth=Math.min(width-~
4,Math.max(40,badge.length*24)),badgeY=Math.max(2,height-38),showLabel=state.layers.Labels!==false&&width>=32&&height>=40;
return`<g transform="translate(${i.x} ${i.y}) rotate(${num(i.rotation)}) ${width/2} ${height/2})" data-object="${esc(i.id)}" class="map-object" ~
tabindex="0" role="button" aria-label="${label}, ${width} by ${height} feet${selected?", selected":""}"><title>${label} · ${width} × ${height} ft · ~
${acresFor(i).toFixed(2)} ac</title>${mapHit(width,height)}<${shape} fill="${c}" fill-~
opacity="${["pasture","eventlawn","grove","orchard","camping","overflow","garden"].includes(i.type)?.42:.86}" stroke="${selected?"#fff":shade(c,-25)}" ~
stroke-width="${selected?5:2}" stroke-dasharray="${i.status==="Needs information"?"10 6":"none"}"/>${detail}${showLabel?`<g pointer-events="none"><rect ~
x="${Math.max(2,(width-badgeWidth)/2)}" y="${badgeY}" width="${badgeWidth}" height="34" rx="5" fill="#07110de8" stroke="#ffffff32"/><text x="${width/2}" ~
y="${badgeY+27}" text-anchor="middle" fill="#f2f7f2" font-size="27" font-weight="700">${esc(badge)}</text></g>`:""}${selected?`<rect x="-5" y="-5" ~
width="${width+10}" height="${height+10}" rx="4" fill="none" stroke="#fff" stroke-width="2" stroke-dasharray="6 4" pointer-~
events="none"/>`:""}</g>`}).join("");
const scaleBar=Math.max(100,Math.round(w/8/100)*100),trees=Array.from({length:Math.min(80,Math.floor((w+h)/90))},(_,i)=>{const ~
perimeter=2*(w+h),p=(i+.5)/Math.min(80,Math.floor((w+h)/90))*perimeter;let x,y;if(p<w){x=p;y=-18}else if(p<w+h){x=w+18;y=p-w}else ~
if(p<2*w+h){x=w-(p-w-h);y=h+18}else{x=-18;y=h-(p-2*w-h)}return`<circle cx="${x}" cy="${y}" r="${7+(i%3)*2}" fill="${i%2?"#315d39":"#3c7044"}" ~
opacity=".8"/><circle cx="${x-2}" cy="${y-2}" r="3" fill="#78a36c" opacity=".55"/>`}).join("");
const boundary=state.project.parcelBoundary,pathFor=poly=>poly.map(ring=>ring.map((p,n)=>`${n?"L":"M"}${p[0]} ${p[1]}`).join(" ")+" Z").join(" ~
"),parcelPath=boundary?.polygons.map(pathFor).join(" ")||"",land=boundary?`<defs><clipPath id="parcelClip"><path d="${parcelPath}" fill-~
rule="evenodd"/></clipPath></defs><path d="${parcelPath}" fill="url(#parcelFill)" fill-rule="evenodd"/><path d="${parcelPath}" fill="url(#grass)" fill-~
rule="evenodd"/><g clip-path="url(#parcelClip)">${lines.join("")}</g>`:`<rect x="0" y="0" width="${w}" height="${h}" fill="url(#parcelFill)"/><rect x="0" ~
y="0" width="${w}" height="${h}" fill="url(#grass)"/>${trees}${lines.join("")}`,mapHit=boundary?`<path id="mapHit" d="${parcelPath}" fill="transparent" ~
fill-rule="evenodd" pointer-events="all"/>`:`<rect id="mapHit" x="0" y="0" width="${w}" height="${h}" fill="transparent" pointer-~
events="all"/>`,boundaryStroke=boundary?`<path d="${parcelPath}" fill="none" stroke="#a7bc9f" stroke-width="9" opacity=".35" fill-rule="evenodd"/><path ~
d="${parcelPath}" fill="none" stroke="#c6d7bc" stroke-width="2.5" stroke-dasharray="18 9" fill-rule="evenodd"/>`:`<rect x="0" y="0" width="${w}" ~
height="${h}" fill="none" stroke="#a7bc9f" stroke-width="9" opacity=".35"/><rect x="0" y="0" width="${w}" height="${h}" fill="none" stroke="#c6d7bc" ~
stroke-width="2.5" stroke-dasharray="18 9"/>`;
return`<svg id="siteMap" viewBox="${viewX} ${viewY} ${viewW} ${viewH}" aria-label="Interactive conceptual property map" role="application"><defs><pattern ~
id="grass" width="90" height="90" patternUnits="userSpaceOnUse"><path d="M12 68l4-8m17 22 4-7m21-29 4-8m15 54 4-8" stroke="#527752" stroke-width="2" ~
opacity=".18"/><circle cx="62" cy="20" r="1.5" fill="#7c9568" opacity=".32"/></pattern><linearGradient id="parcelFill" x2="0" y2="1"><stop stop-~
color="#193522"/><stop offset="1" stop-color="#102219"/></linearGradient><filter id="objectShadow" x="-15%" y="-15%" width="130%" ~
height="140%"><feDropShadow dx="0" dy="3" stdDeviation="3" flood-color="#020806" flood-opacity=".65"/></filter></defs><rect x="${viewX}" y="${viewY}" ~
width="${viewW}" height="${viewH}" fill="#0a120f"/>${land}${mapHit}<g id="mapObjects" ~
```

```
filter="url(#objectShadow)">${objs}</g>${state.layers.boundary===false?"":boundaryStroke}<g pointer-events="none"><path d="M${w-47},-56 l-14,32 h28z" ~
fill="#e1eadf"/><text x="${w-39}" y="-61" fill="#f3f7f1" font-size="20" font-weight="800">N</text><rect x="7" y="${h+14}" width="${scaleBar}" height="7" ~
rx="3" fill="#c6d7bc"/><text x="7" y="${h+39}" fill="#dce8d7" font-size="14">${scaleBar.toLocaleString()} ft</text><text x="${w/2}" y="${h+38}" ~
fill="#dce8d7" font-size="14" text-anchor="middle">${Math.round(w).toLocaleString()} × ${Math.round(h).toLocaleString()} ft · ~
${propertyDims().acres.toFixed(2)} acres${boundary?" · IMPORTED PARCEL":""}</text><rect x="${w*.35}" y="${h+56}" width="${w*.3}" height="21" rx="10" ~
fill="#0a120fee" stroke="#536d56"/><text x="${w/2}" y="${h+71}" fill="#bdd0bd" font-size="10" text-anchor="middle">${boundary?"PARCEL OUTLINE · CONCEPT ~
PLAN":"CONCEPT PLAN · NOT A SURVEY"}</text></g></svg>`}
function drawMap(){
const{w,h}=propertyDims(),view=siteMapBounds(),viewX=view.x,viewY=view.y,viewW=view.w,viewH=view.h,viewRight=viewX+viewW,viewBottom=viewY+viewH;
const boundary=state.project.parcelBoundary,pathFor=poly=>poly.map(ring=>ring.map((p,n)=>`${n?"L":"M"}${p[0]} ${p[1]}`).join(" ")+" Z").join(" ~
"),parcelPath=boundary?.polygons.map(pathFor).join(" ")||"";
const majorStep=gridSize*Math.max(2,Math.ceil(200/gridSize));
const grid=state.layers.Grid===false?"":`<defs><pattern id="gridMinor" width="${gridSize}" height="${gridSize}" patternUnits="userSpaceOnUse"><path ~
d="M${gridSize} 0H0V${gridSize}" fill="none" stroke="#789078" stroke-width="1" opacity=".16"/></pattern><pattern id="gridMajor" width="${majorStep}" ~
height="${majorStep}" patternUnits="userSpaceOnUse"><path d="M${majorStep} 0H0V${majorStep}" fill="none" stroke="#91a88c" stroke-width="1.5" ~
opacity=".24"/></pattern></defs><rect x="${viewX}" y="${viewY}" width="${viewW}" height="${viewH}" fill="url(#gridMinor)"/><rect x="${viewX}" y="${viewY}"~
 width="${viewW}" height="${viewH}" fill="url(#gridMajor)"/>`;
const trees=!boundary?Array.from({length:Math.min(80,Math.floor((w+h)/90))},(_,i)=>{const ~
count=Math.min(80,Math.floor((w+h)/90)),p=(i+.5)/Math.max(1,count)*2*(w+h);let x,y;if(p<w){x=p;y=-18}else if(p<w+h){x=w+18;y=p-w}else ~
if(p<2*w+h){x=w-(p-w-h);y=h+18}else{x=-18;y=h-(p-2*w-h)}return`<circle cx="${x}" cy="${y}" r="${7+(i%3)*2}" fill="${i%2?"#315d39":"#3c7044"}" ~
opacity=".8"/><circle cx="${x-2}" cy="${y-2}" r="3" fill="#78a36c" opacity=".55"/>`}).join(""):"";
const land=boundary?`<defs><clipPath id="parcelClip"><path d="${parcelPath}" fill-rule="evenodd"/></clipPath></defs><path d="${parcelPath}" ~
fill="url(#parcelFill)" fill-rule="evenodd"/><path d="${parcelPath}" fill="url(#grass)" fill-rule="evenodd"/>`:`<rect x="0" y="0" width="${w}" ~
height="${h}" fill="url(#parcelFill)"/><rect x="0" y="0" width="${w}" height="${h}" fill="url(#grass)" opacity=".5"/>${trees}`;
const gridMask=grid;
const objects=items().filter(i=>i.visible!==false&&state.layers[layerFor(i.type)]!==false).map(i=>{
const width=i.w,height=i.h,color=i.color||templates[i.type].color,selected=i.id===selectedId,t=templates[i.type],label=esc(i.name||t.name),status=i.status~
==="Needs information";
let footprint="";
if(i.type==="curvedroad"){const roadWidth=Math.min(width,height)*.64,roadInner=Math.min(width,height)*.5,inset=roadWidth/2,p1=inset+(width-~
2*inset)*.3,p2=inset+(width-2*inset)*.7,d=`M${inset} ${height*.65}C${p1} ${height*.65} ${p2} ${height*.35} ${width-inset} ${height*.35}`;footprint=`<path ~
d="${d}" fill="none" stroke="#4b453c" stroke-width="${roadWidth}" stroke-linecap="round"/><path d="${d}" fill="none" stroke="${color}" stroke-~
width="${roadInner}" stroke-linecap="round"/><path d="${d}" fill="none" stroke="#ded5c2" stroke-opacity=".55" stroke-~
width="${Math.min(width,height)*.035}" stroke-dasharray="${Math.min(width,height)*.22} ${Math.min(width,height)*.16}" stroke-linecap="round"/>`}
else if(i.shape===="circle")footprint=`<ellipse cx="${width/2}" cy="${height/2}" rx="${width/2}" ry="${height/2}" fill="${color}" fill-~
opacity="${{"pasture","eventlawn","grove","orchard","camping","overflow","garden"].includes(i.type)?.42:.86}" stroke="${selected?"#fff":shade(color,-25)}"~
 stroke-width="${selected?5:2}" ${status?'stroke-dasharray="10 6"':""}/>`;
else if(i.shape===="triangle")footprint=`<polygon points="${width/2},0 ${width},${height} 0,${height}" fill="${color}" fill-~
opacity="${{"pasture","eventlawn","grove","orchard","camping","overflow","garden"].includes(i.type)?.42:.86}" stroke="${selected?"#fff":shade(color,-25)}"~
 stroke-width="${selected?5:2}" ${status?'stroke-dasharray="10 6"':""}/>`;
else if(i.shape===="poly"){const cut=Math.min(width,height)*.18;footprint=`<polygon points="${cut},0 ${width-cut},0 ${width},${cut} ${width},${height-cut} ~
${width-cut},${height} ${cut},${height} 0,${height-cut} 0,${cut}" fill="${color}" fill-opacity=".86" stroke="${selected?"#fff":shade(color,-25)}" stroke-~
width="${selected?5:2}" ${status?'stroke-dasharray="10 6"':""}/>`}
else{const radius=i.shape==="roundrect"?Math.min(width,height)*.2:["road","mainroad","farmroad","path","trail","firelane"].includes(i.type)?Math.min(width~
,height)/2:["pasture","eventlawn","camping","grove","orchard"].includes(i.type)?Math.min(15,height*.15):3;footprint=`<rect x="0" y="0" width="${width}" ~
height="${height}" rx="${radius}" fill="${color}" fill-~
opacity="${{"pasture","eventlawn","grove","orchard","camping","overflow","garden"].includes(i.type)?.42:.86}" stroke="${selected?"#fff":shade(color,-25)}"~
 stroke-width="${selected?5:2}" ${status?'stroke-dasharray="10 6"':""}/>`}
const detail=mapObjectDetail(i),badge=String(i.label||t.name.slice(0,2)).slice(0,4),badgeWidth=Math.min(width-~
4,Math.max(40,badge.length*24)),badgeY=Math.max(2,height-38),showLabel=state.layers.Labels!==false&&width>=32&&height>=40;
return`<g transform="translate(${i.x} ${i.y}) rotate(${num(i.rotation)} ${width/2} ${height/2})" data-object="${esc(i.id)}" class="map-object" ~
tabindex="0" role="button" aria-label="${label}, ${width} by ${height} feet, rotated ${num(i.rotation)} degrees${selected?", ~
selected":""}"><title>${label} · ${width} × ${height} ft · ${Math.round(areaSqft(i)).toLocaleString()} sq ft · ${acresFor(i).toFixed(3)} ac · ~
${num(i.rotation)}°</title>${mapHit(width,height)}${footprint}${detail}${showLabel?`<g pointer-events="none"><rect x="${Math.max(2,(width-badgeWidth)/2)}"~
 y="${badgeY}" width="${badgeWidth}" height="34" rx="5" fill="#07110de8" stroke="#ffffff32"/><text x="${width/2}" y="${badgeY+27}" text-anchor="middle" ~
fill="#f2f7f2" font-size="27" font-weight="700">${esc(badge)}</text></g>`:""}${selected?`<rect x="-5" y="-5" width="${width+10}" height="${height+10}" ~
rx="4" fill="none" stroke="#fff" stroke-width="2" stroke-dasharray="6 4" pointer-events="none"/>`:""}</g>`}).join("");
const scaleMagnitude=10**Math.floor(Math.log10(Math.max(1,viewW*.15))),scaleRatio=viewW*.15/scaleMagnitude,scaleBar=(scaleRatio>=5?5:scaleRatio>=2?2:1)*sc~
aleMagnitude,scaleX=viewX+viewW*.04,scaleY=viewY+viewH*.9;
const boundaryStroke=boundary?`<path d="${parcelPath}" fill="none" stroke="#a7bc9f" stroke-width="9" opacity=".35" fill-rule="evenodd"/><path ~
d="${parcelPath}" fill="none" stroke="#d2e4c7" stroke-width="2.5" stroke-dasharray="18 9" fill-rule="evenodd"/>`:`<rect x="0" y="0" width="${w}" ~
height="${h}" fill="none" stroke="#a7bc9f" stroke-width="9" opacity=".35"/><rect x="0" y="0" width="${w}" height="${h}" fill="none" stroke="#d2e4c7" ~
stroke-width="2.5" stroke-dasharray="18 9"/>`;
return`<svg id="siteMap" viewBox="${viewX} ${viewY} ${viewW} ${viewH}" aria-label="Infinite conceptual site plan. One map unit equals one foot. Hold Space~
 and drag to pan; scroll to zoom." role="application"><defs><pattern id="grass" width="90" height="90" patternUnits="userSpaceOnUse"><path d="M12 ~
68l4-8m17 22 4-7m21-29 4-8m15 54 4-8" stroke="#527752" stroke-width="2" opacity=".18"/><circle cx="62" cy="20" r="1.5" fill="#7c9568" ~
opacity=".32"/></pattern><linearGradient id="parcelFill" x2="0" y2="1"><stop stop-color="#193522"/><stop offset="1" stop-~
color="#102219"/></linearGradient><filter id="objectShadow" x="-15%" y="-15%" width="130%" height="140%"><feDropShadow dx="0" dy="3" stdDeviation="3" ~
flood-color="#020806" flood-opacity=".65"/></filter></defs><rect id="mapHit" x="${viewX}" y="${viewY}" width="${viewW}" height="${viewH}" ~
fill="#0a120f"/>${land}${gridMask}<g id="mapObjects" filter="url(#objectShadow)">${objects}</g>${state.layers.boundary===false?"":boundaryStroke}<g ~
pointer-events="none"><path d="M${viewRight-38},${viewY+44}l-12 26h24z" fill="#e1eadf"/><text x="${viewRight-31}" y="${viewY+34}" fill="#f3f7f1" font-~
size="18" font-weight="800">N</text><path d="M${scaleX} ${scaleY}h${scaleBar}" stroke="#d5e2d0" stroke-width="${Math.max(1,viewW*.0015)}"/><path ~
d="M${scaleX} ${scaleY-5}v10M${scaleX+scaleBar} ${scaleY-5}v10" stroke="#d5e2d0" stroke-width="${Math.max(1,viewW*.0015)}"/><text x="${scaleX}" ~
y="${scaleY-9}" fill="#dce8d7" font-size="${Math.max(10,viewW*.012)}">${scaleBar.toLocaleString()} ft</text><text x="${viewX+viewW/2}" ~
y="${viewY+viewH-8}" fill="#dce8d7" font-size="${Math.max(10,viewW*.012)}" text-anchor="middle">${Math.round(w).toLocaleString()} × ~
${Math.round(h).toLocaleString()} ft · ${propertyDims().acres.toFixed(2)} acres${boundary?" · IMPORTED PARCEL":" · CONCEPT EXTENT"}</text></g></svg>`;
}
function mapCenter(){const b=state.project.parcelBoundary;if(b?.center)return b.center;const ~
pick=state.project.locationCoordinates;if(Number.isFinite(pick?.lon)&&Number.isFinite(pick?.lat))return[pick.lon,pick.lat];const ~
url=state.project.locationUrl||"",m=url.match(/@(-?\d+(?:\.\d+)?),(-?\d+(?:\.\d+)?)/);if(m){const ~
lat=+m[1],lon=+m[2];if(Math.abs(lat)<=90&&Math.abs(lon)<=180)return[lon,lat]}return null;}
function drawGlobe(){const b=state.project.parcelBoundary,c=mapCenter(),d=propertyDims(),conus=c&&c[0]>=-125&&c[0]<=-~
66&&c[1]>=24&&c[1]<=50,alaska=c&&c[0]>=-179&&c[0]<=-130&&c[1]>=51&&c[1]<=72,hawaii=c&&c[0]>=-161&&c[0]<=-154&&c[1]>=18&&c[1]<=23;let ~
pin=null;if(conus)pin=[155+(c[0]+125)/59*610,92+(50-c[1])/26*324];else if(alaska)pin=[80+(c[0]+179)/49*115,385+(72-c[1])/24*50];else ~
if(hawaii)pin=[188+(c[0]+161)/8*42,440+(23-c[1])/5*25];const outline="M155 116 L220 116 L258 120 L296 112 L334 117 L372 116 L410 118 L448 122 L480 121 ~
L501 130 L518 125 L530 137 L549 129 L560 142 L581 138 L594 153 L618 150 L630 164 L649 162 L663 177 L681 175 L697 189 L713 193 L720 207 L710 221 L714 236 ~
L703 251 L707 267 L694 278 L687 294 L674 302 L665 318 L652 320 L642 334 L633 348 L621 346 L612 361 L605 384 L596 408 L588 398 L582 377 L575 359 L562 344 ~
L547 339 L534 343 L519 339 L504 344 L490 339 L476 344 L462 337 L447 342 L433 336 L419 340 L405 333 L390 338 L376 330 L361 335 L346 326 L331 329 L316 319 ~
L301 322 L288 310 L273 313 L261 300 L247 299 L233 283 L220 272 L207 258 L199 240 L188 227 L183 209 L172 194 L175 177 L166 161 L171 144 L159 132 ~
Z",stateLines="M218 116V180L235 196V244L222 275M268 119V179L282 197V253L271 299M318 115V178L330 202V266L316 319M368 116V177L376 202V271L361 330M418 ~
118V176L417 206V266L405 333M468 120V168L474 203V267L476 340M515 128V176L511 208V265L519 339M558 139V181L550 211V265L562 344M600 151V190L592 222V270L605 ~
384M646 164V205L637 235V280L621 346M680 180V215L671 248V291L652 320M168 160L225 162M183 209L282 210M207 258L330 258M233 283L417 283M273 310L490 310M331 ~
326L562 326M418 176L600 180M474 203L671 205M511 247L694 247M550 281L687 282",states=[["WA",188,137],["OR",190,173],["CA",187,253],["ID",244,174],["NV",225~
,227],["MT",332,143],["WY",324,201],["UT",278,229],["AZ",285,276],["CO",371,244],["NM",365,286],["ND",458,151],["SD",460,191],["NE",462,228],["KS",464,256~
],["OK",470,284],["TX",446,315],["MN",515,157],["IA",514,218],["MO",533,254],["AR",544,282],["LA",555,319],["WI",553,183],["IL",575,230],["MI",603,186],["~
OH",644,226],["GA",646,301],["FL",677,357],["NC",687,270]],pathFor=poly=>poly.map(r=>r.map((p,n)=>`${n?"L":"M"}${p[0]} ${p[1]}`).join(" ")+" Z").join(" ~
"),shape=b?b.polygons.map(pathFor).join(" "):`M0 0H${d.w}V${d.h}H0Z`,scale=b?Math.min(204/Math.max(1,d.w),105/Math.max(1,d.h)):1,ox=b?664+(204-~
d.w*scale)/2:664,oy=b?440+(105-d.h*scale)/2:440,objects=b?items().filter(i=>i.visible!==false&&state.layers[layerFor(i.type)]!==false).map(i=>`<g ~
transform="translate(${i.x} ${i.y})"><rect width="${i.w}" height="${i.h}" rx="2" fill="${esc(i.color||templates[i.type].color)}" stroke="#edf6ee" stroke-~
width="${Math.max(2,3/scale)}"/></g>`).join(""):"",coords=c?`${c[1].toFixed(5)}° ${c[0].toFixed(5)}°`:"Location not set",mapPin=pin?`<g ~
transform="translate(${pin[0]} ${pin[1]})"><circle r="15" fill="#facc15" opacity=".24"/><path d="M0-12a9 9 0 0 0-9 9c0 7 9 18 9 18S9 4 9-3a9 9 0 0 0-9-9z"~
 fill="#facc15" stroke="#fff6c0" stroke-width="2"/><circle cy="-3" r="3" fill="#102016"/></g>`:"";return`<svg id="usaMap" viewBox="0 0 900 600" role="img"~
 aria-label="United States location overview with property inset"><defs><linearGradient id="usLand" x2="0" y2="1"><stop stop-color="#254d36"/><stop ~
offset="1" stop-color="#163124"/></linearGradient><clipPath id="usClip"><path d="${outline}"/></clipPath><clipPath id="parcelInsetClip"><path d="${shape}"~
```

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 fill-rule="evenodd"/></clipPath></defs><rect width="900" height="600" fill="#0a130f"/><text x="30" y="34" fill="#65e2a6" font-size="12" font-weight="750"~
 letter-spacing="1.2">UNITED STATES · LOCATION OVERVIEW</text><text x="30" y="53" fill="#91a69a" font-size="10">Approximate country and state shapes · not~
 a live basemap</text><path d="${outline}" fill="url(#usLand)" stroke="#9bc9a8" stroke-width="3"/><g clip-path="url(#usClip)" fill="none" stroke="#b6d3bb"~
 stroke-opacity=".42" stroke-width="1.3"><path d="${stateLines}"/></g><g fill="#d3e5d6" font-size="8.5" font-weight="650" text-~
anchor="middle">${states.map(([name,x,y])=>`<text x="${x}" y="${y}">${name}</text>`).join("")}</g><path d="M73 390l20-16 24 5 17-9 21 7 14 13-8 13-22 2-14~
 8-24-4-21-7z" fill="#254d36" stroke="#9bc9a8" stroke-width="2"/><text x="120" y="421" fill="#cbdccc" font-size="9" text-anchor="middle">ALASKA</text><g ~
fill="#4b8159" stroke="#b6d3bb" stroke-width="1"><ellipse cx="208" cy="421" rx="7" ry="3"/><ellipse cx="224" cy="426" rx="6" ry="2.5"/><ellipse cx="239" ~
cy="431" rx="5" ry="2"/></g><text x="224" y="450" fill="#cbdccc" font-size="9" text-anchor="middle">HAWAII</text>${mapPin}<rect x="30" y="475" width="520"~
 height="74" rx="12" fill="#111d17" stroke="#3a5843"/><text x="50" y="500" fill="#65e2a6" font-size="10" font-weight="700">PROPERTY LOCATION</text><text ~
x="50" y="521" fill="#eff7f0" font-size="14" font-weight="700">${esc(c?`${coords}${pin?" · marker shown on map":" · outside displayed map extent"}`:"Add a~
 Google Maps coordinate link or import a parcel boundary")}</text><text x="50" y="538" fill="#a8b9ad" font-size="10">${b?`${b.acres.toFixed(2)} acres · ~
${esc(b.source)}`:`${state.project.acres} acres · conceptual planning area; no parcel boundary imported`}</text>${b?`<rect x="648" y="396" width="236" ~
height="168" rx="11" fill="#111d17" stroke="#3a5843"/><text x="666" y="419" fill="#d6e7d8" font-size="10" font-weight="700">PARCEL &amp;amp; SITE PLAN ~
INSET</text><g transform="translate(${ox} ${oy}) scale(${scale})"><path d="${shape}" fill="#254c30" fill-rule="evenodd" stroke="#b5d7ab" stroke-~
width="${Math.max(2,3/scale)}"/><g clip-path="url(#parcelInsetClip)">${objects}</g></g><text x="666" y="551" fill="#9aafa0" font-size="9">Inset uses ~
parcel geometry; dimensions in feet.</text>`:""}<text x="870" y="584" fill="#708277" font-size="9" text-anchor="end">Illustrative map geometry · not for ~
navigation, survey, or siting.</text></svg>`;}
function drawAccurateUsMap(){const map=window.US_STATE_MAP;if(!map?.states?.length)return`<div class="notice">The bundled Census state-boundary data did ~
not load. Reload the app files together to show the U.S. map.</div>`;const b=state.project.parcelBoundary,c=mapCenter(),d=propertyDims(),conus=c&&c[0]>=-~
125&&c[0]<=-66&&c[1]>=24&&c[1]<=50,alaska=c&&c[0]>=-180&&c[0]<=-129&&c[1]>=50&&c[1]<=72,hawaii=c&&c[0]>=-161&&c[0]<=-154&&c[1]>=18&&c[1]<=29;let ~
pin=null,region=null;if(conus)region="CONUS";else if(alaska)region="AK";else if(hawaii)region="HI";if(c&&region){const rad=Math.PI/180;let ~
x,y;if(region==="CONUS"){const p1=29.5*rad,p2=45.5*rad,p0=23*rad,l0=-~
96*rad,n=(Math.sin(p1)+Math.sin(p2))/2,C=Math.cos(p1)**2+2*n*Math.sin(p1),rho=Math.sqrt(C-2*n*Math.sin(c[1]*rad))/n,rho0=Math.sqrt(C-~
2*n*Math.sin(p0))/n,t=n*(c[0]*rad-l0);[x,y]=[rho*Math.sin(t),rho0-rho*Math.cos(t)];const ~
q=map.bounds.CONUS;pin=[55+(x-q.minX)/(q.maxX-q.minX)*710,78+(q.maxY-y)/(q.maxY-q.minY)*415]}else if(region==="AK"){let lon=c[0];if(lon>0)lon-=360;const ~
q=map.bounds.AK;x=(lon+154)*Math.cos(64*rad);y=c[1];pin=[78+(x-q.minX)/(q.maxX-q.minX)*170,514+(q.maxY-y)/(q.maxY-q.minY)*100]}else{const ~
q=map.bounds.HI;pin=[286+(c[0]-q.minX)/(q.maxX-q.minX)*65,550+(q.maxY-c[1])/(q.maxY-q.minY)*47]}}
const states=map.states.map((s,n)=>`<g><path d="${s.d}" fill="${n%2?"#204332":"#274d39"}" fill-rule="evenodd" stroke="#a9c9ad" stroke-~
width=".9"><title>${esc(s.name)} (${s.abbr}) · Census 2024 generalized boundary</title></path><text x="${s.cx}" y="${s.cy+3}" fill="#ecf4eb" font-~
size="${["DC","CT","DE","MA","MD","NH","NJ","RI","VT"].includes(s.abbr)?6.5:8.5}" font-weight="700" text-anchor="middle" paint-order="stroke" ~
stroke="#13291d" stroke-width="2">${s.abbr}</text></g>`).join(""),marker=pin?`<g transform="translate(${pin[0]} ${pin[1]})"><circle r="14" fill="#fbbf24" ~
opacity=".3"/><path d="M0-11a8 8 0 0 0-8 8c0 6 8 16 8 16S8 3 8-3a8 8 0 0 0-8-8z" fill="#fbbf24" stroke="#fff2b0" stroke-width="2"/><circle cy="-3" r="2.5"~
 fill="#102016"/></g>`:"",pathFor=poly=>poly.map(ring=>ring.map((p,n)=>`${n?"L":"M"}${p[0]} ${p[1]}`).join(" ")+" Z").join(" ~
"),parcelPath=b?pathFor(b.polygons.flat()):`M0 0H${d.w}V${d.h}H0Z`,scale=b?Math.min(205/Math.max(1,d.w),125/Math.max(1,d.h)):1,ox=770+(205-~
d.w*scale)/2,oy=565+(125-d.h*scale)/2,siteItems=b?items().filter(i=>i.visible!==false&&state.layers[layerFor(i.type)]!==false).map(i=>`<g ~
transform="translate(${i.x} ${i.y}) rotate(${num(i.rotation)} ${i.w/2} ${i.h/2})"><rect width="${i.w}" height="${i.h}" rx="2" ~
fill="${esc(i.color||templates[i.type].color)}" stroke="#edf6ee" stroke-width="${Math.max(2,3/scale)}"/></g>`).join(""):"",coords=c?`${c[1].toFixed(5)}° ~
N, ${Math.abs(c[0]).toFixed(5)}° ${c[0]<0?"W":"E"}`:"Location not set",locationText=c?`${coords}${pin?" · marker on map":" · outside displayed U.S. ~
extent"}`:"Add a Google Maps coordinate link or import a parcel boundary";return`<svg id="usaMap" viewBox="0 0 1000 720" role="img" aria-label="U.S. state~
 boundaries from Census Bureau 2024 TIGER/Line cartographic boundary data"><defs><linearGradient id="usPanel" x2="0" y2="1"><stop stop-~
color="#16251c"/><stop offset="1" stop-color="#0e1712"/></linearGradient><clipPath id="parcelInsetClip"><path d="${parcelPath}" fill-~
rule="evenodd"/></clipPath></defs><rect width="1000" height="720" fill="#0a130f"/><text x="30" y="34" fill="#6be2aa" font-size="12" font-weight="750" ~
letter-spacing="1.2">UNITED STATES · STATE BOUNDARIES</text><text x="30" y="53" fill="#9caea1" font-size="10">${esc(map.source)} · Albers equal-area ~
projection</text><g>${states}</g><text x="163" y="625" fill="#d5e5d7" font-size="9" font-weight="700" text-anchor="middle">ALASKA</text><text x="318" ~
y="611" fill="#d5e5d7" font-size="9" font-weight="700" text-anchor="middle">HAWAII</text>${marker}<rect x="30" y="640" width="520" height="62" rx="11" ~
fill="url(#usPanel)" stroke="#3a5843"/><text x="48" y="661" fill="#6be2aa" font-size="9" font-weight="750">PROPERTY LOCATION</text><text x="48" y="680" ~
fill="#eff7f0" font-size="12" font-weight="700">${esc(locationText)}</text><text x="48" y="694" fill="#a8b9ad" font-size="9">${b?`${b.acres.toFixed(2)} ~
acres · ${esc(b.source)}`:`${state.project.acres} acre planning area · no parcel boundary imported`}</text>${b?`<rect x="745" y="535" width="235" ~
height="175" rx="11" fill="#111d17" stroke="#3a5843"/><text x="762" y="555" fill="#d6e7d8" font-size="9" font-weight="700">IMPORTED PARCEL &amp;amp; SITE ~
PLAN</text><g transform="translate(${ox} ${oy}) scale(${scale})"><path d="${parcelPath}" fill="#254c30" fill-rule="evenodd" stroke="#b5d7ab" stroke-~
width="${Math.max(2,3/scale)}"/><g clip-path="url(#parcelInsetClip)">${siteItems}</g></g><text x="762" y="698" fill="#9aafa0" font-size="8">Survey/GIS ~
geometry · inset not to scale.</text>`:""}<text x="980" y="713" fill="#829487" font-size="8" text-anchor="end">Source data are generalized cartographic ~
boundaries; not for navigation, surveying, or legal use.</text></svg>`}
drawGlobe=drawAccurateUsMap;
const stateMapRenderer=drawGlobe;
drawGlobe=function(){const svg=stateMapRenderer(),map=window.US_STATE_MAP;if(!map?.states?.length||!svg.includes("<svg"))return svg;const ~
hits=map.states.map(s=>`<path class="state-hit" data-state-map="${s.abbr}" d="${s.d}" fill-rule="evenodd" role="button" tabindex="0" aria-label="Open ~
${esc(s.name)} terrain map"><title>Open ${esc(s.name)} terrain map</title></path>`).join("");return svg.replace("</g><text x=\"163\" y=\"625\"",`</g><g ~
id="stateHits">${hits}</g><text x="163" y="625"`).replace(" · Albers equal-area projection</text>"," · Albers equal-area projection · SELECT A ~
STATE</text>");};
function mapWorldPoint(lon,lat,zoom){const size=256*2**zoom,latitude=Math.max(-~
85.05112878,Math.min(85.05112878,lat))*Math.PI/180;return{x:(lon+180)/360*size,y:(1-Math.asinh(Math.tan(latitude))/Math.PI)/2*size};}
function mapGeoPoint(x,y,zoom){const ~
size=256*2**zoom,lon=x/size*360-180,n=Math.PI-2*Math.PI*y/size,lat=180/Math.PI*Math.atan(Math.sinh(n));return[lon,lat];}
let terrainOsmEnabled=false,terrainOsmFeatures=[],terrainOsmRequest=null,terrainOsmRefreshTimer=null,terrainOsmMessage="",terrainOsmVisibility={homes:true~
,buildings:true,fields:true,woods:true,roads:true,trees:true};
function normalizeMapFeatures(raw){if(raw==null)return[];if(!Array.isArray(raw)||raw.length>500)throw Error("Map overlays must be a list of no more than ~
500 features.");const kinds=["house","building","land","landmark","clearing"];return raw.map((f,index)=>{if(!f||!kinds.includes(f.kind))throw Error(`Map ~
overlay ${index+1} has an unsupported feature type.`);const point=f.kind==="landmark";let coordinates;if(point){const ~
lon=Number(f.coordinates?.[0]),lat=Number(f.coordinates?.[1]);if(!Number.isFinite(lon)||!Number.isFinite(lat)||Math.abs(lon)>180||Math.abs(lat)>90)throw ~
Error(`Map landmark ${index+1} has invalid WGS84 ~
coordinates.`);coordinates=[lon,lat]}else{if(!Array.isArray(f.coordinates)||f.coordinates.length<3||f.coordinates.length>1000)throw Error(`Map area ~
${index+1} needs between 3 and 1,000 vertices.`);coordinates=f.coordinates.map(p=>{const ~
lon=Number(p?.[0]),lat=Number(p?.[1]);if(!Number.isFinite(lon)||!Number.isFinite(lat)||Math.abs(lon)>180||Math.abs(lat)>90)throw Error(`Map area ~
${index+1} has invalid WGS84 coordinates.`);return[lon,lat]});const first=coordinates[0],last=coordinates.at(-~
1);if(first[0]!==last[0]||first[1]!==last[1])coordinates.push([...first]);if(coordinates.length<4||parcelAreaAcres(coordinates)<=0)throw Error(`Map area ~
${index+1} has no measurable area.`)}return{id:String(f.id||uid()),kind:f.kind,name:String(f.name||({house:"House",building:"Building",land:"Land ~
area",landmark:"Landmark",clearing:"Proposed clearing"}[f.kind])).slice(0,100),coordinates}});}
function openStateTerrain(abbr){const entry=window.US_STATE_MAP?.states.find(s=>s.abbr===abbr),center=STATE_MAP_CENTERS[abbr];if(!entry||!center)return;se~
lectedStateMap=abbr;terrainMapCenter=state.project.locationCoordinates?.state===abbr?[state.project.locationCoordinates.lat,state.project.locationCoordina~
tes.lon]:center;terrainZoom=7;terrainLayer="imagery";terrainDrawMode="navigate";terrainDraft=[];terrainSelectedFeatureId=null;terrain3d=true;mapView="terr~
ain";tool="select";render();}
function drawTerrainMap(){const label=window.US_STATE_MAP?.states.find(s=>s.abbr===selectedStateMap)?.name||selectedStateMap;return`<div id="terrainMap" ~
class="terrain-map" role="application" aria-label="${esc(label)} three-dimensional terrain map. Choose a tool to add measured overlays."><div ~
id="terrainCanvas" class="terrain-canvas"></div><div id="terrainMapStatus" class="terrain-map-status">${esc(label)} · 3D elevation terrain with USGS ~
topographic or aerial imagery<br>Click the map to set a location or choose a drawing tool<br>Measurements use WGS84 coordinates; imagery and elevation are~
 planning references, not survey data.</div><div id="terrainOsmStatus" class="terrain-osm-status" aria-live="polite"></div><div class="terrain-map-~
credit">USGS basemap · AWS Terrain Tiles (open elevation data)<br>${terrainOsmEnabled?'<a href="https://www.openstreetmap.org/copyright" target="_blank" ~
rel="noopener">© OpenStreetMap contributors</a>':""}</div></div>`;}
function geodesicFeet(a,b){const rad=Math.PI/180,lat1=a[1]*rad,lat2=b[1]*rad,dLat=(b[1]-a[1])*rad,dLon=(b[0]-~
a[0])*rad,h=Math.sin(dLat/2)**2+Math.cos(lat1)*Math.cos(lat2)*Math.sin(dLon/2)**2;return ~
6371007.1809*2*Math.atan2(Math.sqrt(h),Math.sqrt(1-h))*3.280839895;}
function mapFeatureStats(f){if(f.kind==="landmark")return`${f.coordinates[1].toFixed(5)}°, ${f.coordinates[0].toFixed(5)}°`;const ring=f.coordinates,areaS~
qft=parcelAreaAcres(ring)*43560,perimeter=ring.slice(1).reduce((sum,p,i)=>sum+geodesicFeet(ring[i],p),0),sides=ring.slice(1).map((p,i)=>`${geodesicFeet(ri~
ng[i],p).toLocaleString(undefined,{maximumFractionDigits:0})} ft`);return`${areaSqft.toLocaleString(undefined,{maximumFractionDigits:0})} sq ft · ~
${(areaSqft/43560).toFixed(3)} ac · ${perimeter.toLocaleString(undefined,{maximumFractionDigits:0})} ft perimeter · sides ${sides.join(" / ")}`;}
function terrainFeatureCollection(){const features=[];const parcel=state.project.parcelBoundary;if(parcel)parcel.coordinates.forEach((rings,index)=>featur~
es.push({type:"Feature",id:`parcel-${index}`,properties:{id:`parcel-${index}`,kind:"parcel",name:"Imported ~
parcel"},geometry:{type:"Polygon",coordinates:rings}}));for(const f of state.project.mapFeatures||[]){const properties={id:f.id,kind:f.kind,name:f.name,se~
lected:f.id===terrainSelectedFeatureId};features.push({type:"Feature",id:f.id,properties,geometry:f.kind==="landmark"?{type:"Point",coordinates:f.coordina~
tes}:{type:"Polygon",coordinates:[f.coordinates]}})}return{type:"FeatureCollection",features};}
function terrainOsmCollection(){return{type:"FeatureCollection",features:terrainOsmFeatures};}
function updateTerrainOsmStatus(message=terrainOsmMessage){terrainOsmMessage=message;const ~
status=$("terrainOsmStatus"),credit=document.querySelector(".terrain-map-credit");if(status)status.textContent=message||(terrainOsmEnabled?"Mapped ~
```

```
features are loading.":"Mapped context is off. Enabling it sends this map area to the public OpenStreetMap Overpass ~
API.");if(credit)credit.innerHTML=`USGS basemap · AWS Terrain Tiles (open elevation data)${terrainOsmEnabled?'<br><a ~
href="https://www.openstreetmap.org/copyright" target="_blank" rel="noopener">© OpenStreetMap contributors</a>':""}`;}
function osmBuildingHeight(tags,kind){const ~
match=String(tags.height||"").trim().match(/^(\d+(?:\.\d+)?)\s*(m|met(?:er|re)s?|ft|feet|')?$/i);if(match){const height=Number(match[1])*(/^(ft|feet|')$/i~
.test(match[2]||"")?.3048:1);if(Number.isFinite(height)&&height>0&&height<=300)return{height,estimated:false}}const levels=Number.parseFloat(tags["buildin~
g:levels"]);if(Number.isFinite(levels)&&levels>0&&levels<=100)return{height:Math.max(3,levels*3),estimated:true};return{height:kind==="home"?6.5:9,estimat~
ed:true};}
function parseTerrainOsm(data){if(!Array.isArray(data?.elements))throw Error("The OpenStreetMap response was not valid.");const features=[];for(const ~
element of data.elements){if(features.length>=4000)break;const tags=element.tags||{};let ~
kind;if(element.type==="node"&&tags.natural==="tree")kind="tree";else if(tags.building){const building=String(tags.building).toLowerCase();kind=["house","~
residential","detached","semidetached_house","terrace","apartments","bungalow","dormitory"].includes(building)?"home":"building"}else ~
if(["farmland","meadow","grass","farmyard","orchard","vineyard","allotments"].includes(tags.landuse))kind="field";else ~
if(tags.landuse==="forest"||tags.natural==="wood")kind="wood";else if(["scrub","grassland","heath"].includes(tags.natural))kind="scrub";else ~
if(tags.highway)kind="road";else continue;let coordinates;if(kind==="tree"){if(!Number.isFinite(element.lon)||!Number.isFinite(element.lat))continue;coord~
inates=[element.lon,element.lat]}else{if(!Array.isArray(element.geometry))continue;coordinates=element.geometry.filter(point=>Number.isFinite(point.lon)&&~
Number.isFinite(point.lat)).map(point=>[point.lon,point.lat]);if(kind==="road"){if(coordinates.length<2)continue}else{if(coordinates.length<3)continue;if(~
coordinates[0][0]!==coordinates.at(-1)[0]||coordinates[0][1]!==coordinates.at(-~
1)[1])coordinates.push([...coordinates[0]]);if(coordinates.length<4)continue}}const ~
label=tags.name||tags.building||tags.highway||tags.landuse||tags.natural||"Mapped feature",height=kind==="home"||kind==="building"?osmBuildingHeight(tags,~
kind):null;features.push({type:"Feature",id:`osm-${element.type}-~
${element.id}`,properties:{kind,name:String(label).slice(0,100),source:"OpenStreetMap",detail:String(tags.building||tags.landuse||tags.natural||tags.highw~
ay||"").slice(0,60),...(height?{height:height.height,heightEstimated:height.estimated}: ~
{})},geometry:kind==="tree"?{type:"Point",coordinates}:kind==="road"?{type:"LineString",coordinates}:{type:"Polygon",coordinates:[coordinates]}})}return ~
features;}
function applyTerrainOsmVisibility(map=terrainMapInstance){if(!map?.getLayer("osm-context-fill"))return;const visibleKinds=["home","building","field","woo~
d","scrub","road","tree"].filter(kind=>terrainOsmVisibility[kind=="home"?"homes":kind=="building"?"buildings":kind=="field"?"fields":kind=="wood"||kin~
d=="scrub"?"woods":kind=="road"?"roads":"trees"]),kindFilter=["in",["get","kind"],["literal",visibleKinds]];map.setFilter("osm-context-~
fill",["all",["==",["geometry-type"],"Polygon"],kindFilter]);map.setFilter("osm-context-line",["all",["in",["geometry-~
type"],["literal",["Polygon","LineString"]]],kindFilter]);if(map.getLayer("osm-context-extrusion"))map.setFilter("osm-context-~
extrusion",["all",["==",["geometry-~
type"],"Polygon"],["in",["get","kind"],["literal",visibleKinds.filter(kind=>kind==="home"||kind==="building")]]]);map.setFilter("osm-context-~
tree",terrainOsmVisibility.trees?["==",["get","kind"],"tree"]:["==",["get","kind"],"__hidden__"]);}
function stopTerrainOsmRequest(){if(terrainOsmRefreshTimer){clearTimeout(terrainOsmRefreshTimer);terrainOsmRefreshTimer=null}if(terrainOsmRequest){terrain~
OsmRequest.abort();terrainOsmRequest=null}}
function scheduleTerrainOsmRefresh(){if(!terrainOsmEnabled||!terrainMapInstance)return;if(terrainOsmRefreshTimer)clearTimeout(terrainOsmRefreshTimer);terr~
ainOsmRefreshTimer=setTimeout(()=>{terrainOsmRefreshTimer=null;loadTerrainOsmFeatures()},800)}
async function loadTerrainOsmFeatures(){const map=terrainMapInstance;if(!terrainOsmEnabled||!map)return;if(map.getZoom()<14){updateTerrainOsmStatus("Zoom ~
in to neighborhood level (14× or closer) to load mapped homes, buildings, trees, fields, and roads.");return}const ~
bounds=map.getBounds(),west=bounds.getWest(),south=bounds.getSouth(),east=bounds.getEast(),north=bounds.getNorth();if(east-west>.12||north-~
south>.12){updateTerrainOsmStatus("Zoom in further so the reference-data request covers a small neighborhood.");return}stopTerrainOsmRequest();const ~
controller=new AbortController();terrainOsmRequest=controller;terrainOsmFeatures=[];map.getSource("osm-~
context")?.setData(terrainOsmCollection());updateTerrainOsmStatus("Loading nearby mapped homes, buildings, trees, fields, woods, and roads…");const box=[s~
outh,west,north,east].map(value=>value.toFixed(5)).join(","),query=`[out:json][timeout:20];(way["building"](${box});way["landuse"~"^(forest|orchard|vineya~
rd|farmland|meadow|grass|farmyard|allotments)$"](${box});way["natural"~"^(wood|scrub|grassland|heath)$"](${box});node["natural"="tree"](${box});way["highw~
ay"~"^(residential|unclassified|service|track|tertiary|secondary|primary)$"](${box}););out tags geom qt;`;try{const response=await ~
fetch("https://overpass-api.de/api/interpreter",{method:"POST",headers:{"Content-Type":"application/x-www-form-~
urlencoded;charset=UTF-8"},body:`data=${encodeURIComponent(query)}`,signal:controller.signal});if(!response.ok)throw Error(`HTTP ~
${response.status}`);const data=await response.json();if(controller.signal.aborted||terrainMapInstance!==map||!terrainOsmEnabled)return;terrainOsmFeatures~
=parseTerrainOsm(data);map.getSource("osm-context")?.setData(terrainOsmCollection());applyTerrainOsmVisibility(map);const ~
counts={home:0,building:0,field:0,wood:0,scrub:0,road:0,tree:0};terrainOsmFeatures.forEach(feature=>counts[feature.properties.kind]++);const ~
total=terrainOsmFeatures.length,plural=(count,singular,pluralForm=`${singular}s`)=>`${count} ~
${count===1?singular:pluralForm}`;updateTerrainOsmStatus(total?`OpenStreetMap references: ${plural(counts.home,"nearby home")}, ~
${plural(counts.building,"other building")}, ${plural(counts.field,"field/orchard")}, ${plural(counts.wood+counts.scrub,"wooded/brush area")}, ~
${plural(counts.road,"road")}, ${plural(counts.tree,"mapped tree")}. Coverage varies; not a complete inventory.`:"No mapped features returned for this ~
view. OSM coverage varies by location.");}catch(error){if(error.name==="AbortError")return;terrainOsmFeatures=[];map.getSource("osm-~
context")?.setData(terrainOsmCollection());updateTerrainOsmStatus(`OpenStreetMap reference lookup failed (${error.message}). Check your connection or ~
retry later.`)}finally{if(terrainOsmRequest===controller)terrainOsmRequest=null}}
function toggleTerrainOsm(){terrainOsmEnabled=!terrainOsmEnabled;if(!terrainOsmEnabled){stopTerrainOsmRequest();terrainOsmFeatures=[];terrainMapInstance?.~
getSource("osm-context")?.setData(terrainOsmCollection());updateTerrainOsmStatus("Mapped context is off. Re-enable it to request nearby map references ~
again.")}else{updateTerrainOsmStatus("Zooming to neighborhood detail. OpenStreetMap coverage is incomplete and may not show every home or ~
tree.");if(terrainMapInstance?.getZoom()<14)terrainMapInstance?.easeTo({zoom:15,duration:700});else scheduleTerrainOsmRefresh()}updateTerrainControls();}
function clearTerrainOsm(){terrainOsmEnabled=false;stopTerrainOsmRequest();terrainOsmFeatures=[];terrainMapInstance?.getSource("osm-~
context")?.setData(terrainOsmCollection());updateTerrainOsmStatus("Temporary mapped context cleared from this map. Your saved parcel and drawn overlays ~
are unchanged.");updateTerrainControls();}
function terrainDraftCollection(){if(!terrainDraft.length)return{type:"FeatureCollection",features:[]};const coords=terrainDraft.map(p=>[p[0],p[1]]);let ~
geometry;if(coords.length>2){const ring=[...coords,[...coords[0]]];geometry={type:"Polygon",coordinates:[ring]}}else ~
geometry={type:"LineString",coordinates:coords};return{type:"FeatureCollection",features:[{type:"Feature",properties:{kind:terrainDrawMode},geometry}]};}
function updateTerrainSources(){const feature=terrainMapInstance?.getSource("estate-features"),draft=terrainMapInstance?.getSource("estate-~
draft"),location=terrainMapInstance?.getSource("estate-~
location");if(feature)feature.setData(terrainFeatureCollection());if(draft)draft.setData(terrainDraftCollection());if(location){const p=state.project.loca~
tionCoordinates;location.setData({type:"FeatureCollection",features:p&&p.state===selectedStateMap?[{type:"Feature",properties:{},geometry:{type:"Point",co~
ordinates:[p.lon,p.lat]}}]:[]})}renderTerrainFeaturePanel();updateTerrainControls();}
function renderTerrainFeaturePanel(){const side=document.querySelector(".map-side");if(!side)return;let ~
panel=$("terrainFeaturePanel");if(!panel){side.insertAdjacentHTML("afterbegin",'<section class="card" id="terrainFeaturePanel"><h3>Map overlays & ~
measurements</h3><div class="terrain-feature-panel"></div></section>');panel=$("terrainFeaturePanel")}const list=panel.querySelector(".terrain-feature-~
panel"),features=state.project.mapFeatures||[];list.innerHTML=features.length?features.map(f=>`<div class="terrain-feature-row ~
${f.id===terrainSelectedFeatureId?"selected":""}"><button data-map-feature="${esc(f.id)}"><strong>${esc(f.name)}</strong><span class="terrain-feature-~
measure">${esc(f.kind)} · ${esc(mapFeatureStats(f))}</span></button>${f.id===terrainSelectedFeatureId?`<button class="toolbtn" data-action="delete-map-~
feature" data-id="${esc(f.id)}" aria-label="Delete ${esc(f.name)}">Delete</button>`:""}</div>`).join(""):'<div class="terrain-hint">Draw a house, ~
building, land area, landmark, or proposed clearing on the 3D map. Shapes are saved with this project.</div>';if(terrainSelectedFeatureId){const ~
f=features.find(item=>item.id===terrainSelectedFeatureId);if(!f)terrainSelectedFeatureId=null;}}
function updateTerrainStatus(){const status=$("terrainMapStatus");if(!status)return;const ~
label=window.US_STATE_MAP?.states.find(s=>s.abbr===selectedStateMap)?.name||selectedStateMap,mode=terrainDrawMode,help=mode==="navigate"?"Click the map to~
 set a location; drag to rotate or pan.":mode==="landmark"?"Click once to place a landmark.":`Click map vertices, then choose Finish shape. Escape ~
cancels.`;const p=state.project.locationCoordinates;status.textContent=`${label} · ${terrainLayer==="topo"?"USGS topographic":"USGS aerial imagery"} · ~
${terrainOsmEnabled?"3D terrain on":"3D terrain off"} · ${help}${p?.state===selectedStateMap?` · Location ${p.lat.toFixed(5)}°, ${p.lon.toFixed(5)}°`:""} · Map ~
buildings may have estimated heights; not survey data.`;}
function updateTerrainControls(){const layer=document.querySelector('[data-action="terrain-layer"]'),relief=document.querySelector('[data-~
action="terrain-3d"]'),osm=document.querySelector('[data-action="terrain-osm"]'),clear=document.querySelector('[data-action="terrain-osm-~
clear"]'),use=document.querySelector('[data-action="use-map-location"]'),finish=document.querySelector('[data-action="finish-map-~
feature"]'),undo=document.querySelector('[data-action="undo-map-vertex"]'),cancel=document.querySelector('[data-action="cancel-map-~
feature"]');if(layer)layer.textContent=terrainLayer==="topo"?"Switch to aerial":"Switch to topo";if(relief)relief.textContent=terrain3d?"3D relief on":"3D~
 relief off";if(osm){osm.textContent=terrainOsmEnabled?"Mapped features on":"Mapped features off";osm.setAttribute("aria-~
pressed",String(terrainOsmEnabled))}if(clear)clear.disabled=!terrainOsmEnabled&&!terrainOsmFeatures.length;document.querySelectorAll("[data-osm-layer]").f~
orEach(input=>{input.checked=terrainOsmVisibility[input.dataset.osmLayer]!==false});if(use)use.disabled=state.project.locationCoordinates?.state!==selecte~
dStateMap;if(finish)finish.disabled=terrainDraft.length<3;if(undo)undo.disabled=!terrainDraft.length;if(cancel)cancel.disabled=!terrainDraft.length;update~
TerrainStatus();updateTerrainOsmStatus();}
function addTerrainLayers(map){map.addSource("estate-features",{type:"geojson",data:terrainFeatureCollection()});map.addSource("estate-~
draft",{type:"geojson",data:terrainDraftCollection()});map.addSource("estate-~
location",{type:"geojson",data:{type:"FeatureCollection",features:[]}});map.addLayer({id:"estate-feature-fill",type:"fill",source:"estate-~
features",filter:["==",["geometry-type"],"Polygon"],paint:{"fill-~
color":["match",["get","kind"],"house","#e6a33a","building","#f2c14e","land","#55a879","clearing","#f97354","parcel","#a9d883","#f2c14e"],"fill-~
opacity":["case",["==",["get","kind"],"parcel"],.06,["==",["get","kind"],"clearing"],.3,.4]}});map.addLayer({id:"estate-feature-~
```

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line",type:"line",source:"estate-features",filter:["==",["geometry-type"],"Polygon"],paint:{"line-~
color":["case",["get","selected"],"#ffffff",["match",["get","kind"],"parcel","#b5e48c","clearing","#ff8b5a","#ffe3a6"]],"line-~
width":["case",["==",["get","kind"],"parcel"],2.5,2]}});map.addLayer({id:"estate-landmark-dot",type:"circle",source:"estate-~
features",filter:["==",["geometry-type"],"Point"],paint:{"circle-radius":7,"circle-color":"#f2c14e","circle-stroke-color":"#fff7df","circle-stroke-~
width":2}});map.addLayer({id:"estate-draft-fill",type:"fill",source:"estate-draft",filter:["==",["geometry-type"],"Polygon"],paint:{"fill-~
color":"#4ade80","fill-opacity":.28}});map.addLayer({id:"estate-draft-line",type:"line",source:"estate-draft",paint:{"line-color":"#d9ffe4","line-~
width":2,"line-dasharray":[2,2]}});map.addLayer({id:"estate-location-dot",type:"circle",source:"estate-location",paint:{"circle-radius":8,"circle-~
color":"#fbbf24","circle-stroke-color":"#fff7df","circle-stroke-width":2}});}
function addTerrainOsmLayers(map){map.addSource("osm-context",{type:"geojson",data:terrainOsmCollection()});map.addLayer({id:"osm-context-~
fill",type:"fill",source:"osm-context",filter:["==",["geometry-type"],"Polygon"],paint:{"fill-~
color":["match",["get","kind"],"home","#f2d6a0","building","#e8d7b2","field","#9aaf61","wood","#32754d","scrub","#6c8b55","#718878"],"fill-~
opacity":["match",["get","kind"],"home",.4,"building",.36,"field",.2,.28]}},"estate-feature-fill");map.addLayer({id:"osm-context-extrusion",type:"fill-~
extrusion",source:"osm-context",minzoom:14,filter:["all",["==",["geometry-~
type"],"Polygon"],["in",["get","kind"],["literal",["home","building"]]]],paint:{"fill-extrusion-~
color":["match",["get","kind"],"home","#f0c878","#d5c6aa"],"fill-extrusion-height":["to-number",["get","height"],6],"fill-extrusion-base":0,"fill-~
extrusion-opacity":.92,"fill-extrusion-vertical-gradient":true}},"estate-feature-fill");map.addLayer({id:"osm-context-line",type:"line",source:"osm-~
context",filter:["in",["geometry-type"],["literal",["Polygon","LineString"]]],paint:{"line-color":["match",["get","kind"],"home","#fff0cc","building","#f0~
dab0","field","#c8d58a","wood","#77aa7c","scrub","#a0bd78","road","#d6cfb7","#a3b0a4"],"line-~
width":["match",["get","kind"],"road",2,"home",1.8,"building",1.5,1]}},"estate-feature-fill");map.addLayer({id:"osm-context-~
tree",type:"circle",source:"osm-context",filter:["==",["get","kind"],"tree"],paint:{"circle-radius":4,"circle-color":"#65a866","circle-stroke-~
color":"#e1efce","circle-stroke-width":1}},"estate-feature-fill");applyTerrainOsmVisibility(map);}
function setTerrainLayer(){if(!terrainMapInstance?.isStyleLoaded()){updateTerrainStatus();return}terrainMapInstance.setLayoutProperty("usgs-~
topo","visibility",terrainLayer==="topo"?"visible":"none");terrainMapInstance.setLayoutProperty("usgs-~
imagery","visibility",terrainLayer==="imagery"?"visible":"none");updateTerrainStatus();}
function wireTerrainOsmMap(map){map.once("load",()=>{if(terrainMapInstance!==map)return;addTerrainOsmLayers(map);updateTerrainControls()});map.on("moveend~
",()=>{if(terrainOsmEnabled)scheduleTerrainOsmRefresh()});}
function mountTerrainMap(){const canvas=$("terrainCanvas");if(!canvas)return;if(!window.maplibregl){canvas.innerHTML='<div class="notice" ~
style="margin:20px">The free MapLibre viewer could not load. Check your internet connection, then reload. The U.S. state map and measured flat planner ~
remain available.</div>';return}const oregon=STATE_MAP_CENTERS.OR,style={version:8,sources:{"usgs-~
topo":{type:"raster",tiles:[`${USGS_TILES.topo}/{z}/{y}/{x}`],tileSize:256,attribution:'<a href="https://www.usgs.gov/the-national-map" target="_blank" ~
rel="noopener">USGS The National Map</a>'},"usgs-imagery":{type:"raster",tiles:[`${USGS_TILES.imagery}/{z}/{y}/{x}`],tileSize:256,attribution:'<a ~
href="https://www.usgs.gov/the-national-map" target="_blank" rel="noopener">USGS The National Map imagery</a>'},"elevation-dem":{type:"raster-~
dem",tiles:["https://s3.amazonaws.com/elevation-tiles-prod/terrarium/{z}/{x}/{y}.png"],tileSize:256,encoding:"terrarium",maxzoom:15,attribution:'<a ~
href="https://registry.opendata.aws/terrain-tiles/" target="_blank" rel="noopener">AWS Terrain Tiles open elevation data</a>'}},layers:[{id:"usgs-~
topo",type:"raster",source:"usgs-topo",layout:{visibility:terrainLayer==="topo"?"visible":"none"}},{id:"usgs-imagery",type:"raster",source:"usgs-~
imagery",layout:{visibility:terrainLayer==="imagery"?"visible":"none"}}]};terrainMapInstance=new maplibregl.Map({container:canvas,style,center:terrainMapC~
enter?[terrainMapCenter[1],terrainMapCenter[0]]:[oregon[1],oregon[0]],zoom:terrainZoom,pitch:terrain3d?55:0,bearing:-~
18,antialias:true,attributionControl:true,cooperativeGestures:false});terrainMapInstance.addControl(new ~
maplibregl.NavigationControl({visualizePitch:true}),"top-right");terrainMapInstance.addControl(new ~
maplibregl.ScaleControl({maxWidth:140,unit:"imperial"}),"bottom-~
left");terrainMapInstance.doubleClickZoom.disable();terrainMapInstance.on("load",()=>{if(terrain3d)terrainMapInstance.setTerrain({source:"elevation-dem",e~
xaggeration:1.15});addTerrainLayers(terrainMapInstance);updateTerrainControls();});terrainMapInstance.on("click",event=>{if(!terrainMapInstance)return;con~
st mode=terrainDrawMode;if(mode==="landmark"){const name=$("terrainFeatureName")?.value.trim()||"Landmark",feature={id:uid(),kind:"landmark",name,coordina~
tes:[event.lngLat.lng,event.lngLat.lat]};state.project.mapFeatures.push(feature);terrainSelectedFeatureId=feature.id;save();updateTerrainSources();return}~
if(["house","building","land","clearing"].includes(mode)){if(terrainDraft.length>=999){alert("A single overlay is limited to 1,000 ~
vertices.");return}terrainDraft.push([event.lngLat.lng,event.lngLat.lat]);const source=terrainMapInstance.getSource("estate-~
draft");if(source)source.setData(terrainDraftCollection());updateTerrainControls();return}const ~
mapped=terrainMapInstance.queryRenderedFeatures(event.point,{layers:["osm-context-extrusion","osm-context-fill","osm-context-line","osm-context-~
tree"]})[0];if(mapped){const properties=mapped.properties||{},categories={home:"Nearby mapped home",building:"Mapped building",field:"Mapped ~
field",wood:"Mapped wooded area",scrub:"Mapped brush area",road:"Mapped road",tree:"Mapped ~
tree"},status=$("terrainMapStatus"),name=String(properties.name||"Unnamed"),detail=properties.detail?` · ~
${String(properties.detail)}`:"",height=properties.height?` · ${properties.heightEstimated?"Approx. ":""}${Number(properties.height).toFixed(1)} m ~
tall`:"";if(status)status.textContent=`${categories[properties.kind]||"Mapped feature"}: ${name}${detail}${height} · OpenStreetMap reference; coverage and~
 heights are incomplete.`;return}const hits=terrainMapInstance.queryRenderedFeatures(event.point,{layers:["estate-feature-fill","estate-feature-~
line","estate-landmark-dot"]}),hit=hits.find(f=>f.properties?.id&&!String(f.properties.id).startsWith("parcel-~
"));if(hit){terrainSelectedFeatureId=String(hit.properties.id);updateTerrainSources();return}const ~
p={lat:event.lngLat.lat,lon:event.lngLat.lng,state:selectedStateMap,source:"USGS 3D map location ~
pick"};state.project.locationCoordinates=p;save();updateTerrainSources();updateTerrainControls();});}
function disposeTerrainMap(){stopTerrainOsmRequest();terrainOsmEnabled=false;terrainOsmFeatures=[];terrainOsmMessage="";if(terrainMapInstance){terrainMapI~
nstance.remove();terrainMapInstance=null;}}
function finishTerrainFeature(){if(!["house","building","land","clearing"].includes(terrainDrawMode)||terrainDraft.length<3)return;const coordinates=terra~
inDraft.map(p=>[...p]),first=coordinates[0],last=coordinates.at(-1);if(first[0]!==last[0]||first[1]!==last[1])coordinates.push([...first]);const ~
name=$("terrainFeatureName")?.value.trim()||({house:"House",building:"Building",land:"Land area",clearing:"Proposed clearing"}[terrainDrawMode]);const fea~
ture={id:uid(),kind:terrainDrawMode,name:name.slice(0,100),coordinates};state.project.mapFeatures.push(feature);terrainSelectedFeatureId=feature.id;terrai~
nDraft=[];save();updateTerrainSources();updateTerrainControls();}
function undoTerrainVertex(){if(!terrainDraft.length)return;terrainDraft.pop();const source=terrainMapInstance?.getSource("estate-~
draft");if(source)source.setData(terrainDraftCollection());updateTerrainControls();}
function cancelTerrainDraft(){terrainDraft=[];const source=terrainMapInstance?.getSource("estate-~
draft");if(source)source.setData(terrainDraftCollection());updateTerrainControls();}
function deleteTerrainFeature(id){if(!id||!state.project.mapFeatures.some(f=>f.id===id))return;if(!confirmAction("Delete this map overlay? This does not ~
alter the underlying map ~
imagery."))return;state.project.mapFeatures=state.project.mapFeatures.filter(f=>f.id!==id);terrainSelectedFeatureId=null;save();updateTerrainSources();}
function draw3d(){const canvas=$("scene3d");if(!canvas)return;const rect=canvas.getBoundingClientRect();if(!rect.width||!rect.height)return;const ~
dpr=Math.min(2,devicePixelRatio||1);canvas.width=Math.round(rect.width*dpr);canvas.height=Math.round(rect.height*dpr);const ~
ctx=canvas.getContext("2d");ctx.setTransform(dpr,0,0,dpr,0,0);const cw=rect.width,ch=rect.height,{w,h}=propertyDims();ctx.clearRect(0,0,cw,ch);const ~
scale=Math.min(cw/(Math.max(w,h)*1.18),ch/(Math.max(w,h)*.7))*sceneZoom,cos=Math.cos(sceneAngle),sin=Math.sin(sceneAngle),cx=cw/2,cy=ch*.56;
ctx.save();ctx.translate(scenePan.x,scenePan.y);
const project=(x,y,z=0)=>{const dx=x-w/2,dy=y-h/2,rx=dx*cos-dy*sin,ry=dx*sin+dy*cos;return[cx+(rx-ry)*scale*.57,cy+(rx+ry)*scale*.29-z*scale*.52]};
const poly=(pts,color)=>{ctx.beginPath();pts.forEach((p,n)=>n?ctx.lineTo(p[0],p[1]):ctx.moveTo(p[0],p[1]));ctx.closePath();ctx.fillStyle=color;ctx.fill();~
ctx.strokeStyle="#ffffff20";ctx.lineWidth=1;ctx.stroke()};
const ~
ground=[[0,0],[w,0],[w,h],[0,h]].map(p=>project(p[0],p[1]));if(terrainData()&&terrainData().show!==false)drawTerrainMesh(ctx,project,w,h,sin,cos);else ~
poly(ground,"#1b3b28");
for(let j=0;j<=Math.floor(h/400);j++){const ~
p1=project(0,j*400),p2=project(w,j*400);ctx.beginPath();ctx.moveTo(...p1);ctx.lineTo(...p2);ctx.strokeStyle="#75a17a22";ctx.stroke()}
const ordered=items().filter(i=>i.visible!==false&&state.layers[layerFor(i.type)]!==false).map(i=>{const ~
m=i.w/2,n=i.h/2,dx=i.x+m-w/2,dy=i.y+n-h/2;return{i,depth:dx*sin+dy*cos}}).sort((a,b)=>a.depth-b.depth);
for(const {i} of ordered){const ww=i.w,hh=i.h,z=Math.max(0,num(i.height,1)),corners=[[i.x,i.y],[i.x+ww,i.y],[i.x+ww,i.y+hh],[i.x,i.y+hh]],ez=terrainElev(i~
.x+ww/2,i.y+hh/2)*terrainExag,top=corners.map(p=>project(p[0],p[1],z+ez)),base=corners.map(p=>project(p[0],p[1],ez)),col=i.color||templates[i.type].color,~
land=["pasture","outdoor","paddock","parking","trailer","yard","pond","septic","reserve","drainage"].includes(i.type),thin=["road","path","fence","utility~
"].includes(i.type);
if(!land&&!thin){poly([base[1],base[2],top[2],top[1]],shade(col,-32));poly([base[2],base[3],top[3],top[2]],shade(col,-52))}
poly(top,land?shade(col,-8):thin?col:shade(col,12));if(!land&&!thin&&z>4){const ridge1=project(i.x+ww/2,i.y,z+ez+Math.min(8,z*.3)),ridge2=project(i.x+ww/2~
,i.y+hh,z+ez+Math.min(8,z*.3));ctx.beginPath();ctx.moveTo(...top[0]);ctx.lineTo(...ridge1);ctx.lineTo(...top[1]);ctx.strokeStyle=shade(col,-~
55);ctx.lineWidth=2;ctx.stroke();ctx.beginPath();ctx.moveTo(...top[2]);ctx.lineTo(...ridge2);ctx.lineTo(...top[3]);ctx.stroke()}
if(sceneCut&&i.id===selectedId&&ww>35&&hh>25){for(let t=.25;t<.9;t+=.25){const a=project(i.x+ww*t,i.y+hh*.18,z+ez+.5),b=project(i.x+ww*t,i.y+hh*.82,z+ez+.~
5);ctx.beginPath();ctx.moveTo(...a);ctx.lineTo(...b);ctx.strokeStyle="#fff9";ctx.lineWidth=1.4;ctx.stroke()}}
if(i.id===selectedId&&ww>50&&hh>30&&!land){const label=String(i.label||i.name||t.name);if(label){const ~
p=project(i.x+ww/2,i.y+hh/2,z+ez+2);ctx.fillStyle="#f3f7f3";ctx.font="bold 10px system-ui";ctx.textAlign="center";ctx.fillText(label,p[0],p[1]+3)}}
}
ctx.restore();ctx.fillStyle="#d5e6d8";ctx.font="10px system-ui";ctx.textAlign="left";ctx.fillText("N ~
↑",9,15);ctx.fillStyle="#a8b7ad";ctx.textAlign="right";ctx.fillText(`${state.project.acres} ac · concept only`,cw-9,ch-8);
}
function shade(hex,amt){const h=hex.replace("#","");if(h.length!==6)return hex;const n=parseInt(h,16),r=Math.max(0,Math.min(255,(n>>16)+amt)),g=Math.max(0~
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,Math.min(255,((n>>8)&255)+amt)),b=Math.max(0,Math.min(255,(n&255)+amt));return"#"+((1<<24)+(r<<16)+(g<<8)+b).toString(16).slice(1);}
function designerPage(){const cats=[["Buildings",Object.keys(templates).filter(k=>templates[k].cat==="Building")],["Land & circulation",["pasture","road",~
"path","fence","parking","trailer","paddock"]],["Infrastructure",Object.keys(templates).filter(k=>["Utility","Water","Operations","Arena"].includes(templa~
tes[k].cat))]];
const sc=scenario(),c=compute(),picked=items().find(i=>i.id===selectedId);
return pageHead("SITE PLAN","Site Designer","Design the whole property—from barns and pasture to an event campus, fairgrounds, guest arrival, landscape, ~
and site infrastructure.",`${btn("Undo","undo")}${btn("Redo","redo")}${btn("Export JSON","export")}${btn("Clear layout","clear","danger")}`)+
`<div class="designer-layout"><div class="card map-card"><div class="map-toolbar"><div class="toolrow"><button class="toolbtn ~
${tool==="select"?"active":""}" data-tool="select">■ Select / move</button><button class="toolbtn ${tool==="erase"?"active":""}" data-tool="erase">■ ~
Erase</button><button class="toolbtn ${tool==="pan"?"active":""}" data-tool="pan">✥ Pan canvas</button><button class="toolbtn ~
${tool==="rotate"?"active":""}" data-tool="rotate">■ Rotate selected</button></div><span class="push"></span><button class="toolbtn" data-action="zoom-in"~
 aria-label="Zoom in">■</button><button class="toolbtn" data-action="zoom-out" aria-label="Zoom out">–</button><button class="toolbtn" data-action="zoom-~
fit">Fit</button><select id="gridSelect" class="toolbtn" aria-label="Planning grid spacing">${selectOpts([[10,"10 ft grid"],[20,"20 ft grid"],[40,"40 ft ~
grid"],[100,"100 ft grid"]],gridSize)}</select><button class="toolbtn ${state.layers.Grid===false?"":"active"}" data-action="toggle-~
grid">Grid</button></div><div class="map-viewport" id="mapViewport">${drawMap()}</div><div class="map-meta"><span>${gridSize} ft planning grid · snap ~
${snapEnabled?"on":"off"}</span><button class="toolbtn" data-action="toggle-snap">Snap ${snapEnabled?"on":"off"}</button><span>Click to place · select and~
 drag to move</span><span>Concept plan · not a survey</span></div></div>
<div class="map-side"><details class="card palette-card designer-details" data-detail="library" ${detailsOpen.library?"open":""}><summary><span>Add an ~
object</span><span class="pill">${Object.keys(templates).length} templates</span></summary><div class="details-body"><input id="paletteSearch" ~
class="input" value="${esc(paletteQuery)}" placeholder="Search trees, arena, camping…" aria-label="Search site objects"><div class="palette-~
filters">${paletteGroups.map(g=>`<button class="palette-filter ${paletteGroup===g?"active":""}" data-palette-group="${esc(g)}">${esc(g)} ~
${paletteCount(g)}</button>`).join("")}</div><div class="palette-grid">${paletteHtml()}</div><div class="palette-hint">${templates[tool]?`Selected: ~
<strong>${esc(templates[tool].name)}</strong> · click the map to place`:"Choose an object, then place it on the map."}</div></details>
<div class="card"><div class="between"><h3>Land summary</h3><span class="pill">${state.project.acres} ac</span></div><div class="key-value"><span>Built / ~
hardscape</span><strong>${c.used.toFixed(2)} ac</strong></div><div class="key-value"><span>Pasture</span><strong>${c.past.toFixed(2)} ~
ac</strong></div><div class="key-value"><span>Open / unassigned</span><strong>${c.open.toFixed(2)} ac</strong></div><div class="key-value"><span>Placed ~
objects</span><strong>${items().length}</strong></div>${c.outOfBounds?`<div class="notice" style="margin-top:10px">${c.outOfBounds} object(s) extend ~
outside the current boundary. Adjust acreage or move/resize them; existing objects were preserved.</div>`:`<div class="notice" style="margin-top:10px">New~
 placements are checked against the current boundary. Map geometry is conceptual.</div>`}</div>
<details class="card designer-details" ${picked?"open":""}><summary><span>${picked?`Edit ${esc(picked.name||templates[picked.type].name)}`:"Selected ~
object"}</span>${picked?`<span class="pill">${esc(picked.phase)} phase</span>`:`<span class="small muted">Select an object to edit ~
details</span>`}</summary><div class="details-body">${picked?objectEditor(picked):`<div class="small muted">Select an object on the map or from the object~
 list.</div>`}</div></details>
<details class="card designer-details" data-detail="objects" ${detailsOpen.objects?"open":""}><summary><span>Site objects</span><span class="small ~
muted">${items().length} placed</span></summary><div class="details-body"><div class="object-list">${items().map(i=>`<button class="object-row ~
${selectedId===i.id?"selected":""}" data-select-object="${esc(i.id)}"><i class="swatch" style="background:${esc(i.color)}"></i><span ~
style="flex:1">${esc(i.name||templates[i.type].name)}</span><span class="muted">${Math.round(i.w)}×${Math.round(i.h)}'</span></button>`).join("")||`<div ~
class="empty">No objects placed yet.</div>`}</div></div></details>
<details class="card designer-details" data-detail="layers" ${detailsOpen.layers?"open":""}><summary><span>Drawing layers</span><span class="small ~
muted">Visibility &amp;amp; labels</span></summary><div class="details-body"><div style="margin-bottom:10px"><button class="toolbtn" data-action="toggle-~
labels">${state.layers.Labels===false?"Labels off":"Labels on"}</button></div><div class="layer-grid">${["boundary","Buildings","Pastures","Roads","Paths"~
,"Fences","Parking","Utilities","Water","Operations","Fairgrounds","Landscape","Farmland","Town","Access"].map(l=>`<label class="layer-chip"><input ~
type="checkbox" data-layer="${esc(l)}" ~
${l==="boundary"?state.layers.boundary!==false?"checked":"":state.layers[l]!==false?"checked":""}>${l==="boundary"?"Parcel ~
boundary":esc(l)}</label>`).join("")}</div></div></details>
</div>
<div class="card scene-card"><div class="between"><h3>Interactive isometric preview</h3><div class="row"><span class="pill">2D canvas · WebGL-~
free</span><button class="toolbtn" data-scene-reset title="Reset isometric view">Reset view</button></div></div><canvas id="scene3d" class="scene-canvas" ~
aria-label="Interactive isometric estate preview. Drag to orbit, Shift-drag or right-drag to pan, and scroll to zoom." tabindex="0"></canvas><div ~
class="between small muted" style="margin-top:8px"><span>Drag to orbit · Shift-drag to pan · wheel to zoom</span><details class="scene-~
options"><summary>Preview options</summary><div class="row"><button class="toolbtn" data-action="toggle-cut">${sceneCut?"Cutaway: on":"Cutaway: ~
off"}</button><button class="toolbtn" data-action="toggle-auto">${sceneAuto?"Auto-rotate: on":"Auto-rotate: ~
off"}</button></div></details></div></div></div></div>${disclaimerShort()}`;}
function objectEditor(i){const t=templates[i.type];return`<div class="field"><label for="objName">Name</label><input id="objName" data-object-field="name"~
 value="${esc(i.name||t.name)}"></div><div class="inline-fields" style="margin-top:8px"><div class="field"><label>Width (ft)</label><input type="number" ~
min="1" max="10000" step="1" data-object-field="w" value="${i.w}"></div><div class="field"><label>Length (ft)</label><input type="number" min="1" ~
max="10000" step="1" data-object-field="h" value="${i.h}"></div><div class="field"><label>Height (ft)</label><input type="number" min="1" max="300" ~
step="1" data-object-field="height" value="${i.height}"></div><div class="field"><label>Rotation (°)</label><input type="number" min="0" max="359" ~
step="1" data-object-field="rotation" value="${num(i.rotation)}"></div><div class="field"><label>Project phase</label><select data-object-~
field="phase">${selectOpts([["1","Phase 1"],["2","Phase 2"],["3","Phase 3"]],String(i.phase||"1"))}</select></div><div class="field"><label>Planning ~
status</label><select data-object-field="status">${selectOpts([["Not evaluated","Not evaluated"],["Needs information","Needs ~
information"],["Preliminary","Preliminary"],["User-confirmed","User-confirmed"],["Professionally verified","Professionally ~
verified"]],i.status||"Preliminary")}</select></div><div class="field"><label>Shape</label><select data-object-~
field="shape">${selectOpts([["rect","Rectangle"],["roundrect","Rounded rectangle"],["circle","Ellipse"],["triangle","Triangle"],["poly","Angled ~
polygon"]],i.shape||"rect")}</select></div><div class="field"><label>Unit cost ($)</label><input type="number" min="0" step="1" data-object-~
field="unitCost" value="${num(i.unitCost,t.cost)}"></div></div><div class="key-value"><span>Footprint</span><strong>${i.w.toLocaleString()} × ~
${i.h.toLocaleString()} ft · ${Math.round(areaSqft(i)).toLocaleString()} sq ft · ${acresFor(i).toFixed(3)} ac</strong></div><div class="key-~
value"><span>Cost method</span><strong>${esc(i.costMethod||t.method)} · ${fmt(i.unitCost)}</strong></div><div class="object-actions" style="margin-~
top:10px"><button class="btn" data-action="duplicate">Duplicate</button><button class="btn" data-action="lock">${i.locked?"Unlock":"Lock"} ~
object</button><button class="btn danger" data-action="delete-selected">Delete</button></div>`;}
function buildingCards(){return Object.keys(templates).filter(k=>templates[k].cat==="Building").map(k=>{const ~
t=templates[k],i=items().find(x=>x.type===k);return`<div class="card"><div class="between"><div class="row"><i class="swatch" ~
style="background:${t.color}"></i><h3 style="margin:0">${esc(t.name)}</h3></div><span class="pill">${i?"Placed":"Not placed"}</span></div><div class="key-~
value"><span>Planning footprint</span><strong>${i?i.w:t.w} × ${i?i.h:t.h} ft</strong></div><div class="key-value"><span>Illustrative cost ~
basis</span><strong>${fmt(i?.unitCost??t.cost)}${t.method==="fixed"?" / unit":t.method==="sqft"?" / sq ~
ft":""}</strong></div>${buildingPlan(k,i)}</div>`}).join("")}
function buildingPlan(k,i){const configs={barn:["Stall count","20"],arena:["Bleacher rows","3"],event:["Tables","24"],shop:["Service ~
bays","3"],stand:["Display counters","2"],yurt:["Units","2"]};const [label,def]=configs[k]||["Units","1"];const val=i?.[k+"Count"]??def;return`<div ~
style="margin-top:10px">${floorPlanSvg(k,num(val),i?.w||templates[k].w,i?.h||templates[k].h)}</div><div class="inline-fields" style="margin-top:9px"><div ~
class="field"><label>${label}</label><input type="number" min="0" data-plan="${k}" value="${val}"></div><div class="field"><label>Displayed ~
area</label><input value="${((i?.w||templates[k].w)*(i?.h||templates[k].h)).toLocaleString()} sq ft" readonly></div></div>`;}
function floorPlanSvg(k,count,w,h){const titles={barn:"Stalls and center aisle",arena:"Footing and viewing",event:"Hall and service zones",shop:"Service ~
bays and support",stand:"Retail display area",yurt:"Two-unit caretaker concept"};let shapes="";
if(k==="barn"){for(let n=0;n<Math.min(20,Math.max(0,count));n++){const x=(n%10)*14.4+1.4,y=n<10?2:27;shapes+=`<rect x="${x}" y="${y}" width="12" ~
height="12" fill="#9a5b2a" stroke="#d39a69"/><text x="${x+6}" y="${y+8}" text-anchor="middle" fill="#fff" font-size="4">S${n+1}</text>`}shapes+=`<rect ~
x="1" y="15" width="142" height="10" fill="#5b6260"/><text x="72" y="22" text-anchor="middle" fill="#fff" font-size="5">12 ft center aisle</text><rect ~
x="130" y="39" width="13" height="8" fill="#52624c"/>`}
else if(k==="arena"){shapes=`<rect x="3" y="3" width="138" height="68" fill="#806c4d" stroke="#d0b489"/><text x="72" y="37" text-anchor="middle" ~
fill="#fff" font-size="8">RIDING FOOTING</text>`;for(let n=0;n<Math.min(8,count);n++)shapes+=`<line x1="10" y1="${10+n*4}" x2="42" y2="${10+n*4}" ~
stroke="#d6d3d1" stroke-width="1"/>`;shapes+=`<rect x="112" y="6" width="25" height="15" fill="#475569"/><text x="124" y="15" text-anchor="middle" ~
fill="#fff" font-size="5">VIEWING</text>`}
else if(k==="event"){shapes=`<rect x="1" y="1" width="98" height="48" fill="#412345" stroke="#d580e3"/><text x="50" y="7" text-anchor="middle" fill="#fff"~
 font-size="5">EVENT HALL</text><rect x="101" y="1" width="42" height="24" fill="#57402d" stroke="#d6a868"/><text x="122" y="14" text-anchor="middle" ~
fill="#fff" font-size="5">KITCHEN</text><rect x="101" y="27" width="20" height="22" fill="#354653" stroke="#8ba9bc"/><text x="111" y="40" text-~
anchor="middle" fill="#fff" font-size="4">REST</text><rect x="123" y="27" width="20" height="22" fill="#354653" stroke="#8ba9bc"/><text x="133" y="40" ~
text-anchor="middle" fill="#fff" font-size="4">STAGE</text>`;for(let n=0;n<Math.min(24,count);n++){const ~
x=10+(n%6)*14,y=15+Math.floor(n/6)*8;shapes+=`<circle cx="${x}" cy="${y}" r="2.5" fill="#eac76e"/>`}}
else if(k==="shop"){shapes=`<rect x="1" y="1" width="108" height="48" fill="#143b47" stroke="#40c8dc"/><rect x="112" y="1" width="31" height="23" ~
fill="#38505a"/><rect x="112" y="26" width="31" height="23" fill="#38505a"/><text x="127" y="14" text-anchor="middle" fill="#fff" font-~
size="4">OFFICE</text><text x="127" y="39" text-anchor="middle" fill="#fff" font-size="4">PARTS</text>`;for(let n=0;n<Math.min(8,count);n++){const ~
x=5+(n%4)*25,y=n<4?5:27;shapes+=`<rect x="${x}" y="${y}" width="20" height="17" fill="none" stroke="#d9f0f4"/><text x="${x+10}" y="${y+10}" text-~
anchor="middle" fill="#d9f0f4" font-size="4">BAY ${n+1}</text>`}}
else if(k==="stand"){shapes=`<rect x="1" y="1" width="141" height="48" fill="#451f2a" stroke="#f08ca7"/><rect x="7" y="8" width="22" height="32" ~
fill="#97374e"/><text x="18" y="26" text-anchor="middle" fill="#fff" font-size="4">ENTRY</text>`;for(let n=0;n<Math.min(8,count);n++){const ~
```

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x=38+(n%4)*25,y=n<4?10:29;shapes+=`<rect x="${x}" y="${y}" width="18" height="8" fill="#c96877"/><text x="${x+9}" y="${y+6}" text-anchor="middle" ~
fill="#fff" font-size="3">DISPLAY</text>`}}
else shapes=`<rect x="10" y="10" width="49" height="28" rx="10" fill="#645219" stroke="#eac957"/><rect x="84" y="10" width="49" height="28" rx="10" ~
fill="#645219" stroke="#eac957"/><text x="34" y="26" text-anchor="middle" fill="#fff" font-size="6">YURT 1</text><text x="108" y="26" text-anchor="middle"~
 fill="#fff" font-size="6">YURT 2</text><path d="M2 45H142" stroke="#83928a" stroke-dasharray="3"/>`;
return`<svg viewBox="0 0 144 52" role="img" aria-label="${esc(titles[k])} conceptual floor plan"><rect x=".5" y=".5" width="143" height="50" ~
fill="#0d1712" stroke="#65756b"/>${shapes}<text x="2" y="51" fill="#a7b8ac" font-size="3">${Math.round(w)} × ${Math.round(h)} ft · conceptual, not to ~
scale</text></svg>`;}
function buildingsPage(){return pageHead("BUILDING DESIGN","Building Design","Conceptual space planning for the principal estate buildings. Dimensions and~
 interior zones are editable planning assumptions.",btn("Open site designer","page:designer","ghost"))+`<div class="grid g2">${buildingCards()}</div><div ~
class="notice" style="margin-top:14px">Floor-plan diagrams are conceptual arrangements—not architectural drawings. Have licensed professionals determine ~
structure, life safety, accessibility, and construction details.</div>`;}
function infrastructurePage(){const list=[["Utilities & access","Plan utility corridors, service capacity, emergency access, fire flow, turnarounds, and ~
maintenance access."],["Water","Confirm source, legal availability, quantity, seasonal reliability, storage, and water quality."],["Wastewater","Confirm ~
site suitability and capacity for domestic, equestrian, food-service, and event loads."],["Stormwater & erosion","Map drainage patterns and assess ~
erosion, disturbance, and permit requirements."],["Pasture & manure","Plan manure storage, nutrient management, fencing, water, rotations, and all-weather~
 access."],["Traffic & parking","Estimate event and daily trips; verify access, parking, emergency circulation, and road approvals."]];return ~
pageHead("SITE SYSTEMS","Infrastructure Planner","Track concept-level infrastructure topics and connect them to due-diligence tasks. No site feasibility ~
determination is implied.")+
`<div class="grid g2">${list.map(([title,desc],n)=>`<div class="card"><div class="between"><h3>${title}</h3><span class="pill ~
warn">Planning</span></div><p class="small muted">${desc}</p><label class="field"><span class="label">Planning note</span><textarea data-infra="${n}" ~
placeholder="Record a question, assumption, or professional follow-up…">${esc(state.notes[n]||"")}</textarea></label></div>`).join("")}</div><div ~
class="notice blueish" style="margin-top:14px">Infrastructure cost allowances are not engineering estimates. Confirm utility availability, permits, ~
connection fees, and construction scope with the relevant providers and licensed professionals.</div>`;}
function financialsPage(){const f=scenario().finance,c=compute();const capRows=Object.keys(templates).map(type=>{const it=items().filter(x=>x.type===type)~
,t=templates[type],qty=it.reduce((s,x)=>s+(t.method==="linear"?Math.max(x.w,x.h):t.method==="acre"?areaSqft(x)/43560:1),0);return{type,t,qty,cost:it.reduc~
e((s,x)=>s+(x.costMethod==="linear"?Math.max(x.w,x.h)*x.unitCost:x.costMethod==="acre"?areaSqft(x)/43560*x.unitCost:x.costMethod==="sqft"||x.costMethod===~
"area"?areaSqft(x)*x.unitCost:x.unitCost),0)}}).filter(x=>x.qty>0);
const phases=[1,2,3].map(p=>({p,cost:items().filter(i=>String(i.phase)===String(p)).reduce((s,i)=>s+estimateItem(i),0)})),projection=[1,2,3,4,5].map(y=>({~
y,rev:c.gross*Math.pow(1.03,y-1),opx:c.opex*Math.pow(1.03,y-1)}));
return pageHead("FINANCIAL MODEL","Financials","Editable inputs drive an illustrative annual operating model and site-linked capital budget.",`<span ~
class="pill">${f.linked?"Layout linked":"Manual assumptions"}</span>`)+
`<div class="grid g2"><div class="card stack"><div class="between"><h3>Operating assumptions</h3><label class="row small"><input type="checkbox" data-~
finance="linked" ${f.linked?"checked":""}> Link to site layout</label></div>${sliderField("Monthly board ~
rate","rate",f.rate,600,1200,25)}${sliderField("Occupied ~
stalls","occupied",f.occupied,0,Math.max(20,f.linked?Math.max(20,items().filter(i=>i.type==="barn").length*20):40),1)}${sliderField("Annual ~
events","events",f.events,0,Math.max(18,f.linked?Math.max(18,items().filter(i=>i.type==="event").length*18):36),1)}${sliderField("Average event site ~
fee","eventFee",f.eventFee,3000,8500,250)}<hr class="divider"><div class="inline-fields">${numberField("Ancillary revenue / ~
year","other",f.other)}${numberField("Base fixed OpEx / year","fixed",f.fixed)}${numberField("Cost / occupied stall / ~
year","stallCost",f.stallCost)}${numberField("Direct cost / event","eventCost",f.eventCost)}${numberField("Contingency ~
(%)","contingency",f.contingency)}<label class="row small"><input type="checkbox" data-finance="payrollOffset" ${f.payrollOffset?"checked":""}> Family ~
labor planning offset (−$90,000)</label></div><div class="small muted">Linked mode uses placed barns/events and selected ancillary categories; unlinked ~
mode uses manual quantities. No barn/event means linked stalls/events are capped at zero.</div></div>
<div class="stack"><div class="grid g2">${kpi("Gross annual revenue",fmt(c.gross),"Boarding + events + ancillary","green")}${kpi("Annual operating ~
expense",fmt(c.opex),"Before financing","amber")}${kpi("Net operating income",fmt(c.noi),"Before financing and tax",c.noi>=0?"green":"rose")}${kpi("Stall ~
breakeven",Number.isFinite(c.breakeven)?`${c.breakeven} stalls`:"Not reached","At current rates and other revenue", "blue")}</div><div ~
class="card"><h3>Transparent calculation</h3><div class="key-value"><span>Boarding</span><strong>${c.occupied} × ${fmt(f.rate)} × 12 = ~
${fmt(c.occupied*f.rate*12)}</strong></div><div class="key-value"><span>Events</span><strong>${c.events} × ${fmt(f.eventFee)} = ~
${fmt(c.events*f.eventFee)}</strong></div><div class="key-value"><span>Ancillary</span><strong>${fmt(c.other)}</strong></div><div class="key-~
value"><span>Revenue − OpEx</span><strong>${fmt(c.gross)} − ${fmt(c.opex)} = ${fmt(c.noi)}</strong></div><div class="key-value"><span>Layout CapEx incl. ~
contingency</span><strong>${fmt(c.capex)}</strong></div><div class="key-value"><span>Simple payback</span><strong>${c.noi>0?(c.capex/c.noi).toFixed(1)+" ~
years":"Not meaningful at current NOI"}</strong></div><p class="tiny muted">Simple payback = illustrative CapEx ÷ annual NOI. Excludes financing, taxes, ~
depreciation, replacement reserves, ramp-up, inflation, and working capital.</p><hr class="divider"><h3>Financing illustration</h3><div class="inline-~
fields">${numberField("Loan principal","financing",f.financing)}${numberField("Annual interest (%)","interest",f.interest)}${numberField("Term ~
(years)","term",f.term)}</div><div class="key-value"><span>Estimated annual debt service</span><strong>${fmt(c.annualDebt)}</strong></div><div class="key-~
value"><span>NOI after estimated debt service</span><strong>${fmt(c.afterDebt)}</strong></div><p class="tiny muted">Fixed-rate, fully amortizing monthly-~
payment estimate. Excludes fees, taxes, covenants, draw schedules, and lender qualification.</p></div></div></div>
<div class="card" style="margin-top:14px"><div class="between"><h3>Site-linked capital budget</h3><span class="pill warn">Placeholder unit ~
costs</span></div><div class="table-wrap"><table><thead><tr><th>Object type</th><th>Quantity</th><th>Cost method</th><th>Estimated ~
subtotal</th></tr></thead><tbody>${capRows.map(r=>`<tr><td>${esc(r.t.name)}</td><td>${r.qty.toFixed(r.t.method===="acre"?2:0)} ~
${r.t.method==="linear"?"ft":r.t.method==="acre"?"ac":"unit(s)"}</td><td>${esc(r.t.method)} · ${fmt(r.t.cost)}${r.t.method==="sqft"||r.t.method==="area"?"~
 / sf":r.t.method==="acre"?" / ac":r.t.method==="linear"?" / ft":""}</td><td>${fmt(r.cost)}</td></tr>`).join("")||`<tr><td colspan="4">Add objects on the ~
site plan to populate the budget.</td></tr>`}</tbody><tfoot><tr><th colspan="3">Subtotal + ${num(f.contingency)}% ~
contingency</th><th>${fmt(c.capex)}</th></tr></tfoot></table></div></div>
<div class="grid g2" style="margin-top:14px"><div class="card"><h3>Phase-level capital allowance</h3>${phases.map(x=>`<div class="phase"><div><h4>Phase ~
${x.p}</h4><p>${x.p===1?"Core site & equestrian":x.p===2?"Support facilities":"Public-facing uses"}</p></div><div class="barline"><i ~
style="width:${c.costs?Math.min(100,x.cost/c.costs*100):0}%"></i></div><strong>${fmt(x.cost*(1+num(f.contingency)/100))}</strong></div>`).join("")}<p ~
class="tiny muted">Phase allocations follow each site's object phase field; contingency is applied proportionally.</p></div><div class="card"><h3>Five-~
year operating illustration</h3><div class="table-wrap"><table><thead><tr><th>Year</th><th>Gross revenue</th><th>Operating expense</th><th>NOI</th></tr></~
thead><tbody>${projection.map(y=>`<tr><td>${y.y}</td><td>${fmt(y.rev)}</td><td>${fmt(y.opx)}</td><td>${fmt(y.rev-~
y.opx)}</td></tr>`).join("")}</tbody></table></div><p class="tiny muted">Illustrative 3% annual growth for revenue and operating costs, held constant; no ~
ramp-up, financing, taxes, or inflation-specific adjustment.</p></div></div>${disclaimerShort()}`;}
function sliderField(label,key,val,min,max,step){return`<div class="field"><div class="between"><label ~
for="fin-${key}">${label}</label><strong>${key==="rate"||key==="eventFee"?fmt(val):Number(val).toLocaleString()}</strong></div><div class="range-~
row"><input id="fin-${key}" type="range" min="${min}" max="${max}" step="${step}" value="${num(val)}" data-finance="${key}"><input aria-label="${label} ~
exact value" type="number" min="${min}" max="${max}" step="${step}" value="${num(val)}" data-finance="${key}"></div></div>`}
function numberField(label,key,val){return`<div class="field"><label>${label}</label><input type="number" min="0" data-finance="${key}" ~
value="${num(val)}"></div>`}
function operationsPage(){const c=compute(),o=state.operations;const staff=[["Barn attendants",`Occupied stalls ÷ ~
${o.barnRatio}`,Math.ceil(c.occupied/Math.max(1,o.barnRatio))],["Barn manager","Baseline planning allowance",items().some(i=>i.type==="barn")?1:0],["Farm ~
& grounds",`Pasture acres ÷ ${o.landRatio}`,Math.ceil(c.past/Math.max(1,o.landRatio))],["Event support",`Events × ~
${o.eventRatio}`,c.events*o.eventRatio],["Farm stand","Seasonal coverage allowance",items().some(i=>i.type==="stand")?1:0]];
return pageHead("OPERATIONS","Operations & Staffing","Illustrative staffing quantities for workload discussion. Adjust the planning ratios; they are not ~
labor or safety guidance.",btn("Review labor due diligence","page:compliance","ghost"))+`<div class="grid g2"><div class="card"><h3>Staffing workload ~
assumptions</h3><div class="inline-fields">${numberField("Stalls per attendant","barnRatio",o.barnRatio).replace('data-finance="barnRatio"','data-~
operation="barnRatio"')}${numberField("Pasture acres per worker","landRatio",o.landRatio).replace('data-finance="landRatio"','data-~
operation="landRatio"')}${numberField("Event support per event","eventRatio",o.eventRatio).replace('data-finance="eventRatio"','data-~
operation="eventRatio"')}</div><p class="tiny muted">Edit ratios here. Actual staffing depends on service scope, schedules, law, and safe ~
operations.</p></div><div class="card"><h3>Illustrative workload result</h3>${staff.map(([role,calc,n])=>`<div class="key-value"><span>${role}<small ~
class="muted" style="display:block">${calc}</small></span><strong>${n} planning units</strong></div>`).join("")}<div class="notice" style="margin-~
top:10px">These are workload indicators, not a staffing plan, headcount commitment, wage estimate, or determination of employment ~
requirements.</div></div></div><div class="card" style="margin-top:14px"><h3>Operating topics</h3><div class="grid g3"><div><h4>Animal care</h4><p ~
class="small muted">Daily care coverage, veterinary coordination, emergency response, feed and bedding, and safe horse ~
handling.</p></div><div><h4>Property operations</h4><p class="small muted">Equipment maintenance, pasture rotation, manure handling, water systems, access~
 roads, and seasonal workloads.</p></div><div><h4>Public-facing activity</h4><p class="small muted">Event staffing, traffic coordination, food-service ~
oversight, security, and customer-facing hours.</p></div></div></div>`;}
function infrastructureTaskGroup(group){return taskSeed.filter(t=>t[3]===group)}
function compliancePage(){const groups=[...new ~
Set(taskSeed.map(t=>t[3]))],total=taskSeed.length,done=taskSeed.filter(t=>state.tasks[t[0]]?.done).length;return pageHead("DUE DILIGENCE","Compliance & ~
Due Diligence","Planning prompts only. Mark items to track progress; completion does not establish compliance or approval.",`<span ~
class="pill">${done}/${total} reviewed</span>${btn("Reset checklist","reset-tasks","ghost")}`)+`<div class="notice blueish" style="margin-~
bottom:14px">Confirm current requirements with county planning, state agencies, and qualified professionals. Regulations and property-specific conditions ~
may change.</div><div class="tabbar">${["All",...groups].map(g=>`<button data-compliance-filter="${g}" ~
class="${complianceFilter===g?"active":""}">${g}</button>`).join("")}</div><div class="grid ~
```

```
g2">${groups.filter(g=>complianceFilter==="All"||complianceFilter===g).map(g=>`<div ~
class="card"><h3>${g}</h3>${infrastructureTaskGroup(g).map(([name,desc,priority])=>`<label class="check-row"><input type="checkbox" data-~
task="${esc(name)}" ${state.tasks[name]?.done?"checked":""}><span class="task-text">${esc(name)}<small>${esc(desc)}</small></span><span class="pill ~
${priority==="High"?"rose":"warn"}">${priority}</span></label>`).join("")}</div>`).join("")}</div>${disclaimerShort()}`;}
function scenariosPage(){const current=scenario(),c=compute();return pageHead("ALTERNATIVES","Scenarios","Create independent planning alternatives and ~
compare their current site and operating assumptions.",btn("New scenario","new-scenario","primary"))+`<div class="grid ~
g2">${state.scenarios.map(sc=>{const keep=state.activeScenario;state.activeScenario=sc.id;const x=compute();state.activeScenario=keep;return`<div ~
class="card"><div class="between"><div><h3>${esc(sc.name)}</h3><span class="small muted">${sc.items.length} site objects</span></div><span ~
class="pill">${sc.id===state.activeScenario?"Active":"Alternative"}</span></div><div class="key-value"><span>Capital ~
estimate</span><strong>${fmt(x.capex)}</strong></div><div class="key-value"><span>Annual revenue</span><strong>${fmt(x.gross)}</strong></div><div ~
class="key-value"><span>Annual NOI</span><strong class="${x.noi>=0?"green":"rose"}">${fmt(x.noi)}</strong></div><div class="row" style="margin-~
top:12px">${sc.id===state.activeScenario?`<button class="btn" data-action="rename-scenario" data-id="${sc.id}">Rename</button>`:`<button class="btn ~
primary" data-action="switch-scenario" data-id="${sc.id}">Open scenario</button>`}<button class="btn" data-action="clone-scenario" data-~
id="${sc.id}">Duplicate</button>${sc.id!==state.activeScenario?`<button class="btn danger" data-action="delete-scenario" data-~
id="${sc.id}">Delete</button>`:""}</div></div>`}).join("")}</div><div class="card" style="margin-top:14px"><h3>Side-by-side operating comparison</h3><div ~
class="table-wrap"><table><thead><tr><th>Scenario</th><th>Placed objects</th><th>Capital</th><th>Gross / yr</th><th>OpEx / yr</th><th>NOI / ~
yr</th></tr></thead><tbody>${state.scenarios.map(sc=>{const keep=state.activeScenario;state.activeScenario=sc.id;const x=compute();state.activeScenario=ke~
ep;return`<tr><td>${esc(sc.name)}</td><td>${sc.items.length}</td><td>${fmt(x.capex)}</td><td>${fmt(x.gross)}</td><td>${fmt(x.opex)}</td><td ~
class="${x.noi>=0?"green":"rose"}">${fmt(x.noi)}</td></tr>`}).join("")}</tbody></table></div><p class="tiny muted">Scenario comparison is based on each ~
scenario's current inputs; it is not a forecast or investment recommendation.</p></div>`;}
function reportsPage(){const c=compute(),sc=scenario();const taskRows=taskSeed.map(t=>`<li>${state.tasks[t[0]]?.done?"■":"■"} ${esc(t[0])} — ~
${state.tasks[t[0]]?.done?"marked complete":"open"} (${esc(t[2])} priority)</li>`).join("");return pageHead("PROJECT OUTPUT","Reports","A print-ready ~
planning summary and portable data export.",`${btn("Print report","print","primary")}${btn("Download JSON","export")}${btn("Export budget ~
CSV","csv")}`)+`<div class="card report-copy"><div class="eyebrow">ILLUSTRATIVE PLANNING ~
SUMMARY</div><h2>${esc(state.project.name)}</h2><p>${esc(state.project.county||"County not specified")}${state.project.community?", ~
"+esc(state.project.community):""} · ${state.project.acres} acres · Scenario: ${esc(sc.name)}</p><div class="grid g3">${kpi("Conceptual ~
CapEx",fmt(c.capex),`${sc.finance.contingency}% contingency`,"amber")}${kpi("Gross annual revenue",fmt(c.gross),"Illustrative","green")}${kpi("Annual ~
NOI",fmt(c.noi),"Before financing / tax",c.noi>=0?"green":"rose")}</div><h3 style="margin-top:20px">Land allocation</h3><p>${c.used.toFixed(2)} acres of ~
conceptual structures/hardscape; ${c.past.toFixed(2)} acres pasture; ${c.open.toFixed(2)} acres open/unassigned.</p><h3>Site object ~
schedule</h3><ul>${items().map(i=>`<li>${esc(i.name||templates[i.type].name)} — ${Math.round(i.w)} × ${Math.round(i.h)} ft; ${acresFor(i).toFixed(3)} ~
acres; phase ${esc(i.phase)}; status: ${esc(i.status)}</li>`).join("")||"<li>No site objects placed.</li>"}</ul><h3>Due-diligence ~
tracker</h3><ul>${taskRows}</ul><p class="banner">${fullDisclaimer()}</p><p class="tiny muted">Generated ${new Date().toLocaleString()} · Values are ~
planning assumptions from this browser's saved project.</p></div>`;}
function manualPage(){const records=manualSeed.filter(r=>(manualFilter==="All"||r[0]===manualFilter)&&r.join(" ~
").toLowerCase().includes(manualQuery.toLowerCase()));return pageHead("REFERENCE","Estate Planning Manual","Search practical planning prompts and ~
conceptual design notes. Verify applicable requirements and assumptions independently.",`<input id="manualQuery" class="input manual-search" ~
placeholder="Search topics…" value="${esc(manualQuery)}" aria-label="Search manual">`)+`<div class="tabbar">${["All",...new ~
Set(manualSeed.map(m=>m[0]))].map(g=>`<button data-manual-filter="${g}" class="${manualFilter===g?"active":""}">${g}</button>`).join("")}</div><div ~
class="grid g2">${records.map(([g,t,d])=>`<article class="card"><div class="eyebrow">${esc(g)}</div><h3 style="margin:5px 0">${esc(t)}</h3><p class="small~
 muted">${esc(d)}</p></article>`).join("")||`<div class="empty">No manual entries match your search.</div>`}</div>`;}
function settingsPage(){const p=state.project;return pageHead("PROJECT CONFIGURATION","Project Settings","Set the planning context, adjust property ~
assumptions, and manage your local project data.",`${btn("Export backup","export")}${btn("Import backup","import")}`)+
`<div class="grid g2"><div class="card"><h3>Project profile</h3><div class="stack"><div class="field"><label>Project name</label><input data-~
project="name" value="${esc(p.name)}"></div><div class="inline-fields"><div class="field"><label>County</label><input data-project="county" ~
value="${esc(p.county)}" placeholder="Optional"></div><div class="field"><label>Nearest community</label><input data-project="community" ~
value="${esc(p.community)}" placeholder="Optional"></div></div><div class="inline-fields"><div class="field"><label>Total acreage (1,000 is the ~
largest)</label><select data-project="acres">${[...new ~
Set([5,10,20,40,80,160,320,640,1000,Math.min(1000,Math.round(num(p.acres,80)))])].sort((a,b)=>a-b).map(a=>`<option value="${a}" ~
${Math.round(num(p.acres,80))===a?"selected":""}>${a.toLocaleString()} acres</option>`).join("")}</select></div><div class="field"><label>Property aspect ~
ratio (width ÷ depth)</label><input type="number" min=".25" max="4" step=".05" data-project="aspect" value="${p.aspect}"></div></div><div ~
class="field"><label>Primary goal</label><input data-project="goal" value="${esc(p.goal)}"></div></div><p class="tiny muted">The quick rectangular ~
property boundary is calculated from acreage and aspect ratio. It does not represent a surveyed boundary.</p></div><div class="card"><h3>Local data & ~
portability</h3><p class="small muted">Autosave stores project data in this browser's local storage. No account or backend is used; browser data is not ~
synced to another device.</p><div class="stack">${btn("Download project JSON","export","primary")}${btn("Import project JSON","import")}${btn("Restore ~
recovery snapshot","restore-recovery")}${btn("Restore starter project","restore-starter")}${btn("Delete local project data","delete-~
data","danger")}</div><div class="notice" style="margin-top:12px">Keep a regular JSON backup. Clearing browser storage or changing browsers may remove ~
unsaved local data.</div><p class="tiny muted">Schema version ${state.schemaVersion} · Last saved ${state.updatedAt?new ~
Date(state.updatedAt).toLocaleString():"not saved yet"}</p></div></div><div class="card" style="margin-top:14px"><h3>Appearance</h3><div ~
class="field"><label for="themePick">Color theme</label><select id="themePick" class="theme-pick" aria-label="Color theme"><option value="cad" ~
${selectedTheme==="cad"?"selected":""}>CAD</option><option value="forest" ${selectedTheme==="forest"?"selected":""}>Forest</option><option value="ranch" ~
${selectedTheme==="ranch"?"selected":""}>Ranch</option><option value="day" ${selectedTheme==="day"?"selected":""}>Daylight</option></select></div><p ~
class="tiny muted">Your theme choice is saved in this browser.</p></div><div class="card" style="margin-top:14px"><h3>Planning disclaimer</h3><p ~
class="small">${fullDisclaimer()}</p></div><div class="card" style="margin-top:14px"><h3>Assumptions & known limits</h3><ul class="small muted"><li>The ~
map is a rectangular conceptual planning grid, not a survey or GIS boundary.</li><li>Unit costs, revenue, operating costs, staffing ratios, and due-~
diligence prompts are illustrative and user-editable where provided.</li><li>The model does not establish legal permissibility, zoning compliance, ~
permits, engineering feasibility, loan eligibility, or tax treatment.</li><li>Scenarios and project data remain on this browser unless you export a ~
backup.</li></ul></div>`;}
function fullDisclaimer(){return"This application provides illustrative planning estimates, conceptual layouts, and organizational prompts only. It is not~
 legal, architectural, engineering, environmental, land-use, zoning, building-code, fire-code, employment, tax, lending, construction, insurance, ~
accessibility, or financial advice. Property-specific conditions, costs, designs, approvals, and requirements must be verified with the appropriate county~
 departments, state agencies, licensed professionals, contractors, lenders, insurers, and legal counsel."}
function disclaimerShort(){return`<div class="banner">${fullDisclaimer()}</div>`}
function exportProject(){const data=JSON.stringify({...state,updatedAt:new ~
Date().toISOString()},null,2),a=document.createElement("a");a.href=URL.createObjectURL(new ~
Blob([data],{type:"application/json"}));a.download=(state.project.name||"oregon-estate-project").toLowerCase().replace(/[^a-z0-9]+/g,"-")+"-"+new ~
Date().toISOString().slice(0,16).replace("T","-").replace(":","")+".json";a.click();setTimeout(()=>URL.revokeObjectURL(a.href),1000);}
function linearLength(i){if(i.type!=="curvedroad")return Math.max(i.w,i.h);const inset=Math.min(i.w,i.h)*.32,span=i.w-~
2*inset,pts=[],p0=[inset,i.h*.65],p1=[inset+span*.3,i.h*.65],p2=[inset+span*.7,i.h*.35],p3=[i.w-inset,i.h*.35];for(let n=0;n<=40;n++){const ~
t=n/40,u=1-t;pts.push([u**3*p0[0]+3*u*u*t*p1[0]+3*u*t*t*p2[0]+t**3*p3[0],u**3*p0[1]+3*u*u*t*p1[1]+3*u*t*t*p2[1]+t**3*p3[1]])}return ~
pts.slice(1).reduce((sum,p,n)=>sum+Math.hypot(p[0]-pts[n][0],p[1]-pts[n][1]),0)}
function exportCsv(){const rows=[["Object","Type","Width ft","Length ft","Rotation deg","Footprint sq ft","Acres","Linear length ft","Phase","Planning ~
status","Estimate USD"],...items().map(i=>[i.name||templates[i.type].name,i.type,i.w,i.h,i.rotation,Math.round(areaSqft(i)),acresFor(i).toFixed(4),templat~
es[i.type].method==="linear"?linearLength(i).toFixed(1):"",i.phase,i.status,estimateItem(i)])];const ~
csv=rows.map(r=>r.map(x=>`"${String(x).replace(/"/g,'""')}"`).join(",")).join("\r\n");const a=document.createElement("a");a.href=URL.createObjectURL(new ~
Blob([csv],{type:"text/csv"}));a.download="estate-site-budget.csv";a.click();setTimeout(()=>URL.revokeObjectURL(a.href),1000);}
function estimateItem(i){const m=i.costMethod||templates[i.type].method;return ~
Math.round(m==="linear"?linearLength(i)*i.unitCost:m==="acre"?acresFor(i)*i.unitCost:m==="area"||m==="sqft"?areaSqft(i)*i.unitCost:i.unitCost)}
function placeObject(id,x,y){const t=templates[id];if(!t)return;const ~
width=t.w,height=t.h,candidate={...makeItem(id,x,y),shape:t.shape||"rect"};if(state.project.parcelBoundary&&objectOutsideParcel(candidate)){alert("Place ~
the whole object inside the imported parcel outline. This planning check is approximate; confirm siting with a survey or qualified ~
professional.");return}const overlap=items().some(o=>x<o.x+o.w&&x+width>o.x&&y<o.y+o.h&&y+height>o.y);if(overlap){if(!confirm("This placement overlaps ~
another object. Place it anyway for planning?"))return}snapshot();items().push(candidate);selectedId=candidate.id;tool="select";save();render();}
function downloadBackup(){exportProject();}
function confirmAction(message){return window.confirm(message)}
function moveSelected(dx,dy){const it=items().find(i=>i.id===selectedId);if(!it||it.locked)return;const ~
next={...it,x:it.x+dx,y:it.y+dy};if(objectOutsideParcel(next)){alert("That move would place the object outside the imported parcel ~
outline.");return}snapshot();it.x=next.x;it.y=next.y;save();render();}
function duplicateSelected(){const it=items().find(i=>i.id===selectedId);if(!it)return;const ~
copy={...it,id:uid(),x:it.x+CELL,y:it.y+CELL};if(objectOutsideParcel(copy)){alert("There is not enough room to duplicate this object inside the imported ~
parcel outline.");return}snapshot();items().push(copy);selectedId=copy.id;save();render();}
function action(name,id){switch(name){
case"page:about":case"page:dashboard":case"page:designer":case"page:buildings":case"page:homes":case"page:infrastructure":case"page:financials":case"page:~
operations":case"page:compliance":case"page:scenarios":case"page:reports":case"page:manual":case"page:settings":{const ~
nextPage=name.slice(5);if(nextPage==="designer"&&activePage!=="designer"){mapZoom=1;mapPan={x:0,y:0}}activePage=nextPage;render();break;}
```

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case"print":window.print();break;case"undo":undo();break;case"redo":redo();break;case"export":exportProject();break;case"csv":exportCsv();break;case"impor~
t":$("importFile").click();break;case"import-parcel":$("parcelFile").click();break;case"toggle-cut":sceneCut=!sceneCut;render();break;case"toggle-~
auto":sceneAuto=!sceneAuto;render();break;case"clear":if(confirmAction("Clear every object from this scenario? This can be undone immediately, but export ~
a backup first if needed.")){snapshot();scenario().items=[];selectedId=null;save();render()}break;
case"delete-selected":if(confirmAction("Delete this site object?")){snapshot();scenario().items=items().filter(i=>i.id!==selectedId);selectedId=null;save(~
);render()}break;case"duplicate":duplicateSelected();break;case"lock":{const ~
x=items().find(i=>i.id===selectedId);if(x)x.locked=!x.locked;save();render();break}
case"toggle-map-view":mapView=mapView==="flat"?"states":"flat";tool="select";render();break;case"terrain-~
back":mapView="states";render();break;case"terrain-layer":terrainLayer=terrainLayer==="topo"?"imagery":"topo";setTerrainLayer();updateTerrainControls();br~
eak;case"terrain-3d":terrain3d=!terrain3d;if(terrainMapInstance?.isStyleLoaded()){terrainMapInstance.setTerrain(terrain3d?{source:"elevation-~
dem",exaggeration:1.15}:null);terrainMapInstance.easeTo({pitch:terrain3d?55:0,duration:500})}updateTerrainControls();break;case"finish-map-~
feature":finishTerrainFeature();break;case"undo-map-vertex":undoTerrainVertex();break;case"cancel-map-feature":cancelTerrainDraft();break;case"delete-map-~
feature":deleteTerrainFeature(id);break;case"use-map-location":if(state.project.locationCoordinates?.state!==selectedStateMap){alert("Click the map to ~
choose a location first.");return}mapView="flat";tool="select";render();break;case"zoom-in":mapZoom=Math.min(4,mapZoom*1.2);render();break;case"zoom-~
out":mapZoom=Math.max(.5,mapZoom/1.2);render();break;case"zoom-fit":mapZoom=1;mapPan={x:0,y:0};render();break;
case"toggle-grid":state.layers.Grid=state.layers.Grid===false;save();render();break;case"toggle-~
labels":state.layers.Labels=state.layers.Labels===false;save();render();break;case"toggle-snap":snapEnabled=!snapEnabled;render();break;
case"clear-parcel":if(confirmAction("Remove the imported parcel outline? The current acreage will remain as the rectangular planning ~
area.")){state.project.parcelBoundary=null;save();render()}break;
case"new-scenario":{const name=prompt("Name this planning alternative:","Alternative "+(state.scenarios.length));if(!name?.trim())return;const sc=JSON.par~
se(JSON.stringify(scenario()));sc.id=uid();sc.name=name.trim();state.scenarios.push(sc);state.activeScenario=sc.id;undoStack=[];save();render();break}
case"switch-scenario":state.activeScenario=id;selectedId=null;undoStack=[];redoStack=[];save();render();break;
case"clone-scenario":{const src=state.scenarios.find(s=>s.id===id);if(!src)return;const cp=JSON.parse(JSON.stringify(src));cp.id=uid();cp.name=src.name+" ~
copy";state.scenarios.push(cp);save();render();break}
case"rename-scenario":{const sc=state.scenarios.find(s=>s.id===id||s.id===state.activeScenario);const n=prompt("Scenario ~
name:",sc?.name);if(n?.trim()&&sc){sc.name=n.trim();save();render()}break}
case"delete-scenario":if(state.scenarios.length<2){alert("At least one scenario is required.");return}if(confirmAction("Delete this scenario and its ~
layout? This cannot be undone.")){state.scenarios=state.scenarios.filter(s=>s.id!==id);if(state.activeScenario===id)state.activeScenario=state.scenarios[0~
].id;save();render()}break;
case"reset-tasks":if(confirmAction("Reset every due-diligence checklist item?")){state.tasks={};save();render()}break;
case"restore-starter":if(confirmAction("Replace this project with a fresh starter concept? Export a backup first if you need the current ~
data.")){state=defaultState();undoStack=[];redoStack=[];selectedId=null;save();render()}break;
case"restore-recovery":{const saved=localStorage.getItem(RECOVERY);if(!saved){alert("No recovery snapshot is available in this ~
browser.");return}if(confirmAction("Replace the current project with the previous autosave ~
snapshot?")){try{state=normalizeState(JSON.parse(saved));selectedId=null;undoStack=[];redoStack=[];save();render()}catch(e){alert("Recovery snapshot could~
 not be restored: "+e.message)}}break}
case"delete-data":if(confirmAction("Delete this browser's saved project data and load a fresh starter ~
concept?")){localStorage.removeItem(STORAGE);localStorage.removeItem(RECOVERY);state=defaultState();save();render()}break;
}}
function zoomViewBox(svg,factor){const [x,y,w,h]=svg.getAttribute("viewBox").split(" ").map(Number),nw=w*factor,nh=h*factor;return`${x+(w-nw)/2} ~
${y+(h-nh)/2} ${nw} ${nh}`;}
function updateObjectField(key,value){const it=items().find(i=>i.id===selectedId);if(!it||it.locked)return;let v=value;if(["w","h","height","unitCost","ro~
tation"].includes(key))v=num(value,it[key]);if(key==="w"||key==="h")v=Math.max(1,Math.min(10000,v));if(key==="height")v=Math.max(1,Math.min(300,v));if(key~
==="unitCost")v=Math.max(0,v);if(key==="rotation")v=((v%360)+360)%360;if(key==="shape"&&!["rect","roundrect","circle","triangle","poly"].includes(v))retur~
n;if(key==="name")v=String(v).slice(0,100);const next={...it,[key]:v};if(state.project.parcelBoundary&&objectOutsideParcel(next)){alert("That footprint ~
would extend outside the imported parcel outline. Measurements are based on the entered dimensions; confirm siting against the source ~
survey.");render();return}snapshot();it[key]=v;save();if(["w","h","shape","name","rotation"].includes(key))render();}
function safeImport(file){const reader=new FileReader();reader.onload=()=>{try{const next=normalizeState(JSON.parse(String(reader.result)));const ~
fmt=d=>d?new Date(d).toLocaleString():"unknown",older=state.updatedAt&&next.updatedAt&&new Date(next.updatedAt)<new ~
Date(state.updatedAt);if(!confirmAction((older?"WARNING: this file is OLDER than your current project.\n\n":"")+"Replace your current project with the ~
imported one?\n\nYours last saved: "+fmt(state.updatedAt)+"\nImported file saved: "+fmt(next.updatedAt)+"\n\nYour current project is kept as a recovery ~
copy, but export it first if unsure."))return;try{localStorage.setItem(RECOVERY,JSON.stringify(state))}catch(e){}state=next;undoStack=[];redoStack=[];sele~
ctedId=null;save();render()}catch(e){alert("Import failed: "+e.message)}};reader.onerror=()=>alert("Could not read the selected ~
file.");reader.readAsText(file);}
function mapLocal(svg,event){const g=$("mapObjects");if(!g)return{x:0,y:0};const p=svg.createSVGPoint();p.x=event.clientX;p.y=event.clientY;const ~
q=p.matrixTransform(g.getScreenCTM().inverse());return{x:q.x,y:q.y};}
function zoomMap(factor,clientX,clientY){const svg=$("siteMap");if(!svg||mapView!=="flat")return;const rect=svg.getBoundingClientRect(),vb=svg.viewBox.bas~
eVal,cx=clientX??rect.left+rect.width/2,cy=clientY??rect.top+rect.height/2,anchor=mapLocal(svg,{clientX:cx,clientY:cy}),nextZoom=Math.max(.15,Math.min(8,m~
apZoom*factor));if(nextZoom===mapZoom)return;const bounds=siteMapBounds(),px=(anchor.x-vb.x)/vb.width,py=(anchor.y-~
vb.y)/vb.height,nextW=bounds.w*mapZoom/nextZoom,nextH=bounds.h*mapZoom/nextZoom,nextX=anchor.x-px*nextW,nextY=anchor.y-~
py*nextH,{w,h}=propertyDims();mapZoom=nextZoom;mapPan={x:nextX-(w/2-nextW/2),y:nextY-(h/2-nextH/2)};render();}
let spacePan=false;
function wireEvents(){
document.addEventListener("toggle",e=>{const detail=e.target.closest?.("[data-detail]");if(detail)detailsOpen[detail.dataset.detail]=detail.open},true);
window.addEventListener("keydown",e=>{const ~
editing=/INPUT|TEXTAREA|SELECT/.test(document.activeElement?.tagName);if(e.code==="Space"&&activePage==="designer"&&!editing&&!e.target.closest?.("[data-o~
bject]")){spacePan=true;e.preventDefault()}},{capture:true});window.addEventListener("keyup",e=>{if(e.code==="Space")spacePan=false},{capture:true});windo~
w.addEventListener("blur",()=>{spacePan=false});
$("content").addEventListener("click",e=>{const button=e.target.closest('[data-action="zoom-in"],[data-action="zoom-~
out"]');if(!button)return;e.preventDefault();e.stopPropagation();zoomMap(button.dataset.action==="zoom-in"?1.2:1/1.2)},{capture:true});
$("content").addEventListener("contextmenu",e=>{if(e.target.closest("#mapViewport"))e.preventDefault()});
$("nav").addEventListener("click",e=>{const b=e.target.closest("[data-page]");if(b){const ~
nextPage=b.dataset.page;if(nextPage==="designer"&&activePage!=="designer"){mapZoom=1;mapPan={x:0,y:0}}activePage=nextPage;render()}});
$("content").addEventListener("click",e=>{
const osmButton=e.target.closest('[data-action="terrain-osm"]');if(osmButton){e.preventDefault();e.stopImmediatePropagation();toggleTerrainOsm();return}
const osmClear=e.target.closest('[data-action="terrain-osm-~
clear"]');if(osmClear){e.preventDefault();e.stopImmediatePropagation();clearTerrainOsm();return}
const stateHit=e.target.closest("[data-state-map]");if(stateHit){openStateTerrain(stateHit.dataset.stateMap);return}
const page=e.target.closest("[data-page]");if(page){const ~
nextPage=page.dataset.page;if(nextPage==="designer"&&activePage!=="designer"){mapZoom=1;mapPan={x:0,y:0}}activePage=nextPage;render();return}
const tb=e.target.closest("[data-tool]");if(tb){tool=tb.dataset.tool;render();return}
const paletteItem=e.target.closest("[data-palette-item]");if(paletteItem){tool=paletteItem.dataset.paletteItem;selectedId=null;render();return}
const paletteButton=e.target.closest("[data-palette-group]");if(paletteButton){paletteGroup=paletteButton.dataset.paletteGroup;render();return}
const filter=e.target.closest("[data-manual-filter]");if(filter){manualFilter=filter.dataset.manualFilter;render();return}
const cf=e.target.closest("[data-compliance-filter]");if(cf){complianceFilter=cf.dataset.complianceFilter;render();return}
const mapFeature=e.target.closest("[data-map-feature]");if(mapFeature){terrainSelectedFeatureId=mapFeature.dataset.mapFeature;const ~
feature=state.project.mapFeatures.find(f=>f.id===terrainSelectedFeatureId);if(feature&&terrainMapInstance){const ~
coords=feature.kind==="landmark"?[feature.coordinates]:feature.coordinates;const bounds=coords.reduce((box,p)=>box.extend(p),new ~
maplibregl.LngLatBounds(coords[0],coords[0]));if(feature.kind==="landmark")terrainMapInstance.flyTo({center:feature.coordinates,zoom:17});else ~
terrainMapInstance.fitBounds(bounds,{padding:70,maxZoom:18})}updateTerrainSources();return}
const sel=e.target.closest("[data-select-object]");if(sel){selectedId=sel.dataset.selectObject;tool="select";render();return}
const act=e.target.closest("[data-action]");if(act){action(act.dataset.action,act.dataset.id);return}
const obj=e.target.closest("[data-object]");if(obj){selectedId=obj.dataset.object;if(tool==="erase"){if(confirmAction("Delete this site ~
object?")){snapshot();scenario().items=items().filter(i=>i.id!==selectedId);selectedId=null;save();render()}return}if(tool==="rotate"){const ~
i=items().find(x=>x.id===selectedId);if(i&&!i.locked){snapshot();i.rotation=(num(i.rotation)+90)%360;save()}tool="select"}render()}
});
$("content").addEventListener("change",e=>{
const el=e.target;
if(el.id==="themePick"){applyTheme(el.value);return}
if(el.id==="terrainDrawMode"){terrainDrawMode=el.value;terrainDraft=[];const source=terrainMapInstance?.getSource("estate-~
draft");if(source)source.setData(terrainDraftCollection());updateTerrainControls();return}
if(el.matches("[data-osm-layer]")){terrainOsmVisibility[el.dataset.osmLayer]=el.checked;applyTerrainOsmVisibility();return}
if(el.matches("[data-project]")){const key=el.dataset.project;if(key==="acres"&&state.project.parcelBoundary){alert("Acreage is calculated from the ~
imported parcel outline. Remove or replace that boundary to change the parcel ~
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area.");render();return}if(key==="locationUrl"&&el.value&&!safeMapReference(el.value)){alert("Use a secure Google Maps or Zillow URL.");render();return}st~
ate.project[key]=key==="acres"?Math.max(1,Math.min(1000,num(el.value,40))):key==="aspect"?Math.max(.25,Math.min(4,num(el.value,1.5))):el.value;save();rend~
er();return}
if(el.matches("[data-finance]")){const key=el.dataset.finance;if(key==="linked"||key==="payrollOffset")scenario().finance[key]=el.checked;else ~
scenario().finance[key]=Math.max(0,num(el.value));save();render();return}
if(el.matches("[data-operation]")){state.operations[el.dataset.operation]=Math.max(1,num(el.value,1));save();render();return}
if(el.matches("[data-layer]")){state.layers[el.dataset.layer]=el.checked;save();render();return}
if(el.matches("[data-task]")){state.tasks[el.dataset.task]={...(state.tasks[el.dataset.task]||{}),done:el.checked};save();return}
if(el.matches("[data-infra]")){state.notes[el.dataset.infra]=el.value;save();return}
if(el.matches("[data-object-field]")){updateObjectField(el.dataset.objectField,el.value);return}
if(el.matches("[data-plan]")){const ~
it=items().find(x=>x.type===el.dataset.plan);if(it){snapshot();it[el.dataset.plan+"Count"]=Math.max(0,num(el.value));save()}return}
});
$("content").addEventListener("input",e=>{
const el=e.target;
if(el.matches("#manualQuery")){manualQuery=el.value;const pos=el.selectionStart;render();const ~
q=$("manualQuery");q.focus();q.setSelectionRange(pos,pos);return}
if(el.matches("#paletteSearch")){paletteQuery=el.value;const pos=el.selectionStart;render();const ~
q=$("paletteSearch");q.focus();q.setSelectionRange(pos,pos);return}
if(el.matches("[data-finance]")&&el.type==="range"){scenario().finance[el.dataset.finance]=num(el.value);const peers=$$("input[data-~
finance='"+el.dataset.finance+"']");peers.forEach(p=>{if(p!==el)p.value=el.value});save()}
if(el.matches("[data-operation]")){state.operations[el.dataset.operation]=Math.max(1,num(el.value,1));save()}
if(el.matches("[data-infra]")){state.notes[el.dataset.infra]=el.value;save()}
});
$("content").addEventListener("pointerdown",e=>{const obj=e.target.closest("[data-object]");if(!obj||tool==="erase"||tool==="pan")return;const it=items().~
find(i=>i.id===obj.dataset.object);if(!it||it.locked)return;selectedId=it.id;dragStart={kind:"object",id:it.id,point:mapLocal($("siteMap"),e),x:it.x,y:it.~
y};$("siteMap")?.setPointerCapture?.(e.pointerId);});
$("content").addEventListener("pointerdown",e=>{const ~
svg=e.target.closest("#siteMap");if(!svg||!(tool==="pan"||spacePan||e.button===1||e.button===2))return;e.preventDefault();const vb=svg.viewBox.baseVal,rec~
t=svg.getBoundingClientRect(),scale=Math.min(rect.width/vb.width,rect.height/vb.height);dragStart={kind:"pan",startX:e.clientX,startY:e.clientY,panX:mapPa~
n.x,panY:mapPan.y,viewX:vb.x,viewY:vb.y,viewW:vb.width,viewH:vb.height,scale};$("mapViewport")?.classList.add("is-~
panning");svg.setPointerCapture?.(e.pointerId);});
$("content").addEventListener("pointermove",e=>{if(dragStart?.kind!=="pan")return;const svg=$("siteMap");if(!svg)return;const dx=(e.clientX-~
dragStart.startX)/dragStart.scale,dy=(e.clientY-dragStart.startY)/dragStart.scale;mapPan={x:dragStart.panX-dx,y:dragStart.panY-~
dy};svg.setAttribute("viewBox",`${dragStart.viewX-dx} ${dragStart.viewY-dy} ${dragStart.viewW} ${dragStart.viewH}`);});
$("content").addEventListener("pointermove",e=>{if(dragStart?.kind!=="object")return;const svg=$("siteMap");if(!svg)return;const q=mapLocal(svg,e),dx=Math~
.round((q.x-dragStart.point.x)/gridSize)*gridSize,dy=Math.round((q.y-dragStart.point.y)/gridSize)*gridSize,src=svg.querySelector(`[data-~
object="${dragStart.id}"]`);if(!src)return;let ~
g=svg.querySelector("#dragGhost");if(!g){g=document.createElementNS("http://www.w3.org/2000/svg","g");g.id="dragGhost";g.setAttribute("pointer-events","no~
ne");g.setAttribute("opacity",".6");g.appendChild(src.cloneNode(true));src.parentNode.appendChild(g);src.setAttribute("opacity",".25")}g.setAttribute("tra~
nsform",`translate(${dx} ${dy})`);const ~
it=items().find(i=>i.id===dragStart.id),bad=it&&objectOutsideParcel({...it,x:dragStart.x+dx,y:dragStart.y+dy});g.style.filter=bad?"drop-shadow(0 0 6px ~
#ff3b3b) sepia(1) saturate(6) hue-rotate(-30deg)":"drop-shadow(0 0 5px #4ade80)"});
$("content").addEventListener("pointerup",e=>{if(!dragStart)return;if(dragStart.kind==="pan"){dragStart=null;render();return}const svg=$("siteMap"),q=svg?~
mapLocal(svg,e):dragStart.point,dx=Math.round((q.x-dragStart.point.x)/gridSize)*gridSize,dy=Math.round((q.y-~
dragStart.point.y)/gridSize)*gridSize;if(dx||dy){const it=items().find(i=>i.id===dragStart.id);if(it){const ~
next={...it,x:dragStart.x+dx,y:dragStart.y+dy};if(objectOutsideParcel(next))alert("That move would place the object outside the imported parcel ~
outline.");else{snapshot();it.x=next.x;it.y=next.y;save()}}}render();dragStart=null;});
$("content").addEventListener("keydown",e=>{const stateHit=e.target.closest("[data-state-map]");if(stateHit&&(e.key==="Enter"||e.key===" ~
")){e.preventDefault();openStateTerrain(stateHit.dataset.stateMap);return}const obj=e.target.closest("[data-object]");if(obj&&(e.key==="Enter"||e.key===" ~
")){e.preventDefault();selectedId=obj.dataset.object;render();return}if(e.target.id==="siteMap"||e.target.closest("#siteMap")){const step=e.shiftKey?200:g~
ridSize;if(e.key==="ArrowLeft"){e.preventDefault();moveSelected(-~
step,0)}if(e.key==="ArrowRight"){e.preventDefault();moveSelected(step,0)}if(e.key==="ArrowUp"){e.preventDefault();moveSelected(0,-~
step)}if(e.key==="ArrowDown"){e.preventDefault();moveSelected(0,step)}if(e.key==="Delete"||e.key==="Backspace"){e.preventDefault();action("delete-~
selected")}}});
$("content").addEventListener("pointerdown",e=>{const svg=e.target.closest("#siteMap");if(!svg||e.target.closest("[data-~
object]")||tool==="select"||tool==="erase"||tool==="rotate"||tool==="pan"||templates[tool]||spacePan||e.button!==0)return;const ~
p=mapLocal(svg,e),x=snapEnabled?Math.round(p.x/gridSize)*gridSize:p.x,y=snapEnabled?Math.round(p.y/gridSize)*gridSize:p.y;placeObject(tool,x,y);});
$("content").addEventListener("pointerdown",e=>{const canvas=e.target.closest("#scene3d");if(!canvas)return;const pan=e.shiftKey||e.button===1||e.button==~
=2;if(e.button!==0&&!pan)return;e.preventDefault();sceneDrag={pointerId:e.pointerId,mode:pan?"pan":"orbit",x:e.clientX,y:e.clientY,angle:sceneAngle,panX:s~
cenePan.x,panY:scenePan.y};canvas.setPointerCapture?.(e.pointerId);canvas.classList.toggle("is-panning",pan);});
$("content").addEventListener("pointermove",e=>{if(!sceneDrag||sceneDrag.pointerId!==e.pointerId||!e.target.closest("#scene3d"))return;if(sceneDrag.mode==~
="pan")scenePan={x:sceneDrag.panX+e.clientX-sceneDrag.x,y:sceneDrag.panY+e.clientY-sceneDrag.y};else sceneAngle=sceneDrag.angle+(e.clientX-~
sceneDrag.x)*.012;draw3d()});
const endSceneDrag=e=>{const ~
canvas=e.target.closest?.("#scene3d");if(sceneDrag&&(!e.pointerId||sceneDrag.pointerId===e.pointerId)){sceneDrag=null;canvas?.classList.remove("is-~
panning")}};
$("content").addEventListener("pointerup",endSceneDrag);$("content").addEventListener("pointercancel",endSceneDrag);
$("content").addEventListener("contextmenu",e=>{if(e.target.closest("#scene3d"))e.preventDefault()});
$("content").addEventListener("click",e=>{if(!e.target.closest("[data-scene-reset]"))return;sceneAngle=.65;sceneZoom=1;scenePan={x:0,y:0};draw3d()});
document.addEventListener("wheel",e=>{if(!e.target.closest("#siteMap"))return;e.preventDefault();zoomMap(e.deltaY<0?1.12:1/1.12,e.clientX,e.clientY)},{pas~
sive:false});
document.addEventListener("wheel",e=>{if(!e.target.closest("#scene3d"))return;e.preventDefault();const r=e.target.closest("#scene3d").getBoundingClientRec~
t(),nz=Math.max(.65,Math.min(80,sceneZoom*(e.deltaY<0?1.15:1/1.15))),k=nz/sceneZoom,ox=e.clientX-r.left-r.width/2,oy=e.clientY-r.top-~
r.height*.56;scenePan={x:ox-(ox-scenePan.x)*k,y:oy-(oy-scenePan.y)*k};sceneZoom=nz;draw3d()},{passive:false});
$("content").addEventListener("change",e=>{if(e.target.id==="placeType"){tool=e.target.value||"select";render();return}if(e.target.id==="gridSelect"){grid~
Size=Math.max(1,num(e.target.value,40));render()}});
$("importFile").addEventListener("change",e=>{if(e.target.files?.[0])safeImport(e.target.files[0]);e.target.value="";});
$("parcelFile").addEventListener("change",e=>{if(e.target.files?.[0])importParcelFile(e.target.files[0]);e.target.value="";});
document.addEventListener("keydown",e=>{const editing=/INPUT|TEXTAREA|SELECT/.test(document.activeElement?.tagName);if(activePage==="designer"&&mapView===~
"terrain"&&!editing){if(e.key==="Escape"&&terrainDraft.length){e.preventDefault();cancelTerrainDraft();return}if(e.key==="Enter"&&terrainDraft.length>=3){~
e.preventDefault();finishTerrainFeature();return}}if((e.ctrlKey||e.metaKey)&&e.key.toLowerCase()==="z"&&!editing){e.preventDefault();e.shiftKey?redo():und~
o()}if((e.ctrlKey||e.metaKey)&&e.key.toLowerCase()==="y"&&!editing){e.preventDefault();redo()}if(e.key.toLowerCase()==="r"&&!editing&&activePage==="design~
er"){const i=items().find(x=>x.id===selectedId);if(i&&!i.locked){snapshot();i.rotation=(num(i.rotation)+90)%360;save();render()}}});
}
const renderBase=render;render=function(){disposeTerrainMap();renderBase();if(activePage!=="designer")return;const ~
viewport=$("mapViewport"),toggle=document.querySelector('[data-action="toggle-map-view"]'),hint=document.querySelector(".map-meta span:nth-of-~
type(2)");if(mapView==="flat"){const panButton=document.querySelector('[data-tool="pan"]'),fitButton=document.querySelector('[data-action="zoom-~
fit"]');if(panButton){panButton.textContent="✥ Pan · Space + drag";panButton.title="Pan by dragging, or hold Space and drag anywhere on the ~
map"}if(fitButton){fitButton.textContent="Fit plan";fitButton.title="Center and fit the property planning area"}if(hint)hint.textContent="Hold Space + ~
drag or use Pan · scroll to zoom";if(viewport){const hud=document.createElement("div");hud.className="map-hud";hud.innerHTML=`<span>1 map unit = 1 ~
ft</span><span>Grid: ${gridSize} ft</span><span>Zoom: ${Math.round(mapZoom*100)}%</span><span>Plan: ${state.project.parcelBoundary?"Imported ~
boundary":"300-acre planning extent"}</span>`;viewport.append(hud)}return}const ~
plan=$("siteMap");if(plan)plan.style.display="none";if(mapView==="states"){if(toggle){toggle.textContent="Flat plan";toggle.dataset.action="toggle-map-~
view"}viewport?.insertAdjacentHTML("beforeend",drawGlobe());if(hint)hint.textContent="Select a state to open its 3D terrain and add measured map ~
overlays";return}if(toggle){toggle.textContent="← U.S. states";toggle.dataset.action="terrain-back"}if(hint)hint.textContent="Draw georeferenced map ~
overlays or select a location";const modes=[["navigate","Navigate / location"],["house","House footprint"],["building","Other building"],["land","Land ~
area"],["landmark","Landmark"],["clearing","Proposed clearing"]];document.querySelector(".map-toolbar")?.insertAdjacentHTML("beforeend",`<button ~
class="toolbtn" data-action="terrain-layer">${terrainLayer==="topo"?"Switch to aerial":"Switch to topo"}</button><button class="toolbtn" data-~
action="terrain-3d">${terrain3d?"3D relief on":"3D relief off"}</button><label class="toolbtn">Draw <select id="terrainDrawMode" aria-label="Map drawing ~
tool">${modes.map(([id,label])=>`<option value="${id}" ${terrainDrawMode===id?"selected":""}>${label}</option>`).join("")}</select></label><input ~
class="input" id="terrainFeatureName" style="width:150px;padding:6px 8px" aria-label="Overlay name" placeholder="Overlay name"><button class="toolbtn" ~
data-action="undo-map-vertex" disabled>Undo point</button><button class="toolbtn" data-action="finish-map-feature" disabled>Finish shape</button><button ~
class="toolbtn" data-action="cancel-map-feature" disabled>Cancel draw</button><button class="toolbtn" data-action="use-map-location" ~
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${state.project.locationCoordinates?.state===selectedStateMap?"":"disabled"}>Use location in ~
plan</button>`);viewport?.insertAdjacentHTML("beforeend",drawTerrainMap());renderTerrainFeaturePanel();mountTerrainMap();updateTerrainControls();};
const renderWithOsmControl=render;
render=function(){renderWithOsmControl();if(activePage!=="designer"||mapView!=="terrain")return;const toolbar=document.querySelector(".map-~
toolbar");if(toolbar&&!toolbar.querySelector('[data-action="terrain-osm"]'))toolbar.insertAdjacentHTML("beforeend",'<button class="toolbtn" data-~
action="terrain-osm" aria-pressed="false" title="Sends the visible map area to the public OpenStreetMap Overpass API when enabled">Mapped features ~
off</button><details class="terrain-reference-options"><summary>Reference layers</summary><div class="terrain-reference-checks"><label><input ~
type="checkbox" data-osm-layer="homes" checked> Nearby homes</label><label><input type="checkbox" data-osm-layer="buildings" checked> Other ~
buildings</label><label><input type="checkbox" data-osm-layer="fields" checked> Fields and orchards</label><label><input type="checkbox" data-osm-~
layer="woods" checked> Woods and brush</label><label><input type="checkbox" data-osm-layer="roads" checked> Roads</label><label><input type="checkbox" ~
data-osm-layer="trees" checked> Mapped trees</label><button class="toolbtn" data-action="terrain-osm-clear" type="button">Clear reference ~
context</button><small>Hides temporary OpenStreetMap references only; it does not edit imagery, saved parcel boundaries, or your drawn ~
overlays.</small></div></details><span class="terrain-guidance">Import your parcel boundary for your land. “Proposed clearing” is an annotation only; it ~
does not change real land cover.</span>');if(terrainMapInstance)wireTerrainOsmMap(terrainMapInstance);updateTerrainControls();};
const renderWithDynamicPlanExtent=render;
render=function(){renderWithDynamicPlanExtent();if(activePage==="designer"&&mapView==="flat"){const ~
viewport=$("mapViewport"),aspect=viewport?.clientWidth/viewport?.clientHeight;if(Number.isFinite(aspect)&&aspect>0&&Math.abs(aspect-~
mapAspect)>.01){mapAspect=aspect;renderWithDynamicPlanExtent()}}const extent=document.querySelector(".map-hud span:last-~
child");if(extent&&!state.project.parcelBoundary)extent.textContent=`Plan: ${state.project.acres.toLocaleString()}-acre planning extent`;};
window.addEventListener("resize",()=>{if(activePage!=="designer"||mapView!=="flat")return;const ~
viewport=$("mapViewport"),aspect=viewport?.clientWidth/viewport?.clientHeight;if(Number.isFinite(aspect)&&aspect>0&&Math.abs(aspect-~
mapAspect)>.01){mapAspect=aspect;render()}});
/* Editable terrain: a vertex grid of elevations (ft) stored in state.project.terrain */
let terrainBrush="raise",terrainRadius=4,terrainStrength=2,terrainExag=8,terrainPaint=null,terrainUndo=[],terrainRedo=[],terrainRaf=0;
document.head.insertAdjacentHTML("beforeend","<style>.terrain-tool #mapObjects{pointer-events:none}.terrain-tool svg{cursor:crosshair}.terrain-panel ~
.row{flex-wrap:wrap;gap:6px}.terrain-panel label{display:block;font-size:12px;margin-top:8px}.terrain-panel input[type=range]{width:100%}</style>");
function terrainData(){const t=state.project.terrain;if(!t||!Number.isInteger(t.cols)||!Number.isInteger(t.rows)||t.cols<4||t.rows<4||t.cols>200||t.rows>2~
00||!Array.isArray(t.h)||t.h.length!==t.cols*t.rows||!t.h.every(Number.isFinite))return null;return t}
function terrainEnsure(){let t=terrainData();if(t)return t;const{w,h}=propertyDims(),cols=40,rows=Math.max(8,Math.min(120,Math.round(cols*h/w)));return ~
state.project.terrain={cols,rows,h:new Array(cols*rows).fill(0),show:true}}
function terrainElev(x,y){const t=terrainData();if(!t||t.show===false)return 0;const{w,h}=propertyDims(),gx=Math.max(0,Math.min(t.cols-1,x/w*(t.cols-~
1))),gy=Math.max(0,Math.min(t.rows-1,y/h*(t.rows-1))),i=Math.min(t.cols-2,Math.floor(gx)),j=Math.min(t.rows-2,Math.floor(gy)),fx=gx-i,fy=gy-~
j,v=(a,b)=>t.h[b*t.cols+a];return(v(i,j)*(1-fx)+v(i+1,j)*fx)*(1-fy)+(v(i,j+1)*(1-fx)+v(i+1,j+1)*fx)*fy}
function mixHex(a,b,t){const ~
p=s=>[1,3,5].map(k=>parseInt(s.substr(k,2),16)),x=p(a),y=p(b);return"#"+x.map((v,k)=>Math.round(v+(y[k]-v)*t).toString(16).padStart(2,"0")).join("")}
function terrainColor(v,m){const t=Math.max(-1,Math.min(1,v/m));return t>=0?mixHex("#1b3b28","#e4dbb4",t):mixHex("#1b3b28","#2a6a9a",-t)}
function terrainMax(t){let m=1;for(const v of t.h)m=Math.max(m,Math.abs(v));return m}
function drawTerrainMesh(ctx,project,w,h,sin,cos){const t=terrainData(),m=terrainMax(t),quads=[];for(let j=0;j<t.rows-1;j++)for(let ~
i=0;i<t.cols-1;i++){const xs=[i,i+1,i+1,i],ys=[j,j,j+1,j+1],pts=xs.map((x,k)=>[x/(t.cols-1)*w,ys[k]/(t.rows-~
1)*h,t.h[ys[k]*t.cols+x]]);quads.push({pts,avg:(pts[0][2]+pts[1][2]+pts[2][2]+pts[3][2])/4,slope:(pts[1][2]-pts[0][2]+pts[2][2]-~
pts[3][2])/2,depth:((pts[0][0]+pts[2][0])/2-w/2)*sin+((pts[0][1]+pts[2][1])/2-h/2)*cos})}quads.sort((a,b)=>a.depth-b.depth);for(const q of quads){const p=~
q.pts.map(a=>project(a[0],a[1],a[2]*terrainExag));ctx.beginPath();p.forEach((a,n)=>n?ctx.lineTo(a[0],a[1]):ctx.moveTo(a[0],a[1]));ctx.closePath();ctx.fill~
Style=ctx.strokeStyle=shade(terrainColor(q.avg,m),Math.max(-35,Math.min(35,Math.round(-q.slope*6))));ctx.lineWidth=.7;ctx.fill();ctx.stroke()}}
function terrainOverlaySvg(){const t=terrainData();if(!t||t.show===false)return'<g ~
id="terrainOverlay"></g>';const{w,h}=propertyDims(),cw=w/(t.cols-1),ch=h/(t.rows-1),m=terrainMax(t);let s="";for(let j=0;j<t.rows;j++)for(let ~
i=0;i<t.cols;i++){const v=t.h[j*t.cols+i];if(Math.abs(v)<.25)continue;s+=`<rect x="${(i*cw-cw/2).toFixed(1)}" y="${(j*ch-ch/2).toFixed(1)}" ~
width="${cw.toFixed(1)}" height="${ch.toFixed(1)}" fill="${terrainColor(v,m)}"/>`}return`<g id="terrainOverlay" opacity=".6" pointer-events="none" ~
${state.project.parcelBoundary?'clip-path="url(#parcelClip)"':""}>${s}</g>`}
function refreshTerrainView(){const ~
g=$("terrainOverlay");if(g)g.outerHTML=terrainOverlaySvg();if(!terrainRaf)terrainRaf=requestAnimationFrame(()=>{terrainRaf=0;draw3d()})}
function terrainApply(p){const t=terrainEnsure(),{w,h}=propertyDims(),gx=p.x/w*(t.cols-1),gy=p.y/h*(t.rows-1),r=terrainRadius,out=t.h;for(let ~
j=Math.max(0,Math.floor(gy-r));j<=Math.min(t.rows-1,Math.ceil(gy+r));j++)for(let ~
i=Math.max(0,Math.floor(gx-r));i<=Math.min(t.cols-1,Math.ceil(gx+r));i++){const d=Math.hypot(i-gx,j-gy);if(d>r)continue;const ~
f=.5*(1+Math.cos(Math.PI*d/r)),k=j*t.cols+i;let v=out[k];if(terrainBrush==="raise")v+=terrainStrength*f;else ~
if(terrainBrush==="lower")v-=terrainStrength*f;else if(terrainBrush==="smooth"){let s=0,n=0;for(let dj=-1;dj<=1;dj++)for(let di=-1;di<=1;di++){const ~
a=i+di,b=j+dj;if(a<0||b<0||a>=t.cols||b>=t.rows)continue;s+=out[b*t.cols+a];n++}v+=(s/n-v)*.5*f}else ~
v+=(terrainPaint.level-v)*.35*f;out[k]=Math.max(-300,Math.min(300,v))}}
function terrainPanel(){const t=terrainData(),hs=t?t.h:[0],lo=Math.min(...hs),hi=Math.max(...hs),b=(id,l)=>`<button class="toolbtn ~
${terrainBrush===id?"active":""}" data-tbrush="${id}">${l}</button>`;return`<div class="card terrain-panel"><div class="between"><h3>Terrain ~
editor</h3><span class="pill">Relief ${(hi-lo).toFixed(0)} ft</span></div><div class="row">${b("raise","▲ Raise")}${b("lower","▼ Lower")}${b("smooth","≈ ~
Smooth")}${b("flat","■ Flatten")}</div><label>Brush size <input type="range" id="tRadius" min="2" max="14" ~
value="${terrainRadius}"></label><label>Strength (ft per stroke step) <input type="range" id="tStrength" min="0.5" max="10" step="0.5" ~
value="${terrainStrength}"></label><label>Preview exaggeration <select id="tExag" class="toolbtn">${[1,4,8,16,32].map(v=>`<option value="${v}" ~
${terrainExag===v?"selected":""}>${v}x</option>`).join("")}</select></label><div class="row" style="margin-top:10px"><button class="toolbtn" data-~
tact="undo" ${terrainUndo.length?"":"disabled"}>Undo</button><button class="toolbtn" data-tact="redo" ~
${terrainRedo.length?"":"disabled"}>Redo</button><button class="toolbtn" data-tact="hills">Gentle hills</button><button class="toolbtn" data-tact="reset" ~
${t?"":"disabled"}>Flatten all</button><button class="toolbtn" data-tact="toggle">${t&&t.show===false?"Show":"Hide"} terrain</button></div><div ~
class="small muted" style="margin-top:8px">Drag on the map to sculpt. Blue = low, tan = high. The isometric preview lifts objects with the ground. Concept~
 only; confirm with a survey. Family: use Export / Import project to share edits.</div></div>`}
function terrainHistory(fn){const ~
t=terrainEnsure();terrainUndo.push(t.h.slice());if(terrainUndo.length>30)terrainUndo.shift();terrainRedo=[];fn(t);t.show=true;save();render()}
const drawMapBase=drawMap;drawMap=function(){return drawMapBase().replace('<g id="mapObjects"',terrainOverlaySvg()+'<g id="mapObjects"')};
const renderBeforeTerrain=render;
render=function(){renderBeforeTerrain();if(activePage!=="designer"||mapView!=="flat")return;document.querySelector(".toolrow")?.insertAdjacentHTML("before~
end",`<button class="toolbtn ${tool==="terrain"?"active":""}" data-tool="terrain">■ Terrain</button>`);$("mapViewport")?.classList.toggle("terrain-~
tool",tool==="terrain");if(tool==="terrain")document.querySelector(".map-side")?.insertAdjacentHTML("afterbegin",terrainPanel())};
$("content").addEventListener("pointerdown",e=>{const ~
svg=e.target.closest("#siteMap");if(!svg||tool!=="terrain"||spacePan||e.button!==0)return;e.preventDefault();const ~
t=terrainEnsure();t.show=true;terrainUndo.push(t.h.slice());if(terrainUndo.length>30)terrainUndo.shift();terrainRedo=[];const p=mapLocal(svg,e);terrainPai~
nt={pointerId:e.pointerId,level:terrainElev(p.x,p.y)};svg.setPointerCapture?.(e.pointerId);terrainApply(p);refreshTerrainView()});
$("content").addEventListener("pointermove",e=>{if(!terrainPaint||terrainPaint.pointerId!==e.pointerId)return;const ~
svg=$("siteMap");if(!svg)return;terrainApply(mapLocal(svg,e));refreshTerrainView()});
const endTerrainPaint=e=>{if(!terrainPaint||(e.pointerId&&terrainPaint.pointerId!==e.pointerId))return;terrainPaint=null;save();render()};
$("content").addEventListener("pointerup",endTerrainPaint);$("content").addEventListener("pointercancel",endTerrainPaint);
$("content").addEventListener("click",e=>{const b=e.target.closest("[data-tbrush]");if(b){terrainBrush=b.dataset.tbrush;render();return}const ~
a=e.target.closest("[data-tact]");if(!a)return;const k=a.dataset.tact,t=terrainData();if(k==="undo"&&terrainUndo.length){terrainRedo.push(terrainEnsure().~
h.slice());terrainEnsure().h=terrainUndo.pop();save();render()}else ~
if(k==="redo"&&terrainRedo.length){terrainUndo.push(terrainEnsure().h.slice());terrainEnsure().h=terrainRedo.pop();save();render()}else ~
if(k==="reset"&&t&&confirm("Flatten all terrain back to zero elevation?"))terrainHistory(x=>x.h.fill(0));else ~
if(k==="toggle"&&t){t.show=t.show===false;save();render()}else if(k==="hills")terrainHistory(x=>{for(let j=0;j<x.rows;j++)for(let ~
i=0;i<x.cols;i++)x.h[j*x.cols+i]=Math.round((12*Math.sin(i/5)+9*Math.cos(j/4+i/9)+6*Math.sin((i+j)/3.5))*2)/2})});
$("content").addEventListener("input",e=>{if(e.target.id==="tRadius")terrainRadius=+e.target.value;if(e.target.id==="tStrength")terrainStrength=+e.target.~
value});
$("content").addEventListener("change",e=>{if(e.target.id==="tExag"){terrainExag=+e.target.value;draw3d()}});
/* Style themes */
const CADCSS="[data-theme=cad]{color-scheme:dark;--bg:#1b1f24;--panel:#23282e;--panel2:#2b3138;--line:#3d4650;--text:#d9dee3;--muted:#8d98a3;--~
green:#3b9cff;--amber:#f2b134;--blue:#3b9cff;--rose:#ff5c5c}[data-theme=cad] body{background:#1b1f24;font-family:Segoe UI,Arial,sans-serif;font-~
size:13px}[data-theme=cad] *{border-radius:2px!important}[data-theme=cad] aside,[data-theme=cad] .topbar{background:#23282e;border-color:#3d4650}[data-~
theme=cad] .nav button{color:#b4bcc4;border-left:3px solid transparent}[data-theme=cad] .nav button:hover{background:#2f363e}[data-theme=cad] .nav ~
button.active{background:#2f363e;color:#fff;border-left-color:#3b9cff}[data-theme=cad] .card,[data-theme=cad] .map-card{background:#23282e;border:1px ~
solid #3d4650;box-shadow:none}[data-theme=cad] h1,[data-theme=cad] h2,[data-theme=cad] h3{letter-spacing:.06em;text-transform:uppercase;font-~
weight:600}[data-theme=cad] h3{font-size:12px;color:#9fb4c8;border-bottom:1px solid #3d4650;padding-bottom:6px}[data-theme=cad] .btn,[data-theme=cad] ~
.toolbtn,[data-theme=cad] .palette-item,[data-theme=cad] .palette-filter,[data-theme=cad] button{background:linear-~
gradient(#323a42,#2a3037);color:#d9dee3;border:1px solid #4a545f}[data-theme=cad] .btn:hover,[data-theme=cad] .toolbtn:hover,[data-theme=cad] .palette-~
```

```
item:hover{background:#3a444e;border-color:#3b9cff}[data-theme=cad] .toolbtn.active,[data-theme=cad] .palette-filter.active,[data-theme=cad] .palette-~
item.active{background:#16436f;border-color:#3b9cff;color:#fff}[data-theme=cad] input,[data-theme=cad] select,[data-theme=cad] ~
textarea{background:#171b20;border:1px solid #444f5a;color:#e6eaee;font-family:Consolas,monospace}[data-theme=cad] input:focus,[data-theme=cad] ~
select:focus{outline:1px solid #3b9cff}[data-theme=cad] th{background:#2b3138;color:#9fb4c8;text-transform:uppercase;font-size:11px;letter-~
spacing:.05em}[data-theme=cad] td,[data-theme=cad] th{border-color:#3d4650}[data-theme=cad] .map-viewport,[data-theme=cad] ~
#usaMap{background:#101316;border:1px solid #4a545f;box-shadow:inset 0 0 0 1px #000}[data-theme=cad] #siteMap{cursor:crosshair;background-~
color:#101316}[data-theme=cad] #siteMap [data-object]{cursor:move}[data-theme=cad] .scene-canvas{background:linear-gradient(#2a3037,#14171a)}[data-~
theme=cad] .pill{background:#16436f;color:#cfe6ff}[data-theme=cad] .pubbar{background:#14171a}[data-theme=cad] .notice{background:#2b3138;border-~
color:#4a545f;color:#c8d0d8}[data-theme=cad] ::-webkit-scrollbar{width:10px;height:10px}[data-theme=cad] ::-webkit-scrollbar-~
thumb{background:#4a545f}[data-theme=cad] .map-card::after{content:\"MODEL SPACE  |  UNITS: FT  |  ORTHO / SNAP\";display:block;margin-top:6px;padding:4px~
 8px;background:#14171a;border:1px solid #3d4650;color:#7fb8f5;font:11px Consolas,monospace;letter-spacing:.08em}";const ~
THEMES={forest:"",cad:CADCSS,ranch:"[data-~
theme=ranch]{--bg:#150f0a;--panel:#211710;--panel2:#2b1f15;--line:#4a3826;--text:#f5ece0;--muted:#b9a58d;--green:#e0a458}",day:"[data-theme=day]{color-sch~
eme:light;--bg:#f3f1ea;--panel:#fffdf8;--panel2:#ece8dc;--line:#d3ccb9;--text:#1d2a22;--muted:#5d6b61;--green:#1f7a4d;--amber:#a86a00;--blue:#1d5fa8;--~
rose:#b4233a}[data-theme=day] body{background:var(--bg)}[data-theme=day] aside,[data-theme=day] .topbar{background:#fffdf8f2;border-~
color:var(--line)}[data-theme=day] .nav button{color:#3b4a40}[data-theme=day] .nav button:hover{background:#ece8dc}[data-theme=day] .nav ~
button.active{background:#dcefe3;color:#14623b}[data-theme=day] .card,[data-theme=day] .map-card{background:#fffdf8;box-shadow:0 1px 4px #0001}[data-~
theme=day] .btn,[data-theme=day] .toolbtn,[data-theme=day] .palette-item,[data-theme=day] .palette-filter,[data-theme=day] .scene-options ~
summary{background:#f3f0e6;color:var(--text);border-color:var(--line)}[data-theme=day] .btn:hover,[data-theme=day] .palette-item:hover,[data-theme=day] ~
.object-row:hover,[data-theme=day] .object-row.selected{background:#e4eadb}[data-theme=day] .toolbtn.active,[data-theme=day] .palette-filter.active,[data-~
theme=day] .palette-item.active{background:#d5ecdd;color:#14623b;border-color:#1f7a4d}[data-theme=day] .field input,[data-theme=day] .field select,[data-~
theme=day] .field textarea,[data-theme=day] .input,[data-theme=day] td input,[data-theme=day] td select{background:#fff;border-~
color:var(--line);color:var(--text)}[data-theme=day] th{background:#ece8dc;color:#3b4a40}[data-theme=day] th,[data-theme=day] td,[data-theme=day] .key-~
value,[data-theme=day] .check-row{border-color:var(--line)}[data-theme=day] .field label,[data-theme=day] .field .label,[data-theme=day] .layer-~
chip{color:#3b4a40}[data-theme=day] .pill{background:#dcefe3;color:#14623b}[data-theme=day] .pill.warn{background:#f7e8c2;color:#7a4e00}[data-theme=day] ~
.pill.rose{background:#f8d9de;color:#9a1c30}[data-theme=day] .notice{background:#f7f0cf;color:#5a5320;border-color:#d9cf8f}[data-theme=day] ~
.notice.greenish{background:#e0f1e6;color:#14623b;border-color:#a8d3b8}[data-theme=day] .notice.blueish{background:#e0ebf7;color:#1d4f86;border-~
color:#a9c4e3}[data-theme=day] .progress,[data-theme=day] .barline{background:#e2dccb}[data-theme=day] .map-viewport,[data-theme=day] ~
#usaMap{background:#e8efe3;border-color:var(--line)}[data-theme=day] .scene-canvas{background:radial-gradient(ellipse at 50% 68%,#dfeadb,#eef1e6 ~
74%)}[data-theme=day] .banner,[data-theme=day] .palette-head{grid-column:1/-1;font-size:9px;font-weight:700;letter-spacing:.08em;text-~
transform:uppercase;color:#8de9bc;padding:8px 2px 2px;position:sticky;top:0;background:#101912;z-index:1}.palette-head span{color:#819287;font-~
weight:400}.palette-hint{background:#f3f0e6;color:#3b4a40;border-color:var(--line)}[data-theme=day] .btn.danger{background:#fbe4e8;color:#9a1c30;border-~
color:#e3a9b3}[data-theme=day] .theme-pick{background:#fff}";
let selectedTheme="cad";
function applyTheme(value,persist=true){const ~
next=Object.prototype.hasOwnProperty.call(THEMES,value)?value:"cad";selectedTheme=next;if(next===="forest")document.documentElement.removeAttribute("data-~
theme");else document.documentElement.dataset.theme=next;const ~
select=$("themePick");if(select)select.value=next;if(persist){try{localStorage.setItem("oregon-estate-planner-theme",next)}catch(e){}}}
document.head.insertAdjacentHTML("beforeend",`<style>${THEMES.ranch}${THEMES.day}${THEMES.cad}.theme-pick{display:block;width:100%;max-~
width:260px;background:var(--panel2);color:var(--text);border:1px solid var(--line);border-radius:8px;padding:8px 10px}.about-~
hero{padding:clamp(24px,5vw,52px);border:1px solid var(--line);border-radius:14px;background:radial-gradient(ellipse at 85% 0%,color-mix(in ~
srgb,var(--blue) 18%,transparent),transparent 42%),linear-gradient(135deg,var(--panel),var(--panel2));}.about-hero h2{font-size:clamp(30px,5vw,54px);line-~
height:1.08;letter-spacing:-.035em;margin:14px 0}.about-hero .about-lead{font-size:clamp(17px,2vw,22px);line-height:1.5;max-~
width:760px;color:var(--text)}.about-hero>p:not(.about-lead){max-width:760px;line-height:1.7}.about-hero .actions{justify-content:flex-start;flex-~
wrap:wrap;margin-top:22px}.about-features{margin-top:14px}.about-features .card{min-height:150px}</style>`);
(function(){let saved="cad";try{saved=localStorage.getItem("oregon-estate-planner-theme")||"cad"}catch(e){}applyTheme(saved,false)})();
(()=>{const FLAG="oregon-estate-edit",q=new ~
URLSearchParams(location.search);if(q.get("edit")==="1")localStorage.setItem(FLAG,"1");if(q.get("edit")==="0")localStorage.removeItem(FLAG);const ~
local=["localhost","127.0.0.1",""].includes(location.hostname),EDIT=q.get("edit")!=="0";
document.head.insertAdjacentHTML("beforeend","<style>.pubbar{display:flex;gap:10px;align-items:center;flex-wrap:wrap;padding:8px ~
14px;background:var(--panel2);border-bottom:1px solid var(--line);font-size:13px;color:var(--text)}.pubbar button{border:1px solid ~
var(--line);background:transparent;color:var(--text);border-radius:8px;padding:5px 10px;cursor:pointer}.credits{font-size:11px;opacity:.7;padding:10px ~
14px 70px}body.viewonly #mapObjects{pointer-events:none}</style>");
const bar=document.createElement("div");bar.className="pubbar";document.body.prepend(bar);
const credits=document.createElement("div");credits.className="credits";credits.innerHTML='© 2026 AcreForge. All rights reserved. · <a ~
href="support.html">Project status</a> · <a href="data-notice.html">Data &amp;amp; privacy</a> · Map data: <a href="https://www.usgs.gov/the-national-map" ~
target="_blank" rel="noopener">USGS The National Map</a> · <a href="https://registry.opendata.aws/terrain-tiles/" target="_blank" rel="noopener">AWS ~
Terrain Tiles</a> · © <a href="https://www.openstreetmap.org/copyright" target="_blank" rel="noopener">OpenStreetMap</a> contributors · U.S. Census ~
boundaries · MapLibre GL. Conceptual planning only, not a survey.';document.body.append(credits);
if(EDIT){const b=document.createElement("button");b.textContent="Public export";b.title="Download published-project.json with the parcel boundary, ~
location, notes and finances removed";b.onclick=()=>{const p=JSON.parse(JSON.stringify(state)),d=defaultState();["parcelBoundary","locationCoordinates","l~
ocationUrl","address","county","community"].forEach(k=>delete ~
p.project[k]);p.project.mapFeatures=[];p.notes=[];p.tasks={};p.scenarios.forEach(s=>s.finance=d.scenarios[0].finance);delete p._recoveryWarning;const ~
a=document.createElement("a");a.href=URL.createObjectURL(new Blob([JSON.stringify(p,null,2)],{type:"application/json"}));a.download="published-~
project.json";a.click();setTimeout(()=>URL.revokeObjectURL(a.href),1000);alert("Saved published-project.json. Put it next to index.html and redeploy to ~
update the public view.")};if(local)bar.append("Edit mode ",b)}
else{document.body.classList.add("viewonly");save=function(){};bar.append("View-only copy of the estate plan. ");const ~
c=document.createElement("button");c.textContent="Make my own editable ~
copy";c.onclick=()=>{localStorage.setItem(STORAGE,JSON.stringify(state));localStorage.setItem(FLAG,"1");location.href=location.pathname};bar.append(c);
const hide=/^(■ Erase|■ Rotate selected|■ Terrain|Undo|Redo|Clear layout|Import parcel boundary|Snap on)/;const r0=render;render=function(){r0();document.~
querySelectorAll("button").forEach(x=>{if(hide.test(x.textContent.trim()))x.style.display="none"});document.querySelectorAll("h3,h2").forEach(h=>{if(/^(Add~
d an object|Selected object)/.test(h.textContent.trim()))h.closest(".card")?.setAttribute("style","display:none")})};
fetch("published-project.json",{cache:"no-store"}).then(r=>r.ok?r.json():null).then(j=>{if(j){state=normalizeState(j);render()}}).catch(()=>{})})();
(()=>{const bar=document.querySelector(".pubbar");if(bar){const status=document.createElement("span");status.textContent="Paid plans are not ~
active";bar.append(status);const PAY_LINK="",a=document.createElement("a");a.className="hbb";a.target="_blank";a.rel="noopener";a.style.cssText="margin-~
left:8px;text-decoration:none";if(PAY_LINK){a.href=PAY_LINK;a.textContent="Upgrade to Pro"}else{a.href="https://github.com/jacobbaltins/acreforge/issues/n~
ew?title=Pro%20waitlist&body=I%27d%20pay%20for%20AcreForge%20Pro.%20I%20want%3A%20(saved%20projects%2C%20PDF%20export%2C%20parcel%20import%2C%20other)%20(~
this%20post%20is%20public)";a.textContent="Join Pro waitlist"}bar.append(a)}})();
(()=>{const prevRender=render,CT={cols:3,rows:2,bw:300,bh:200,sw:36,lw:60,ld:100};
const tcat=it=>(templates[it.type]||{}).cat;
const limitOf=it=>{if(["road","citystreet"].includes(it.type))return 8;if(["parking","citylot"].includes(it.type))return 5;const ~
c=tcat(it);if(c==="Arena")return 3;if(c==="Building"||(c==="City"&&!["citypark","plaza"].includes(it.type)))return 10;return null};
function sg(){const t=terrainData();if(!t||t.show===false)return null;const{w,h}=propertyDims(),cw=w/(t.cols-1),ch=h/(t.rows-~
1),H=(i,j)=>t.h[Math.max(0,Math.min(t.rows-1,j))*t.cols+Math.max(0,Math.min(t.cols-1,i))],s=new Float32Array(t.cols*t.rows);for(let ~
j=0;j<t.rows;j++)for(let i=0;i<t.cols;i++)s[j*t.cols+i]=100*Math.hypot((H(i+1,j)-H(i-1,j))/(2*cw),(H(i,j+1)-H(i,j-1))/(2*ch));return{t,w,h,s}}
const slopeAt=(g,x,y)=>g.s[Math.max(0,Math.min(g.t.rows-1,Math.round(y/g.h*(g.t.rows-1))))*g.t.cols+Math.max(0,Math.min(g.t.cols-~
1,Math.round(x/g.w*(g.t.cols-1))))];
function analyze(){const g=sg();if(!g)return null;const bins=[0,0,0,0];for(const v of g.s)bins[v<5?0:v<10?1:v<15?2:3]++;const n=g.s.length,rows=[];let ~
cut=0,fill=0;for(const it of items()){const lim=limitOf(it);if(lim===null)continue;let mx=0,hs=[];const N=6;for(let a=0;a<=N;a++)for(let ~
b=0;b<=N;b++){const x=it.x+it.w*a/N,y=it.y+it.h*b/N;mx=Math.max(mx,slopeAt(g,x,y));hs.push(terrainElev(x,y))}const ~
avg=hs.reduce((p,c)=>p+c,0)/hs.length,cell=it.w*it.h/hs.length;let c1=0,f1=0;for(const v of hs){if(v>avg)c1+=(v-avg)*cell;else f1+=(avg-~
v)*cell}if(!["road","citystreet","parking","citylot"].includes(it.type)||true){cut+=c1;fill+=f1}rows.push({it,mx,lim,cy:Math.max(c1,f1)/27})}return{bins,n~
,rows,cut:cut/27,fill:fill/27}}
const slopePanel=()=>{const a=analyze();if(!a)return"";const ~
pc=i=>(100*a.bins[i]/a.n).toFixed(0)+"%",bad=a.rows.filter(r=>r.mx>r.lim).sort((p,q)=>q.mx-p.mx);return`<section class="card" id="slopePanel"><div ~
class="between"><h3>Slope &amp;amp; earthwork</h3><span class="pill ${bad.length?"rose":""}">${bad.length?bad.length+" warning"+(bad.length>1?"s":""):"No ~
warnings"}</span></div><div class="small">Land by slope: <b>${pc(0)}</b> under 5% · <b>${pc(1)}</b> 5–10% · <b>${pc(2)}</b> 10–15% · <b>${pc(3)}</b> over ~
15%</div><div class="small" style="margin-top:6px">Rough earthwork to level every building and road footprint: cut ~
<b>${Math.round(a.cut).toLocaleString()}</b> cu yd · fill <b>${Math.round(a.fill).toLocaleString()}</b> cu yd</div>${bad.slice(0,8).map(r=>`<div ~
class="small" style="margin-top:4px;color:#fca5a5">■ ${esc(r.it.name)}: ${r.mx.toFixed(0)}% slope (suggested max ${r.lim}%), about ~
${Math.round(r.cy).toLocaleString()} cu yd to level</div>`).join("")}<div class="small muted" style="margin-top:6px">Concept-level estimate from your ~
sculpted terrain. Not a grading plan; get a survey and civil engineer.</div></section>`};
const cityPanel=()=>{const f=(k,l)=>`<label class="small">${l}<input class="input" type="number" min="1" data-ct="${k}" ~
value="${CT[k]}"></label>`;return`<section class="card" id="cityPanel"><h3>City tools</h3><div class="grid g2">${f("cols","Blocks ~
```

```
across")}${f("rows","Blocks down")}${f("bw","Block width ft")}${f("bh","Block depth ft")}${f("sw","Street width ft")}${f("lw","Lot width ~
ft")}${f("ld","Lot depth ft")}</div><div class="row" style="margin-top:8px"><button class="toolbtn" data-city="grid">Streets only</button><button ~
class="toolbtn" data-city="lots">Streets + house lots</button><button class="toolbtn" data-city="clear">Remove generated</button></div><div class="small ~
muted" style="margin-top:6px">Builds a street grid centered on the property, then splits each block into lots. Everything stays editable and ~
undoable.</div></section>`};
render=function(){prevRender();if(activePage==="designer"&&mapView==="flat"){const side=document.querySelector(".map-~
side");if(side)side.insertAdjacentHTML("beforeend",slopePanel()+cityPanel())}};
function build(withLots){const{w,h}=propertyDims(),{cols,rows,bw,bh,sw,lw,ld}=CT,W=cols*bw+(cols+1)*sw,H=rows*bh+(rows+1)*sw;if([cols,rows,bw,bh,sw,lw,ld]~
.some(v=>!(v>0)))return alert("Enter positive numbers.");if(W>w||H>h)return alert(`That grid is ${Math.round(W)} x ${Math.round(H)} ft, larger than the ~
property (${Math.round(w)} x ${Math.round(h)} ft). Reduce blocks or sizes.`);const x0=(w-W)/2,y0=(h-H)/2,out=[];const add=(type,x,y,iw,ih,name)=>{const ~
it=makeItem(type,x,y);it.w=iw;it.h=ih;if(name)it.name=name;it.gen="city";out.push(it)};for(let ~
k=0;k<=cols;k++)add("citystreet",x0+k*(bw+sw),y0,sw,H,"Street");for(let k=0;k<=rows;k++)add("citystreet",x0,y0+k*(bh+sw),W,sw,"Street");if(withLots){const~
 nx=Math.floor(bw/lw),ny=bh>=2*ld?2:1,off=(bw-nx*lw)/2;for(let c=0;c<cols;c++)for(let r=0;r<rows;r++){const bx=x0+sw+c*(bw+sw),by=y0+sw+r*(bh+sw);for(let ~
a=0;a<nx;a++)for(let b=0;b<ny;b++)add("homelot",bx+off+a*lw,b?by+bh-ld:by,lw,ld,"House lot")}}if(out.length>600)return alert("That would create ~
"+out.length+" objects. Use fewer blocks.");let kept=out;if(state.project.parcelBoundary){kept=out.filter(it=>!objectOutsideParcel(it))}const ~
skipped=out.length-kept.length;snapshot();kept.forEach(it=>items().push(it));save();render();alert(`Added ${kept.length} objects${skipped?` ($skipped} ~
skipped outside the parcel)`:""}.`)}
$("content").addEventListener("change",e=>{const k=e.target.dataset&&e.target.dataset.ct;if(k)CT[k]=Number(e.target.value)});
$("content").addEventListener("click",e=>{const b=e.target.closest("[data-city]");if(!b)return;const m=b.dataset.city;if(m==="clear"){const ~
n=items().filter(i=>i.gen==="city").length;if(!n)return alert("No generated city objects to remove.");if(!confirm("Remove "+n+" generated street and lot ~
objects?"))return;snapshot();for(let i=items().length-1;i>=0;i--)if(items()[i].gen==="city")items().splice(i,1);selectedId=null;save();render()}else ~
build(m==="lots")});
})();
(()=>{const MG={grid:true,plan:true,sp:0},SPS=[0,50,100,250,500,1000],R=6371007.1809,FT=3.280839895,empty={type:"FeatureCollection",features:[]};
function anchor(){const d=propertyDims(),b=state.project.parcelBoundary,ky=R*FT*Math.PI/180;if(b){const[lon0,lat0]=b.center,kx=ky*Math.max(.01,Math.cos(la~
t0*Math.PI/180)),p=b.polygons[0][0][0],c=b.coordinates[0][0][0];let ~
dx=c[0]-lon0;if(dx>180)dx-=360;if(dx<-180)dx+=360;return{d,kx,ky,lon0,lat0,minX:dx*kx-p[0],minY:(c[1]-lat0)*ky-p[1],n:1}}let ~
c=state.project.locationCoordinates?[state.project.locationCoordinates.lon,state.project.locationCoordinates.lat]:null;if(!c&&terrainMapInstance){const ~
m=terrainMapInstance.getCenter();c=[m.lng,m.lat]}if(!c)return ~
null;return{d,kx:ky*Math.max(.01,Math.cos(c[1]*Math.PI/180)),ky,lon0:c[0],lat0:c[1],minX:-d.w/2,minY:-d.h/2,n:-1}}
const geo=(a,x,y)=>[a.lon0+(x+a.minX)/a.kx,a.lat0+a.n*(y+a.minY)/a.ky];
const spacing=d=>MG.sp?MG.sp:SPS.slice(1).find(s=>Math.max(d.w,d.h)/s<=60)||1000;
function gridData(a){const{w,h}=a.d,s=spacing(a.d),f=[],ln=(p,q,major)=>f.push({type:"Feature",properties:{major},geometry:{type:"LineString",coordinates:~
[geo(a,...p),geo(a,...q)]}});for(let x=0,k=0;x<=w+1e-6;x+=s,k++)ln([x,0],[x,h],k%5===0);for(let ~
y=0,k=0;y<=h+1e-6;y+=s,k++)ln([0,y],[w,y],k%5===0);return{type:"FeatureCollection",features:f}}
function planData(a){const f=[],vis=items().filter(i=>i.visible!==false&&state.layers[layerFor(i.type)]!==false);let out=0;for(const it of vis){const ~
o=objectOutsideParcel(it);if(o)out++;const ring=footprintPoints(it).map(p=>geo(a,...p));ring.push(ring[0]);f.push({type:"Feature",properties:{color:it.col~
or||(templates[it.type]||{}).color||"#f2c14e",out:o},geometry:{type:"Polygon",coordinates:[ring]}})}const{w,h}=a.d,box=[[0,0],[w,0],[w,h],[0,h],[0,0]].map~
(p=>geo(a,...p));f.push({type:"Feature",properties:{color:"#000",out:false,box:true},geometry:{type:"Polygon",coordinates:[box]}});return{data:{type:"Feat~
ureCollection",features:f},count:vis.length,out}}
function status(a,pd){const el=$("mgStatus");if(!el)return;el.innerHTML=a?`Grid every <b>${spacing(a.d)} ft</b> (thick line every ${spacing(a.d)*5} ft). ~
Plan overlay: <b>${pd?pd.count:0}</b> objects${pd&&pd.out?`, <b style="color:#fca5a5">${pd.out} outside the parcel ~
(red)</b>`:pd&&state.project.parcelBoundary?", all inside the parcel":""}. ${state.project.parcelBoundary?"Placed using the imported parcel ~
coordinates.":"No parcel imported: plan is centered on the map location (set one with Navigate, or import a parcel for exact placement)."}`:"Click the map~
 to set a location, or import a parcel boundary, then the grid and plan appear here."}
function refresh(){const m=terrainMapInstance;if(!m||!m.getSource("mg-grid"))return;const a=anchor();let pd=null;if(a){m.getSource("mg-~
grid").setData(gridData(a));pd=planData(a);m.getSource("mg-plan").setData(pd.data)}else{m.getSource("mg-grid").setData(empty);m.getSource("mg-~
plan").setData(empty)}m.setLayoutProperty("mg-grid-line","visibility",MG.grid?"visible":"none");["mg-plan-fill","mg-plan-~
line"].forEach(id=>m.setLayoutProperty(id,"visibility",MG.plan?"visible":"none"));status(a,pd);const g=document.querySelector('[data-~
mg="grid"]'),p=document.querySelector('[data-mg="plan"]'),s=document.querySelector('[data-mg="spacing"]');if(g)g.textContent=MG.grid?"Grid on":"Grid ~
off";if(p)p.textContent=MG.plan?"Plan overlay on":"Plan overlay off";if(s&&a)s.textContent="Grid "+spacing(a.d)+" ft"}
const baseAdd=addTerrainLayers;addTerrainLayers=function(m){baseAdd(m);m.addSource("mg-grid",{type:"geojson",data:empty});m.addSource("mg-~
plan",{type:"geojson",data:empty});m.addLayer({id:"mg-grid-line",type:"line",source:"mg-grid",paint:{"line-color":"#ffffff","line-~
opacity":["case",["get","major"],.75,.4],"line-width":["case",["get","major"],1.6,.8]}});m.addLayer({id:"mg-plan-fill",type:"fill",source:"mg-~
plan",filter:["!",["has","box"]],paint:{"fill-color":["get","color"],"fill-opacity":.6}});m.addLayer({id:"mg-plan-line",type:"line",source:"mg-~
plan",paint:{"line-color":["case",["has","box"],"#38bdf8",["get","out"],"#ef4444","#ffffff"],"line-width":["case",["has","box"],2,1.2]}});refresh()};
const baseUpd=updateTerrainSources;updateTerrainSources=function(){baseUpd();refresh()};
const prev=render;render=function(){prev();if(activePage!=="designer"||mapView!=="terrain")return;document.querySelector(".map-~
toolbar")?.insertAdjacentHTML("beforeend",`<button class="toolbtn" data-mg="grid">Grid on</button><button class="toolbtn" data-~
mg="spacing">Grid</button><button class="toolbtn" data-mg="plan">Plan overlay on</button><button class="toolbtn" data-mg="fit">Zoom to ~
plan</button>`);$("terrainMap")?.insertAdjacentHTML("afterend",'<div id="mgStatus" class="small" style="margin-top:8px"></div>');status(anchor(),null)};
$("content").addEventListener("click",e=>{const b=e.target.closest("[data-mg]");if(!b)return;const k=b.dataset.mg;if(k==="grid")MG.grid=!MG.grid;else ~
if(k==="plan")MG.plan=!MG.plan;else if(k==="spacing"){const a=anchor(),cur=MG.sp?SPS.indexOf(MG.sp):0;MG.sp=SPS[(cur+1)%SPS.length]}else ~
if(k==="fit"){const a=anchor(),m=terrainMapInstance;if(!a||!m)return alert("Set a location or import a parcel first.");const cs=[[0,0],[a.d.w,0],[a.d.w,a.~
d.h],[0,a.d.h]].map(p=>geo(a,...p)),xs=cs.map(c=>c[0]),ys=cs.map(c=>c[1]);m.fitBounds([[Math.min(...xs),Math.min(...ys)],[Math.max(...xs),Math.max(...ys)]~
],{padding:40,duration:600});return}refresh()});
})();
(()=>{const esc=s=>String(s).replace(/[&<>"]/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;"}[c]));
const layerOf=it=>String(it.type||it.category||it.name||"OBJECT").toUpperCase().replace(/[^A-Z0-9]+/g,"_").slice(0,30)||"OBJECT";
function dxf(){const L=[];const ~
add=(c,v)=>L.push(String(c),String(v));add(0,"SECTION");add(2,"HEADER");add(9,"$INSUNITS");add(70,2);add(0,"ENDSEC");add(0,"SECTION");add(2,"ENTITIES");
const b=state.project&&state.project.parcelBoundary;if(b&&b.polygons)b.polygons.forEach(poly=>(Array.isArray(poly[0][0])?poly:[poly]).forEach(r=>{add(0,"L~
WPOLYLINE");add(8,"PARCEL");add(90,r.length);add(70,1);r.forEach(p=>{add(10,p[0].toFixed(3));add(20,(-p[1]).toFixed(3))})}));
items().forEach(it=>{const pts=footprintPoints(it);add(0,"LWPOLYLINE");add(8,layerOf(it));add(90,pts.length);add(70,1);pts.forEach(p=>{add(10,p[0].toFixed~
(3));add(20,(-p[1]).toFixed(3))});const c=pts.reduce((a,p)=>[a[0]+p[0]/pts.length,a[1]+p[1]/pts.length],[0,0]);add(0,"TEXT");add(8,"LABELS");add(10,c[0].t~
oFixed(3));add(20,(-c[1]).toFixed(3));add(40,Math.max(4,Math.min(it.w,it.h)*.12).toFixed(2));add(1,String(it.name||"").replace(/[\r\n]/g," "))});
add(0,"ENDSEC");add(0,"EOF");return L.join("\r\n")}
function download(name,type,data){const a=document.createElement("a");a.href=URL.createObjectURL(new ~
Blob([data],{type}));a.download=name;a.click();setTimeout(()=>URL.revokeObjectURL(a.href),1500)}
function sheet(){const svg=document.getElementById("siteMap");if(!svg){alert("Open the flat plan first.");return}const ~
s=svg.cloneNode(true);s.removeAttribute("style");s.setAttribute("width","100%");s.setAttribute("height","100%");s.querySelector("#dragGhost")?.remove();
const P=state.project||{},d=new Date().toLocaleDateString(),dims=typeof propertyDims==="function"?propertyDims():null;
const html=`<!doctype html><title>Site plan sheet</title><style>@page{size:landscape;margin:8mm}body{margin:0;font:11px Segoe UI,Arial,sans-~
serif;color:#000}.sheet{border:2px solid #000;display:grid;grid-template-columns:1fr 62mm;height:calc(100vh - 6mm)}.draw{border-right:2px solid ~
#000;background:#fff;padding:4mm;overflow:hidden}.draw svg{background:#fff}.tb{display:flex;flex-direction:column}.tb div{border-bottom:1px solid ~
#000;padding:2mm 3mm}.tb small{display:block;font-size:8px;letter-spacing:.08em;text-transform:uppercase;color:#444}.tb b{font-~
size:13px}.fill{flex:1}</style><div class="sheet"><div class="draw">${s.outerHTML}</div><div ~
class="tb"><div><small>Project</small><b>${esc(P.name||"AcreForge")}</b></div><div><small>Sheet</small><b>SITE ~
PLAN</b></div><div><small>Extent</small>${dims?Math.round(dims.w||0)+" × "+Math.round(dims.h||0)+" ft":"n/a"}</div><div><small>Grid</small>${gridSize} ~
ft</div><div><small>Date</small>${d}</div><div><small>Revision</small>${esc(String(state.revision||"A"))}</div><div><small>Drawn ~
by</small> </div><div><small>Checked by</small> </div><div class="fill"><small>Notes</small>Conceptual planning only. Not a survey. Verify with ~
licensed professionals and county requirements.</div><div><small>Scale</small>Fit to sheet · see ~
grid</div></div></div><script>onload=()=>setTimeout(()=>print(),300)<\/script>`;
const w=window.open("","_blank");if(!w){alert("Allow pop-ups to print the sheet.");return}w.document.write(html);w.document.close()}
const r0=render;render=function(){r0();const bar=document.querySelector(".map-toolbar");if(bar&&!bar.querySelector("[data-~
sx]")&&document.getElementById("siteMap")){[["ws","■ CAD workspace"],["sheet","Print sheet"],["dxf","Export DXF"]].forEach(([k,t])=>{const ~
b=document.createElement("button");b.className="toolbtn";b.dataset.sx=k;b.textContent=t;bar.appendChild(b)})}};
document.addEventListener("click",e=>{const b=e.target.closest("[data-sx]");if(!b)return;if(b.dataset.sx==="dxf"){download("site-~
plan.dxf","application/dxf",dxf())}else if(b.dataset.sx==="sheet")sheet()})})();
(function(){
const css=`
.cadhud{display:none;position:fixed;z-index:960;font:12px/1.2 Consolas,"Segoe UI",monospace;color:#d7dde6;background:#2b3038;box-sizing:border-box}
body.cadws .cadhud{display:flex}
body.cadws{overflow:hidden}
```

```
#cadTop{left:0;right:0;top:0;height:64px;flex-direction:column;border-bottom:1px solid #000;background:#262a31}
#cadTop .r1{display:flex;align-items:center;height:28px;background:#1d2025;border-bottom:1px solid #000;padding:0 6px;gap:2px}
#cadTop .logo{background:#1f6fd0;color:#fff;font-weight:700;padding:3px 7px;margin-right:8px;letter-spacing:1px}
#cadTop .menu{position:relative}
#cadTop .mb{background:none;border:0;color:#d7dde6;padding:6px 10px;cursor:pointer;font:inherit}
#cadTop .mb:hover,#cadTop .menu.open .mb{background:#3a4150}
#cadTop .dd{display:none;position:absolute;left:0;top:100%;min-width:230px;background:#2b3038;border:1px solid #000;box-shadow:0 6px 18px #0009;z-index:5}
#cadTop .menu.open .dd{display:block}
#cadTop .dd button{display:flex;justify-content:space-between;gap:20px;width:100%;background:none;border:0;color:#d7dde6;padding:7px 12px;text-~
align:left;cursor:pointer;font:inherit}
#cadTop .dd button:hover{background:#1f6fd0;color:#fff}
#cadTop .dd i{opacity:.6;font-style:normal}
#cadTop .ttl{flex:1;text-align:center;color:#9aa5b4}
#cadTop .r2{display:flex;align-items:center;height:35px;padding:0 6px;gap:3px}
#cadTop .qb{background:#343a45;border:1px solid #444c5a;color:#d7dde6;padding:4px 9px;cursor:pointer;font:inherit}
#cadTop .qb:hover{background:#1f6fd0;border-color:#5ea0f0;color:#fff}
#cadTop .sep{width:1px;height:20px;background:#4a5262;margin:0 6px}
#cadTop .tabs{margin-left:auto;display:flex;gap:2px}
#cadTop .tab{padding:5px 12px;background:#1d2025;border:1px solid #444c5a;color:#9aa5b4}
#cadTop .tab.on{background:#3a4150;color:#fff;border-bottom:2px solid #1f6fd0}
#cadTools{left:0;top:64px;bottom:58px;width:44px;flex-direction:column;align-items:center;gap:4px;padding:8px 0;border-right:1px solid #000}
#cadTools button,#cadNav button{width:34px;height:34px;background:#343a45;border:1px solid #444c5a;color:#d7dde6;cursor:pointer;font-size:16px;padding:0}
#cadTools button:hover,#cadNav button:hover{background:#1f6fd0;color:#fff}
#cadTools button.on{background:#1f6fd0;border-color:#8fc1ff;color:#fff}
#cadNav{top:96px;right:332px;width:40px;flex-direction:column;gap:3px;padding:3px;background:#2b3038cc;border:1px solid #000}
#cadNav .cmp{width:34px;height:34px;border-radius:50%;border:1px solid #6f7a8c;display:flex;align-items:center;justify-content:center;color:#ff6a6a;font-~
weight:700;margin-bottom:3px}
#cadBot{left:0;right:0;bottom:0;height:58px;flex-direction:column;border-top:1px solid #000;background:#262a31}
#cadBot .cmd{display:flex;align-items:center;height:30px;padding:0 8px;gap:8px;background:#1d2025;border-bottom:1px solid #000}
#cadBot .cmd b{color:#8fc1ff}
#cadBot .cmd input{flex:1;background:transparent;border:0;outline:0;color:#fff;font:inherit}
#cadBot .cmd .msg{color:#9aa5b4;max-width:46%;overflow:hidden;white-space:nowrap;text-overflow:ellipsis}
#cadBot .st{display:flex;align-items:center;height:28px;padding:0 8px;gap:4px}
#cadBot .st .xy{min-width:230px;color:#8fc1ff}
#cadBot .st button{background:#343a45;border:1px solid #444c5a;color:#9aa5b4;padding:3px 9px;cursor:pointer;font:inherit}
#cadBot .st button.on{background:#1f6fd0;color:#fff;border-color:#8fc1ff}
#cadBot .st .sp{flex:1}
#cadCrossH,#cadCrossV{position:fixed;z-index:955;pointer-events:none;background:#7fb4ff55;display:none}
#cadCrossH{height:1px;left:0;right:0}#cadCrossV{width:1px;top:0;bottom:0}
body.cadws .designer-layout{position:fixed;inset:64px 0 58px 44px;z-index:900;display:grid;grid-template-columns:1fr ~
320px;gap:0;margin:0;padding:0;background:#15171b}
body.cadws .designer-layout>.map-card{grid-column:1;display:flex;flex-direction:column;border-radius:0;margin:0;height:100%;min-height:0;border:0}
body.cadws .designer-layout .map-viewport{flex:1;height:auto!important;min-height:0}
body.cadws .designer-layout>.map-side{grid-column:2;overflow:auto;height:100%;padding:8px;background:#23272e;border-left:1px solid #000}
body.cadws .designer-layout>.scene-card{display:none}
body.cadws .designer-layout .map-toolbar{display:none}
body.cadws .designer-layout>.map-card>*:not(.map-viewport):not(.map-toolbar){display:none!important}
body.cadws .cadhide{display:none!important}
body.cadws .designer-layout{grid-template-rows:minmax(0,1fr)}
@media(max-width:900px){body.cadws .designer-layout{grid-template-columns:minmax(0,1fr)}body.cadws .designer-layout>.map-~
side{display:none}#cadNav{right:6px}#cadTop .ttl,#cadTop .tabs,#cadTop .sep,#cadTop .r2>span{display:none}#cadTop .qb,#cadTop .mb{padding:4px 6px;font-~
size:11px}#cadBot .st .xy{min-width:0}#cadInfo{display:none}#cadBot .cmd .msg{display:none}}
#cadDrop{position:fixed;inset:64px 0 58px 44px;z-index:970;display:none;align-items:center;justify-content:center;background:#0b3a7acc;color:#fff;font:700~
 22px "Segoe UI",sans-serif;border:3px dashed #8fc1ff;pointer-events:none}`;
const st=document.createElement("style");st.textContent=css;document.head.appendChild(st);
const mk=(id,html)=>{const d=document.createElement("div");d.id=id;d.className="cadhud";d.innerHTML=html;document.body.appendChild(d);return d};
const top=mk("cadTop",`<div class="r1"><span class="logo">OEP</span>
<div class="menu"><button class="mb">File</button><div class="dd"><button data-c="OPEN">Open project… <i>OPEN</i></button><button data-c="IMPORT">Load ~
parcel / survey (GeoJSON, KML)… <i>IMPORT</i></button><button data-c="SAVE">Save project (JSON) <i>SAVE</i></button><button data-c="DXF">Export DXF for ~
AutoCAD <i>DXF</i></button><button data-c="PRINT">Print sheet with title block <i>PRINT</i></button><button data-c="EXIT">Close workspace ~
<i>EXIT</i></button></div></div>
<div class="menu"><button class="mb">Edit</button><div class="dd"><button data-c="UNDO">Undo <i>U</i></button><button data-c="REDO">Redo ~
<i>REDO</i></button><button data-c="ERASE">Erase <i>E</i></button><button data-c="ROTATE">Rotate selected <i>RO</i></button></div></div>
<div class="menu"><button class="mb">View</button><div class="dd"><button data-c="ZOOM E">Zoom extents <i>Z E</i></button><button data-c="ZOOM IN">Zoom in~
 <i>Z IN</i></button><button data-c="ZOOM OUT">Zoom out <i>Z OUT</i></button><button data-c="GRID">Toggle grid <i>GRID</i></button><button ~
data-c="TERRAIN">Terrain <i>TERRAIN</i></button><button data-c="MAP">USA map <i>MAP</i></button></div></div>
<div class="menu"><button class="mb">Help</button><div class="dd"><button data-c="HELP">Command list <i>HELP</i></button></div></div>
<span class="ttl" id="cadTtl"></span><button class="mb" data-c="EXIT" title="Close workspace">✕ Close</button></div>
<div class="r2"><button class="qb" data-c="OPEN">■ Open</button><button class="qb" data-c="SAVE">■ Save</button><button class="qb" data-c="IMPORT">■ Load ~
parcel</button><span class="sep"></span><button class="qb" data-c="UNDO">■</button><button class="qb" data-c="REDO">■</button><span ~
class="sep"></span><button class="qb" data-c="DXF">DXF</button><button class="qb" data-c="PRINT">Print sheet</button><span ~
class="sep"></span><span>Grid</span><select id="cadGrid" ~
class="qb"><option>10</option><option>20</option><option>40</option><option>100</option></select><div class="tabs"><span class="tab">Start</span><span ~
class="tab on" id="cadTab">Site plan*</span></div></div>`);
const tools=mk("cadTools",`<button data-c="SELECT" data-t="select" title="Select / move (S)">■</button><button data-c="PAN" data-t="pan" title="Pan ~
(P)">✥</button><button data-c="ERASE" data-t="erase" title="Erase (E)">■</button><button data-c="ROTATE" data-t="rotate" title="Rotate ~
(RO)">■</button><button data-c="ZOOM IN" title="Zoom in">■</button><button data-c="ZOOM OUT" title="Zoom out">–</button><button data-c="ZOOM E" ~
title="Zoom extents">■</button><button data-c="GRID" title="Toggle grid">■</button>`);
const nav=mk("cadNav",`<div class="cmp">N</div><button data-c="PAN" title="Pan">✥</button><button data-c="ZOOM E" title="Zoom extents">■</button><button ~
data-c="ZOOM IN" title="Zoom in">■</button><button data-c="ZOOM OUT" title="Zoom out">–</button>`);
const bot=mk("cadBot",`<div class="cmd"><b>Command:</b><input id="cadIn" placeholder="Type a command (HELP for list) and press Enter" autocomplete="off" ~
spellcheck="false"><span class="msg" id="cadMsg"></span></div><div class="st"><span class="xy" id="cadXY">X: —   Y: —</span><button ~
class="on">MODEL</button><button id="cadSnap" class="on">SNAP</button><button id="cadGridB" class="on">GRID</button><button class="on">ORTHO</button><span~
 class="sp"></span><span id="cadInfo"></span></div>`);
const drop=document.createElement("div");drop.id="cadDrop";drop.textContent="Drop a project .json, GeoJSON or KML to ~
load";document.body.appendChild(drop);
const ch=document.createElement("div"),cv=document.createElement("div");ch.id="cadCrossH";cv.id="cadCrossV";document.body.append(ch,cv);
const $=id=>document.getElementById(id),msg=t=>{$("cadMsg").textContent=t};
const hist=[];let hi=0;
const clickSel=s=>{const e=document.querySelector(s);if(e)e.click();return !!e};
const gridBtn=()=>[...document.querySelectorAll(".map-toolbar button")].find(b=>b.textContent.trim()==="Grid");
function setWs(on){document.body.classList.toggle("cadws",on);if(on){if(typeof state!=="undefined")msg("Workspace ready. Type HELP for ~
commands.");sync()}else{ch.style.display=cv.style.display="none"}}
function sync(){const g=document.getElementById("gridSelect");if(g)$("cadGrid").value=g.value;tools.querySelectorAll("[data-~
t]").forEach(b=>b.classList.toggle("on",!!document.querySelector('.map-toolbar [data-tool="'+b.dataset.t+'"].active')));$("cadTtl").textContent="AcreForge~
 — "+((state.project&&state.project.name)||"Site plan");const ~
gb=gridBtn();$("cadGridB").classList.toggle("on",!!gb&&gb.classList.contains("active"));const z=document.querySelector(".map-~
status");$("cadInfo").textContent=(z?z.textContent.replace(/\s+/g," ").trim().slice(0,110):"")+" · UNITS: FT"}
function run(raw){const s=raw.trim().toUpperCase();if(!s)return;const [c,a]=s.split(/\s+/);
const t={S:"select",SELECT:"select",P:"pan",PAN:"pan",E:"erase",ERASE:"erase",RO:"rotate",ROTATE:"rotate"};
if(t[c]){clickSel('.map-toolbar [data-tool="'+t[c]+'"]');msg(c+" — tool set")}
else if(c==="Z"||c==="ZOOM"){if(!a||a==="E"||a==="F"||a==="EXTENTS"||a==="ALL")clickSel('[data-action="zoom-fit"]');else ~
if(a==="IN"||a==="+")clickSel('[data-action="zoom-in"]');else if(a==="OUT"||a==="-")clickSel('[data-action="zoom-out"]');else msg("ZOOM [E|IN|OUT]")}
else if(c==="GRID"){const n=+a;if(n){const g=document.getElementById("gridSelect");if([10,20,40,100].includes(n)&&g){g.value=n;g.dispatchEvent(new ~
```

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Event("change",{bubbles:true}));msg("Grid spacing "+n+" ft")}else msg("GRID 10|20|40|100")}else{const b=gridBtn();if(b)b.click();msg("Grid toggled")}}
else if(c==="SNAP"){clickSel('[data-snap],.snap-btn')||[...document.querySelectorAll("button")].find(b=>/^Snap ~
(on|off)$/.test(b.textContent.trim()))?.click();msg("Snap toggled")}
else if(c==="U"||c==="UNDO"){undo();msg("Undo")}
else if(c==="REDO"){redo();msg("Redo")}
else if(c==="OPEN"){document.getElementById("importFile").click();msg("Choose a project .json")}
else if(c==="IMPORT"){document.getElementById("parcelFile").click();msg("Choose a GeoJSON / KML parcel")}
else if(c==="SAVE"){exportProject();msg("Project saved as JSON")}
else if(c==="DXF"){download("site-plan.dxf","application/dxf",dxf());msg("DXF exported")}
else if(c==="PRINT"){sheet();msg("Print sheet opened")}
else if(c==="TERRAIN"){[...document.querySelectorAll(".map-toolbar button")].find(b=>/Terrain/.test(b.textContent))?.click();msg("Terrain toggled")}
else if(c==="MAP"){[...document.querySelectorAll(".map-toolbar button")].find(b=>/USA map/.test(b.textContent))?.click()}
else if(c==="EXIT"||c==="QUIT"||c==="CLOSE"){setWs(false)}
else if(c==="HELP"||c==="?"){msg("S P E RO | Z E/IN/OUT | GRID [n] | SNAP | U REDO | OPEN IMPORT SAVE DXF PRINT | TERRAIN MAP | EXIT")}
else msg('Unknown command "'+raw.trim()+'". Type HELP.');
setTimeout(sync,80)}
$("cadIn").addEventListener("keydown",e=>{if(e.key==="Enter"){const ~
v=e.target.value;if(v.trim()){hist.push(v);hi=hist.length}run(v);e.target.value=""}else ~
if(e.key==="ArrowUp"&&hist.length){hi=Math.max(0,hi-1);e.target.value=hist[hi];e.preventDefault()}else ~
if(e.key==="ArrowDown"){hi=Math.min(hist.length,hi+1);e.target.value=hist[hi]||"";e.preventDefault()}else ~
if(e.key==="Escape"){e.target.value="";e.target.blur()}e.stopPropagation()});
document.addEventListener("click",e=>{const ws=e.target.closest('[data-sx="ws"]');if(ws){setWs(true);return}
const mb=e.target.closest("#cadTop .mb");if(mb&&mb.parentElement.classList.contains("menu")){const ~
m=mb.parentElement,was=m.classList.contains("open");document.querySelectorAll("#cadTop ~
.menu").forEach(x=>x.classList.remove("open"));if(!was)m.classList.add("open");return}
const c=e.target.closest("#cadTop [data-c],#cadTools [data-c],#cadNav [data-c]");document.querySelectorAll("#cadTop ~
.menu").forEach(x=>x.classList.remove("open"));if(c)run(c.dataset.c)});
$("cadGrid").addEventListener("change",e=>run("GRID "+e.target.value));
$("cadSnap").addEventListener("click",()=>{run("SNAP");$("cadSnap").classList.toggle("on")});
$("cadGridB").addEventListener("click",()=>run("GRID"));
document.addEventListener("pointermove",e=>{if(!document.body.classList.contains("cadws"))return;const ~
svg=e.target.closest&&e.target.closest("#siteMap");if(!svg){ch.style.display=cv.style.display="none";return}
try{const p=mapLocal(svg,e);$("cadXY").textContent="X: "+p.x.toFixed(1)+"   Y: "+(-p.y).toFixed(1)+" ft"}catch(_){}
ch.style.display=cv.style.display="block";ch.style.top=e.clientY+"px";cv.style.left=e.clientX+"px"});
document.addEventListener("keydown",e=>{if(!document.body.classList.contains("cadws"))return;const ~
tg=e.target.tagName;if(tg==="INPUT"||tg==="TEXTAREA"||tg==="SELECT")return;if(e.key==="Escape"){const m=document.querySelector("#cadTop ~
.menu.open");if(m)m.classList.remove("open");return}if(e.key.length===1&&/[a-z?]/i.test(e.key)&&!e.ctrlKey&&!e.metaKey&&!e.altKey&&!/^[ ~
]$/.test(e.key)){$("cadIn").focus()}});
let dd=0;window.addEventListener("dragenter",e=>{if(document.body.classList.contains("cadws")&&e.dataTransfer&&[...e.dataTransfer.types].includes("Files")~
){dd++;drop.style.display="flex"}});
window.addEventListener("dragleave",()=>{dd=Math.max(0,dd-1);if(!dd)drop.style.display="none"});
window.addEventListener("dragover",e=>{if(document.body.classList.contains("cadws"))e.preventDefault()});
window.addEventListener("drop",e=>{if(!document.body.classList.contains("cadws"))return;e.preventDefault();dd=0;drop.style.display="none";const ~
f=e.dataTransfer.files&&e.dataTransfer.files[0];if(!f)return;if(/\.json$/i.test(f.name)&&!/geo/i.test(f.name)){safeImport(f);msg("Loaded project ~
"+f.name)}else{importParcelFile(f);msg("Loaded "+f.name)}});
const r1=render;render=function(){r1();if(document.body.classList.contains("cadws")){if(!document.getElementById("siteMap"))setWs(false);else sync()}};
if(location.hash==="#cad")setTimeout(()=>{if(document.getElementById("siteMap"))setWs(true)},400);
})();
/* HOME BUILDER (game-style) */
(function(){
const KEY="oep-home",LH=9.5;
const RT={Living:"#fde7c3",Kitchen:"#d8f0d2",Bedroom:"#d9e4fb",Bath:"#cfeef2",Dining:"#f9d9d9",Office:"#e8dcf5",Garage:"#d9d9d9",Hall:"#f3eed9",Stall:"#e8~
d9b5",Tack:"#d9c3a5",Storage:"#e0e0e0",Shop:"#d6dbe0",Retail:"#f7e6c9",Lobby:"#efe3f5",Restroom:"#cfeef2",Laundry:"#e1ecf7",Porch:"#e8e2d0",Mechanical:"#c~
fcfcf"};
const F={chair:[1.5,1.5,"Chair",2.5,"#8b5a2b"],armchair:[3,3,"Armchair",3,"#a0522d"],sofa:[7,3,"Sofa",3,"#6b7fa8"],bed:[5,6.7,"Queen ~
bed",2,"#9fb4d6"],bed1:[3.2,6.5,"Twin bed",2,"#9fb4d6"],table:[6,3,"Table",2.5,"#b98b54"],round:[4,4,"Round table",2.5,"#b98b54"],desk:[5,2.5,"Desk",2.5,"~
#b98b54"],toilet:[1.5,2.3,"Toilet",1.4,"#fff"],tub:[5,2.5,"Bathtub",1.8,"#fff"],sink:[2,1.5,"Sink",3,"#fff"],stove:[2.5,2.5,"Stove",3,"#777"],fridge:[3,2.~
5,"Fridge",6,"#ccc"],counter:[6,2,"Counter",3,"#a68a64"],wardrobe:[4,2,"Wardrobe",6.5,"#8b5a2b"],plant:[1.5,1.5,"Plant",3,"#4a9a4a"],rug:[6,4,"Rug",.1,"#c~
75b5b"],stairs:[3.5,10,"Stairs",4,"#bbb"],tv:[4,1,"TV",3,"#222"],coffee:[3.5,2,"Coffee ~
table",1.5,"#b98b54"],fireplace:[5,1.5,"Fireplace",4,"#8a5a44"],bookshelf:[4,1,"Bookshelf",6.5,"#6b4423"],kingbed:[6.3,6.7,"King ~
bed",2,"#9fb4d6"],dresser:[4,1.7,"Dresser",3,"#8b5a2b"],nightstand:[1.5,1.5,"Nightstand",2,"#8b5a2b"],island:[6,3,"Kitchen island",3,"#a68a64"],dishwasher~
:[2,2,"Dishwasher",3,"#bbb"],shower:[3,3,"Shower",7,"#bfe3ee"],vanity:[4,1.8,"Vanity",3,"#a68a64"],washer:[2.3,2.3,"Washer",3,"#ddd"],dryer:[2.3,2.3,"Drye~
r",3,"#ccc"],heater:[2,2,"Water heater",5,"#999"],hvac:[2.5,2.5,"HVAC unit",6,"#888"],car:[6.5,15,"Car bay",5,"#7a8aa0"],truck:[8,20,"Truck / trailer ~
bay",7,"#6a7a90"],workbench:[8,2.5,"Workbench",3,"#8b6b3a"],shelving:[6,1.5,"Shelving",6,"#777"],ohdoor:[16,.6,"Overhead door 16 ~
ft",7,"#d0d0d0"],stall:[12,12,"Horse stall 12x12",.5,"#c9a27a"],washrack:[10,12,"Wash rack",.3,"#9bb7c9"],tackrack:[6,2,"Tack / saddle ~
rack",5,"#6b4423"],hay:[6,3,"Hay stack",5,"#d9c06a"],feeder:[3,2,"Feed bin",3,"#8b5a2b"],trough:[8,2,"Water trough",2,"#4aa3d9"],arena:[100,200,"Riding ~
arena 100x200",.2,"#c9b38a"],register:[3,2,"Register counter",3.5,"#555"],reception:[8,3,"Reception desk",3.5,"#7a5a3a"],display:[6,2,"Display ~
rack",5,"#6b7fa8"],conference:[10,4,"Conference table",2.5,"#b98b54"],pool:[16,32,"Pool",.2,"#38bdf8"],patio:[16,12,"Patio / ~
deck",.5,"#b8a58c"],driveway:[12,60,"Driveway",.1,"#8a8a8a"],parking:[9,18,"Parking ~
space",.1,"#777"],lawn:[40,40,"Lawn",.1,"#5fa85f"],garden:[8,16,"Garden bed",.5,"#6b4a2b"],tree:[12,12,"Shade ~
tree",25,"#2f7d3a"],pine:[8,8,"Evergreen",35,"#1f5d33"],pond:[30,20,"Pond",.1,"#3b82b8"],septic:[25,40,"Septic ~
drainfield",.1,"#9b8a5f"],well:[3,3,"Well",4,"#555"],shed:[10,12,"Storage shed",8,"#b08a5a"],firepit:[5,5,"Fire ~
pit",1.5,"#a0522d"],mailbox:[1,1,"Mailbox",4,"#333"],tank:[8,8,"Propane / water tank",8,"#ccc"],solar:[20,10,"Solar ~
array",4,"#1e3a8a"],gate:[16,1,"Gate",4,"#8b5a2b"]};
const ROUND=new Set(["round","tree","pine","pond","firepit","well","tank"]);
const CAT={"Living":["sofa","armchair","coffee","tv","rug","fireplace","bookshelf","plant","stairs"],"Dining":["table","round","chair"],"Bedroom":["bed","~
bed1","kingbed","wardrobe","dresser","nightstand"],"Kitchen":["stove","fridge","counter","island","dishwasher","sink"],"Bath":["toilet","tub","shower","va~
nity"],"Office & utility":["desk","conference","washer","dryer","heater","hvac"],"Garage & shop":["car","truck","workbench","shelving","ohdoor"],"Barn & ~
equestrian":["stall","washrack","tackrack","hay","feeder","trough","arena"],"Commercial":["register","reception","display","conference","shelving"],"Lot: ~
structures":["shed","tank","solar","mailbox","gate"],"Lot: paving & water":["driveway","parking","patio","pool","pond"],"Lot: land & ~
utilities":["lawn","garden","tree","pine","septic","well","firepit"]};
const mk=()=>({walls:[],doors:[],wins:[],furn:[],rooms:[]});
let H=load(),L=0,tool="move",rtype="Living",sel=-1,selR=-1,undoS=[],view="2d",ang=.7,auto=true,drag=null,prev=null,timer=null;
function load(){try{const h=JSON.parse(localStorage.getItem(KEY));if(h&&h.levels&&h.levels.length)return h}catch(e){}return{name:"My ~
Home",psf:275,levels:[mk()]}}
function store(){try{localStorage.setItem(KEY,JSON.stringify(H))}catch(e){}}
function push(){undoS.push(JSON.stringify(H.levels));if(undoS.length>60)undoS.shift()}
function lv(){return H.levels[L]}
let roof=false;
function sn(v,h){return h?Math.round(v*2)/2:Math.round(v)}
function addRoom(l,x,y,w,h,t){l.rooms.push({x,y,w,h,t});[[x,y,x+w,y],[x+w,y,x+w,y+h],[x,y+h,x+w,y+h],[x,y,x,y+h]].forEach(s=>l.walls.push(s))}
function addDine(l,x,y){l.furn.push({t:"table",x,y,r:0});[[-2,-2.3],[0,-2.3],[2,-2.3],[-~
2,2.3],[0,2.3],[2,2.3]].forEach((o,i)=>l.furn.push({t:"chair",x:x+o[0],y:y+o[1],r:i<3?0:180}))}
function starter(){push();const l=mk();addRoom(l,0,0,20,16,"Living");addRoom(l,20,0,14,16,"Kitchen");addRoom(l,0,16,14,14,"Bedroom");addRoom(l,14,16,12,14~
,"Bedroom");addRoom(l,26,16,8,14,"Bath");
l.doors.push({x:10,y:0,o:"h"},{x:20,y:8,o:"v"},{x:7,y:16,o:"h"},{x:23,y:16,o:"h"},{x:30,y:16,o:"h"});
l.wins.push({x:4,y:0,o:"h"},{x:16,y:0,o:"h"},{x:0,y:8,o:"v"},{x:28,y:0,o:"h"},{x:34,y:8,o:"v"},{x:4,y:30,o:"h"},{x:20,y:30,o:"h"},{x:30,y:30,o:"h"});
l.furn.push({t:"sofa",x:8,y:12,r:0},{t:"tv",x:8,y:1.5,r:0},{t:"rug",x:8,y:9,r:0},{t:"armchair",x:15,y:11,r:270},{t:"counter",x:27,y:1.5,r:0},{t:"stove",x:~
31.5,y:2,r:0},{t:"fridge",x:21.8,y:2,r:0},{t:"bed",x:7,y:24,r:0},{t:"wardrobe",x:2.5,y:29,r:0},{t:"bed1",x:17,y:24,r:0},{t:"desk",x:22,y:28.5,r:0},{t:"toi~
let",x:28,y:28.5,r:0},{t:"tub",x:31,y:18,r:0},{t:"sink",x:28.5,y:17,r:0});
addDine(l,26,9);H.levels=[l];L=0;sel=-1;store();draw()}
const sides=r=>[[r.x,r.y,r.x+r.w,r.y],[r.x+r.w,r.y,r.x+r.w,r.y+r.h],[r.x,r.y+r.h,r.x+r.w,r.y+r.h],[r.x,r.y,r.x,r.y+r.h]];
function dropWalls(l,r){sides(r).forEach(s=>{const i=l.walls.findIndex(w=>w.every((v,j)=>v===s[j]));if(i>=0)l.walls.splice(i,1)})}
function delRoom(l,i){dropWalls(l,l.rooms[i]);l.rooms.splice(i,1)}
```

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function roomEdit(l,i,k,v){const r=l.rooms[i];if(k==="t"){r.t=v;return}const n=parseFloat(v);if(isNaN(n)||((k==="w"||k==="h")&&n<2))return;const ~
o={...r};dropWalls(l,r);r[k]=n;if(k==="x"||k==="y"){const dx=r.x-o.x,dy=r.y-o.y,inb=(x,y)=>x>=o.x-.1&&x<=o.x+o.w+.1&&y>=o.y-~
.1&&y<=o.y+o.h+.1;l.furn.forEach(f=>{if(inb(f.x,f.y)){f.x+=dx;f.y+=dy}});[l.doors,l.wins].forEach(a=>a.forEach(d=>{if(inb(d.x,d.y)){d.x+=dx;d.y+=dy}}))}si~
des(r).forEach(s=>l.walls.push(s))}
function preset(n){H.W=72;if(n==="home"){starter();return}push();const l=mk();
if(n==="barn"){for(let ~
i=0;i<5;i++){addRoom(l,i*12,0,12,12,"Stall");l.doors.push({x:6+12*i,y:12,o:"h"});l.wins.push({x:6+12*i,y:0,o:"h"})}addRoom(l,0,12,60,12,"Hall");for(let i=~
0;i<3;i++){addRoom(l,i*12,24,12,12,"Stall");l.doors.push({x:6+12*i,y:24,o:"h"})}addRoom(l,36,24,12,12,"Tack");addRoom(l,48,24,12,12,"Storage");l.doors.pus~
h({x:42,y:24,o:"h"},{x:54,y:24,o:"h"},{x:0,y:18,o:"v"},{x:60,y:18,o:"v"});l.furn.push({t:"tackrack",x:42,y:35,r:0},{t:"hay",x:54,y:34,r:0},{t:"feeder",x:5~
0,y:26,r:0},{t:"trough",x:10,y:18,r:0})}
else if(n==="shop"){addRoom(l,0,0,40,30,"Shop");addRoom(l,40,0,16,15,"Office");addRoom(l,40,15,16,15,"Restroom");l.doors.push({x:0,y:15,o:"v"},{x:40,y:8,o~
:"v"},{x:48,y:15,o:"h"});l.furn.push({t:"ohdoor",x:20,y:29.7,r:0},{t:"truck",x:12,y:14,r:90},{t:"workbench",x:30,y:2,r:0},{t:"shelving",x:4,y:2,r:0},{t:"d~
esk",x:46,y:6,r:0},{t:"toilet",x:44,y:28,r:0},{t:"sink",x:52,y:28,r:0})}
else if(n==="retail"){addRoom(l,0,0,36,36,"Retail");addRoom(l,36,0,14,18,"Storage");addRoom(l,36,18,14,9,"Restroom");addRoom(l,36,27,14,9,"Office");l.door~
s.push({x:18,y:36,o:"h"},{x:36,y:9,o:"v"},{x:36,y:22,o:"v"},{x:36,y:31,o:"v"});l.wins.push({x:6,y:36,o:"h"},{x:30,y:36,o:"h"},{x:0,y:18,o:"v"});l.furn.pus~
h({t:"register",x:8,y:30,r:0},{t:"display",x:10,y:12,r:0},{t:"display",x:20,y:12,r:0},{t:"display",x:28,y:22,r:0},{t:"shelving",x:42,y:2,r:0},{t:"toilet",~
x:38.5,y:25,r:0},{t:"desk",x:43,y:31,r:0})}
else if(n==="lot"){H.W=300;addRoom(l,100,40,30,24,"Living");addRoom(l,130,40,20,24,"Kitchen");addRoom(l,100,64,20,16,"Bedroom");addRoom(l,120,64,18,16,"Be~
droom");addRoom(l,138,64,12,16,"Bath");addRoom(l,150,40,24,24,"Garage");l.doors.push({x:115,y:40,o:"h"},{x:162,y:64,o:"h"},{x:150,y:52,o:"v"});l.wins.push~
({x:110,y:80,o:"h"},{x:128,y:80,o:"h"});
l.furn.push({t:"driveway",x:162,y:107,r:0,w:14,h:86},{t:"patio",x:114,y:90,r:0,w:28,h:12},{t:"pool",x:140,y:112,r:0},{t:"lawn",x:120,y:125,r:0,w:100,h:30}~
,{t:"septic",x:60,y:100,r:0},{t:"well",x:190,y:60,r:0},{t:"shed",x:195,y:85,r:0},{t:"garden",x:205,y:125,r:0},{t:"tree",x:80,y:45,r:0},{t:"tree",x:215,y:5~
0,r:0},{t:"pine",x:70,y:70,r:0},{t:"pine",x:85,y:150,r:0},{t:"mailbox",x:172,y:150,r:0})}
H.levels=[l];L=0;sel=-1;selR=-1;store();lvBar();const sc=document.getElementById("hbscale");if(sc)sc.value=H.W;draw()}
function near(p){let best=null,bd=2.5;for(const w of lv().walls){const[a,b,c,d]=w;if(b===d){const lo=Math.min(a,c),hi=Math.max(a,c);if(hi-~
lo<3||p.x<lo-1||p.x>hi+1)continue;const dist=Math.abs(p.y-b);if(dist<bd){bd=dist;best={x:Math.min(hi-1.5,Math.max(lo+1.5,sn(p.x,1))),y:b,o:"h"}}}else ~
if(a===c){const lo=Math.min(b,d),hi=Math.max(b,d);if(hi-lo<3||p.y<lo-1||p.y>hi+1)continue;const ~
dist=Math.abs(p.x-a);if(dist<bd){bd=dist;best={x:a,y:Math.min(hi-1.5,Math.max(lo+1.5,sn(p.y,1))),o:"v"}}}}return best}
function pt(e){const s=document.getElementById("hbsvg"),p=s.createSVGPoint();p.x=e.clientX;p.y=e.clientY;const ~
q=p.matrixTransform(s.getScreenCTM().inverse());return{x:q.x,y:q.y}}
function area(){let a=0;H.levels.forEach(l=>l.rooms.forEach(r=>a+=r.w*r.h));if(!a){const b=bbox();a=b?b.w*b.h*H.levels.length:0}return a}
function bbox(){let x0=1e9,y0=1e9,x1=-1e9,y1=-~
1e9;H.levels.forEach(l=>l.walls.forEach(w=>{x0=Math.min(x0,w[0],w[2]);x1=Math.max(x1,w[0],w[2]);y0=Math.min(y0,w[1],w[3]);y1=Math.max(y1,w[1],w[3])}));ret~
urn x1<x0?null:{x:x0,y:y0,w:x1-x0,h:y1-y0}}
function furnSvg(f,i,l){const d=F[f.t];if(!d)return"";const w=f.w||d[0],h=f.h||d[1],n=d[2],c=d[4],rot=f.r||0;let inner=`<rect x="${-w/2}" y="${-h/2}" ~
width="${w}" height="${h}" rx=".25" fill="${c}" stroke="#333" stroke-width=".12"/>`;
if(f.t==="chair"||f.t==="armchair"||f.t==="sofa")inner+=`<rect x="${-w/2}" y="${-h/2}" width="${w}" height="${Math.min(.6,h/3)}" fill="#0004"/>`;
if(ROUND.has(f.t))inner=`<circle r="${w/2}" fill="${c}" stroke="#333" stroke-width=".12"/>`;
if(f.t==="bed"||f.t==="bed1")inner+=`<rect x="${-w/2+.3}" y="${-h/2+.3}" width="${w-.6}" height="1.4" rx=".3" fill="#fff"/>`;
if(f.t==="stairs"){for(let k=1;k<10;k++)inner+=`<line x1="${-w/2}" x2="${w/2}" y1="${-h/2+k}" y2="${-h/2+k}" stroke="#555" stroke-width=".08"/>`}
return`<g data-k="furn" data-i="${i}" transform="translate(${f.x} ${f.y}) rotate(${rot})" ~
class="${sel===i&&l===lv()?"hbsel":""}">${inner}<title>${n}</title></g>`}
function draw2(){const l=lv();let s="";const ~
W=H.W||72,K=Math.max(1,W/72),vh=(W+4)*48/76-4,gs=W<=80?1:W<=160?5:W<=320?10:25,g0=-Math.floor(4/gs)*gs;for(let x=g0;x<=W;x+=gs)s+=`<line x1="${x}" y1="-4"~
 x2="${x}" y2="${vh}" stroke="${x%(gs*5)?"#2a3a4d":"#3e5470"}" stroke-width="${(x%(gs*5)?.04:.08)*K}"/>`;for(let y=g0;y<=vh;y+=gs)s+=`<line x1="-4" ~
y1="${y}" x2="${W}" y2="${y}" stroke="${y%(gs*5)?"#2a3a4d":"#3e5470"}" stroke-width="${(y%(gs*5)?.04:.08)*K}"/>`;
if(L>0)H.levels[L-1].walls.forEach(w=>s+=`<line x1="${w[0]}" y1="${w[1]}" x2="${w[2]}" y2="${w[3]}" stroke="#fff3" stroke-width=".4"/>`);
l.rooms.forEach((r,i)=>s+=`<g data-k="room" data-i="${i}"><rect x="${r.x}" y="${r.y}" width="${r.w}" height="${r.h}" fill="${RT[r.t]||"#eee"}" ~
stroke="${selR===i?"#facc15":"none"}" stroke-width="${.5*K}"/><text x="${r.x+r.w/2}" y="${r.y+r.h/2}" font-size="${1.3*K}" text-anchor="middle" ~
fill="#0008" pointer-events="none">${r.t} ${r.w*r.h} sf</text></g>`);
l.furn.forEach((f,i)=>s+=furnSvg(f,i,l));
l.walls.forEach((w,i)=>s+=`<line data-k="wall" data-i="${i}" x1="${w[0]}" y1="${w[1]}" x2="${w[2]}" y2="${w[3]}" stroke="#1d2733" stroke-~
width="${.55*Math.sqrt(K)}" stroke-linecap="square"/><line data-k="wall" data-i="${i}" x1="${w[0]}" y1="${w[1]}" x2="${w[2]}" y2="${w[3]}" ~
stroke="transparent" stroke-width="${1.6*K}"/>`);
l.doors.forEach((d,i)=>s+=`<g data-k="door" data-i="${i}" transform="translate(${d.x} ${d.y}) rotate(${d.o===="v"?90:0})"><rect x="-1.5" y="-.45" width="3"~
 height=".9" fill="#fff"/><path d="M-1.5 0 L-1.5 3 M-1.5 3 A3 3 0 0 0 1.5 0" fill="none" stroke="#a0522d" stroke-width=".18"/></g>`);
l.wins.forEach((d,i)=>s+=`<g data-k="win" data-i="${i}" transform="translate(${d.x} ${d.y}) rotate(${d.o===="v"?90:0})"><rect x="-2" y="-.45" width="4" ~
height=".9" fill="#fff"/><line x1="-2" x2="2" y1="0" y2="0" stroke="#38bdf8" stroke-width=".3"/></g>`);
const dimSvg=(q,i)=>{const D=Math.hypot(q[2]-q[0],q[3]-q[1]),ft=Math.floor(D),inch=Math.round((D-~
ft)*12),mx=(q[0]+q[2])/2,my=(q[1]+q[3])/2,ang=Math.atan2(q[3]-q[1],q[2]-q[0])*180/Math.PI,fl=ang>90||ang<-~
90?ang+180:ang,col=i<0?"#facc15":"#22d3ee";return`<g ${i<0?"":`data-k="dim" data-i="${i}"`}><line x1="${q[0]}" y1="${q[1]}" x2="${q[2]}" y2="${q[3]}" ~
stroke="${col}" stroke-width=".18"/><line x1="${q[0]}" y1="${q[1]}" x2="${q[2]}" y2="${q[3]}" stroke="transparent" stroke-width="1.2"/><circle ~
cx="${q[0]}" cy="${q[1]}" r=".35" fill="${col}"/><circle cx="${q[2]}" cy="${q[3]}" r=".35" fill="${col}"/><text transform="translate(${mx} ${my}) ~
rotate(${fl})" y="-.5" font-size="${1.2*(Math.max(1,(H.W||72)/72))}" fill="${col}" text-anchor="middle" pointer-events="none">${ft} ft${inch?" "+inch+" ~
in":""}</text></g>`};
(l.dims||[]).forEach((q,i)=>s+=dimSvg(q,i));
if(prev){const p=prev;if(p.k==="dim")s+=dimSvg([p.a,p.b,p.c,p.d],-1);else if(p.k==="wall")s+=`<line x1="${p.a}" y1="${p.b}" x2="${p.c}" y2="${p.d}" ~
stroke="#facc15" stroke-width=".5" stroke-dasharray=".6 .4"/><text x="${(p.a+p.c)/2}" y="${(p.b+p.d)/2-.8}" font-size="1.2" fill="#facc15" text-~
anchor="middle">${Math.hypot(p.c-p.a,p.d-p.b)} ft</text>`;else if(p.k==="room")s+=`<rect x="${Math.min(p.a,p.c)}" y="${Math.min(p.b,p.d)}" ~
width="${Math.abs(p.c-p.a)}" height="${Math.abs(p.d-p.b)}" fill="#facc1530" stroke="#facc15" stroke-width=".3" stroke-dasharray=".6 .4"/><text ~
x="${(p.a+p.c)/2}" y="${(p.b+p.d)/2}" font-size="1.4" fill="#facc15" text-anchor="middle">${Math.abs(p.c-p.a)} × ${Math.abs(p.d-p.b)} ft</text>`;else ~
if(p.k==="ghost")s+=`<g opacity=".6" style="pointer-events:none">${p.html}</g>`}
return s}
function proj(x,y,z,c,sc){const ~
dx=x-c.x,dy=y-c.y,rx=dx*Math.cos(ang)-dy*Math.sin(ang),ry=dx*Math.sin(ang)+dy*Math.cos(ang);return{X:400+rx*sc,Y:310+ry*sc*.5-z*sc*.85,d:ry}}
function draw3(){const ~
b=bbox()||{x:0,y:0,w:30,h:30},c={x:b.x+b.w/2,y:b.y+b.h/2},sc=Math.min(16,560/Math.max(b.w,b.h,10)/(roof?1.45:1.1)),floors=[],solids=[];
const Q=(pts,fill,key,list,st)=>list.push({k:key,h:`<polygon points="${pts.map(p=>p.X.toFixed(1)+","+p.Y.toFixed(1)).join(" ")}" fill="${fill}" ~
stroke="${st||"#0004"}" stroke-width=".6"/>`});
H.levels.forEach((l,li)=>{const z=li*LH;l.rooms.forEach(r=>{const ~
P=[[r.x,r.y],[r.x+r.w,r.y],[r.x+r.w,r.y+r.h],[r.x,r.y+r.h]].map(q=>proj(q[0],q[1],z,c,sc));Q(P,RT[r.t]||"#eee",z,floors)});
l.walls.forEach(w=>{const a=proj(w[0],w[1],z,c,sc),b2=proj(w[2],w[3],z,c,sc),wh=(a.d+b2.d)/2>0&&!roof?1:7.5,a2=proj(w[0],w[1],z+wh,c,sc),b3=proj(w[2],w[3]~
,z+wh,c,sc);Q([a,b2,b3,a2],"#f4efe6",(a.d+b2.d)/2+z*.01,solids,"#8a7f70")});
l.doors.forEach(d=>{if(proj(d.x,d.y,0,c,sc).d>0)return;const dx=d.o===="h"?1.5:0,dy=d.o===="v"?1.5:0,P=[proj(d.x-dx,d.y-~
dy,z,c,sc),proj(d.x+dx,d.y+dy,z,c,sc),proj(d.x+dx,d.y+dy,z+6.8,c,sc),proj(d.x-dx,d.y-dy,z+6.8,c,sc)];Q(P,"#8b5a2b",P[0].d+z*.01+.05,solids)});
l.wins.forEach(d=>{if(proj(d.x,d.y,0,c,sc).d>0)return;const dx=d.o===="h"?2:0,dy=d.o===="v"?2:0,P=[proj(d.x-dx,d.y-~
dy,z+3,c,sc),proj(d.x+dx,d.y+dy,z+3,c,sc),proj(d.x+dx,d.y+dy,z+6.5,c,sc),proj(d.x-dx,d.y-dy,z+6.5,c,sc)];Q(P,"#7dd3fc",P[0].d+z*.01+.05,solids)});
l.furn.forEach(f=>{const d=F[f.t];if(!d)return;const fw=f.w||d[0],fh=f.h||d[1],hw=(f.r%180?fh:fw)/2,hh=(f.r%180?fw:fh)/2,ht=f.z||d[3],g=[[f.x-hw,f.y-~
hh],[f.x+hw,f.y-hh],[f.x+hw,f.y+hh],[f.x-~
hw,f.y+hh]],B=g.map(q=>proj(q[0],q[1],z,c,sc)),T=g.map(q=>proj(q[0],q[1],z+ht,c,sc)),k=proj(f.x,f.y,z,c,sc).d+z*.01+.02;for(let i=0;i<4;i++){const ~
j=(i+1)%4;Q([B[i],B[j],T[j],T[i]],d[4],k,solids)}Q(T,d[4],k+.01,solids)})});
if(roof&&bbox()){const ~
o=1.5,x0=b.x-o,x1=b.x+b.w+o,y0=b.y-o,y1=b.y+b.h+o,zt=(H.levels.length-1)*LH+8,rz=zt+Math.min(b.w,b.h)*.2,P=(x,y,z)=>proj(x,y,z,c,sc);if(b.w>=b.h){const ym~
=(y0+y1)/2,A=[P(x0,y0,zt),P(x1,y0,zt),P(x1,ym,rz),P(x0,ym,rz)],B2=[P(x0,y1,zt),P(x1,y1,zt),P(x1,ym,rz),P(x0,ym,rz)];Q(A,"#7a3b2e",1e6+A[0].d,solids,"#3a1a~
12");Q(B2,"#8f4a3a",1e6+B2[0].d,solids,"#3a1a12");Q([P(x0,y0,zt),P(x0,y1,zt),P(x0,ym,rz)],"#e8dfd0",1e6,solids);Q([P(x1,y0,zt),P(x1,y1,zt),P(x1,ym,rz)],"#~
e8dfd0",1e6,solids)}else{const xm=(x0+x1)/2,A=[P(x0,y0,zt),P(x0,y1,zt),P(xm,y1,rz),P(xm,y0,rz)],B2=[P(x1,y0,zt),P(x1,y1,zt),P(xm,y1,rz),P(xm,y0,rz)];Q(A,"~
#7a3b2e",1e6+A[0].d,solids,"#3a1a12");Q(B2,"#8f4a3a",1e6+B2[0].d,solids,"#3a1a12");Q([P(x0,y0,zt),P(x1,y0,zt),P(xm,y0,rz)],"#e8dfd0",1e6,solids);Q([P(x0,y~
1,zt),P(x1,y1,zt),P(xm,y1,rz)],"#e8dfd0",1e6,solids)}}
floors.sort((a,b)=>a.k-b.k);solids.sort((a,b)=>a.k-b.k);return floors.map(p=>p.h).join("")+solids.map(p=>p.h).join("")}
function stats(){const a=area(),bd=H.levels.reduce((n,l)=>n+l.rooms.filter(r=>r.t==="Bedroom").length,0),ba=H.levels.reduce((n,l)=>n+l.rooms.filter(r=>r.t~
==="Bath").length,0),fn=H.levels.reduce((n,l)=>n+l.furn.length,0);return`<div class="hbs"><b>${Math.round(a).toLocaleString()}</b> sq ft</div><div ~
class="hbs"><b>${bd}</b> bed · <b>${ba}</b> bath</div><div class="hbs"><b>${H.levels.length}</b> level(s) · <b>${fn}</b> items</div><div class="hbs">Build~
 cost <b>$${Math.round(a*H.psf).toLocaleString()}</b></div>`}
function insp(){const l=lv(),fld=(k,n,v)=>`<label class="hbf">${n}<input type="number" step=".5" data-ins="${k}" value="${v}"></label>`;
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if(sel>=0&&l.furn[sel]){const f=l.furn[sel],d=F[f.t];return`<h4>Edit: ${d[2]}</h4>`+fld("x","X ft",f.x)+fld("y","Y ft",f.y)+fld("r","Rotate ~
°","f.r||0)+fld("w","Width ft",f.w||d[0])+fld("h","Depth ft",f.h||d[1])+fld("z","Height ft",f.z||d[3])+`<button class="hbb" data-~
act="dup">Duplicate</button><button class="hbb" data-act="del">Delete</button>`}
if(selR>=0&&l.rooms[selR]){const r=l.rooms[selR];return`<h4>Edit room · ${r.w*r.h} sf</h4><label class="hbf">Type<select data-~
ins="t">${Object.keys(RT).map(k=>`<option ${k===r.t?"selected":""}>${k}</option>`).join("")}</select></label>`+fld("x","X ft",r.x)+fld("y","Y ~
ft",r.y)+fld("w","Width ft",r.w)+fld("h","Depth ft",r.h)+`<button class="hbb" data-act="dup">Duplicate</button><button class="hbb" data-~
act="del">Delete</button>`}
return`<div class="hbhint">Move tool: click any room or item to edit its size, position, rotation and type.</div>`}
function draw(){const el=document.getElementById("hbsvg");if(!el)return;const is3=view==="3d";el.setAttribute("viewBox",is3?"0 0 800 560":`-4 -4 ~
${(H.W||72)+4} ${((H.W||72)+4)*48/76}`);{const l0=lv();if(selR>=l0.rooms.length)selR=-1;if(sel>=l0.furn.length)sel=-1}el.innerHTML=is3?`<rect width="800" ~
height="560" fill="#0f1b2a"/>`+draw3():draw2();const s=document.getElementById("hbstats");if(s)s.innerHTML=stats();const ~
ib=document.getElementById("hbinsp");if(ib&&!ib.contains(document.activeElement))ib.innerHTML=insp();document.querySelectorAll("#hb [data-~
tool]").forEach(b=>b.classList.toggle("on",b.dataset.tool===tool));document.querySelectorAll("#hb [data-~
lv]").forEach(b=>b.classList.toggle("on",+b.dataset.lv===L));const v=document.getElementById("hbview");if(v)v.textContent=is3?"Back to 2D plan":"3D view"}
function css(){if(document.getElementById("hbcss"))return;const s=document.createElement("style");s.id="hbcss";s.textContent=`#hb{display:grid;grid-~
template-columns:200px 1fr;gap:10px;color:#e6edf6}#hb .hbp{background:#101a28;border:1px solid #2b3d55;border-radius:10px;padding:8px;max-~
height:78vh;overflow:auto}#hb h4{margin:8px 0 4px;font-size:11px;letter-spacing:.08em;color:#7fa6d6;text-transform:uppercase}#hb button,#hb select,#hb ~
input{font:inherit}#hb .hbb{display:inline-block;background:#1a283a;color:#e6edf6;border:1px solid #33496a;border-radius:6px;padding:5px ~
8px;margin:2px;cursor:pointer;font-size:12px}#hb .hbb.on{background:#2563eb;border-color:#93c5fd;box-shadow:0 0 8px #3b82f6}#hb .hbb:hover{border-~
color:#93c5fd}#hb .hbtop{display:flex;flex-wrap:wrap;gap:6px;align-items:center;margin-bottom:8px}#hb .hbs{background:#101a28;border:1px solid ~
#2b3d55;border-radius:8px;padding:5px 10px;font-size:12px}#hb .hbs b{color:#facc15;font-size:14px}#hbsvg{width:100%;aspect-~
ratio:76/48;background:#0f1b2a;border:1px solid #2b3d55;border-radius:10px;touch-action:none;cursor:crosshair;display:block}#hbsvg .hbsel rect,#hbsvg ~
.hbsel circle{stroke:#facc15;stroke-width:.35}#hb .hbhint{font-size:12px;color:#9fb4d0;margin-top:6px}#hb summary{cursor:pointer;color:#9ec5f5;font-~
size:12px;padding:4px 2px;font-weight:600}#hb details{border-top:1px solid #243449;margin-top:2px}#hb .hbf{display:flex;justify-content:space-~
between;align-items:center;font-size:11px;color:#9fb4d0;margin:2px 0}#hb .hbf input,#hb .hbf select{width:78px;background:#1a283a;color:#fff;border:1px ~
solid #33496a;border-radius:4px;padding:2px}@media(max-width:800px){#hb{grid-template-columns:1fr}}`;document.head.appendChild(s)}
function mount(){css();clearInterval(timer);const used=new Set(),fb=Object.entries(CAT).map(([c,ks],i)=>`<details ~
${i<2?"open":""}><summary>${c}</summary>${ks.filter(k=>F[k]).map(k=>{used.add(k);return`<button class="hbb" data-~
tool="f:${k}">${F[k][2]}</button>`}).join("")}</details>`).join("");
document.getElementById("content").innerHTML=`<div id="hb"><div class="hbp"><div id="hbinsp"></div><h4>Build</h4><button class="hbb" data-tool="move">■ ~
Move (1)</button><button class="hbb" data-tool="room">■ Room (2)</button><button class="hbb" data-tool="wall">■ Wall (3)</button><button class="hbb" data-~
tool="door">■ Door (4)</button><button class="hbb" data-tool="win">■ Window (5)</button><button class="hbb" data-tool="dim">■ Dimension ~
(7)</button><button class="hbb" data-tool="erase">■ Erase (6)</button><h4>Room type</h4><select id="hbrt" class="hbb" ~
style="width:100%">${Object.keys(RT).map(k=>`<option ${k===rtype?"selected":""}>${k}</option>`).join("")}</select><h4>Library folders</h4><button ~
class="hbb" data-tool="set:dine">Dining set + 6 chairs</button>${fb}</div><div><div class="hbtop"><input id="hbname" class="hbb" ~
value="${(H.name||"").replace(/"/g,"&quot;")}" style="width:140px"><span id="hblv"></span><button class="hbb" id="hbaddlv">■ Level</button><button ~
class="hbb" id="hbview">3D view</button><button class="hbb" id="hbroof">Roof: off</button><button class="hbb" id="hbundo">■ Undo</button><button ~
class="hbb" id="hbdxf">■ DXF</button><select id="hbpreset" class="hbb"><option value="">Load starter…</option><option value="home">3-bed ~
home</option><option value="barn">Horse barn</option><option value="shop">Shop / garage</option><option value="retail">Retail / commercial</option><option~
 value="lot">Full lot (home, garage, pool, septic)</option></select><select id="hbscale" class="hbb" title="Drawing area">${[[72,"Building area 72 ~
ft"],[150,"Large building 150 ft"],[300,"Lot 300 ft"],[600,"Large lot 600 ft"]].map(([v,n])=>`<option value="${v}" ~
${(H.W||72)===v?"selected":""}>${n}</option>`).join("")}</select><button class="hbb" id="hbclear">Clear level</button><button class="hbb" ~
id="hbsite">Place on site plan →</button><label class="hbs">$/sf <input id="hbpsf" type="number" value="${H.psf}" ~
style="width:60px;background:#1a283a;color:#fff;border:1px solid #33496a"></label></div><div class="hbtop" id="hbstats"></div><svg id="hbsvg" ~
xmlns="http://www.w3.org/2000/svg"></svg><div class="hbhint">Pick a tool (keys 1–7; 7 = dimension line), then click/drag on the grid. 1 square = 1 ft. ~
Furniture: click to place, Move tool to drag, R rotates, Delete removes. In 3D drag to spin the house.</div></div></div>`;lvBar();draw();
timer=setInterval(()=>{if(activePage!=="homes"){clearInterval(timer);return}if(view==="3d"&&auto&&!drag){ang+=.01;draw()}},60)}
function lvBar(){const e=document.getElementById("hblv");if(e)e.innerHTML=H.levels.map((_,i)=>`<button class="hbb" data-lv="${i}">Level ~
${i+1}</button>`).join("")}
function ghost(p){const t=tool;if(t.startsWith("f:"))return furnSvg({t:t.slice(2),x:p.x,y:p.y,r:0},-9,null);return""}
document.addEventListener("pointerdown",e=>{const ~
svg=e.target.closest&&e.target.closest("#hbsvg");if(!svg||view==="3d"){if(svg&&view==="3d"){drag={x:e.clientX,a:ang}}return}const ~
p=pt(e),l=lv(),k=e.target.closest("[data-k]");
if(tool==="move"){if(k&&k.dataset.k==="furn"){sel=+k.dataset.i;selR=-1;push();drag={mv:true,dx:p.x-l.furn[sel].x,dy:p.y-l.furn[sel].y};draw()}else ~
if(k&&k.dataset.k==="room"){selR=+k.dataset.i;sel=-1;draw()}else{sel=-1;selR=-1;draw()}return}
if(tool==="erase"){if(!k)return;const i=+k.dataset.i,m={dim:"dims",furn:"furn",door:"doors",win:"wins",wall:"walls",room:"rooms"}[k.dataset.k];push();l[m]~
.splice(i,1);sel=-1;store();draw();return}
if(tool==="room"||tool==="wall"||tool==="dim"){drag={k:tool,a:sn(p.x),b:sn(p.y)};prev={k:tool,a:drag.a,b:drag.b,c:drag.a,d:drag.b};draw();return}
if(tool==="door"||tool==="win"){const n=near(p);if(n){push();(tool==="door"?l.doors:l.wins).push(n);store();draw()}return}
if(tool.startsWith("f:")){push();l.furn.push({t:tool.slice(2),x:sn(p.x,1),y:sn(p.y,1),r:0});sel=l.furn.length-1;store();draw();return}
if(tool==="set:dine"){push();addDine(l,sn(p.x,1),sn(p.y,1));store();draw()}});
document.addEventListener("pointermove",e=>{const ~
svg=document.getElementById("hbsvg");if(!svg||activePage!=="homes")return;if(view==="3d"){if(drag&&drag.a!=null){ang=drag.a+(e.clientX-~
drag.x)*.01;draw()}return}const over=e.target.closest&&e.target.closest("#hbsvg");const p=over||drag?pt(e):null;if(!p)return;const l=lv();
if(drag&&drag.mv&&sel>=0){const f=l.furn[sel];f.x=sn(p.x-drag.dx,1);f.y=sn(p.y-drag.dy,1);draw()}
else if(drag&&(drag.k==="wall")){let c=sn(p.x),d=sn(p.y);if(Math.abs(c-drag.a)>=Math.abs(d-drag.b))d=drag.b;else ~
c=drag.a;prev={k:"wall",a:drag.a,b:drag.b,c,d};draw()}
else if(drag&&drag.k==="dim"){prev={k:"dim",a:drag.a,b:drag.b,c:sn(p.x),d:sn(p.y)};draw()}
else if(drag&&drag.k==="room"){prev={k:"room",a:drag.a,b:drag.b,c:sn(p.x),d:sn(p.y)};draw()}
else if(tool.startsWith("f:")&&over){prev={k:"ghost",html:ghost({x:sn(p.x,1),y:sn(p.y,1)})};draw()}});
document.addEventListener("pointerup",()=>{if(view==="3d"){drag=null;return}const l=lv();if(drag&&prev&&activePage==="homes"){const ~
p=prev;if(p.k==="wall"&&(p.a!==p.c||p.b!==p.d)){push();l.walls.push([p.a,p.b,p.c,p.d])}else ~
if(p.k==="dim"&&(p.a!==p.c||p.b!==p.d)){push();(l.dims=l.dims||[]).push([p.a,p.b,p.c,p.d])}else if(p.k==="room"&&p.a!==p.c&&p.b!==p.d){push();addRoom(l,Ma~
th.min(p.a,p.c),Math.min(p.b,p.d),Math.abs(p.c-p.a),Math.abs(p.d-~
p.b),rtype)}}if(drag&&activePage==="homes")store();drag=null;if(prev&&prev.k!=="ghost"){prev=null;draw()}});
document.addEventListener("click",e=>{const t=e.target.closest&&e.target.closest("#hb [data-tool],#hb [data-lv],#hb button[id]");if(!t||activePage!=="home~
s")return;if(t.dataset.tool){tool=t.dataset.tool;prev=null;draw();return}if(t.dataset.lv!=null){L=+t.dataset.lv;sel=-1;draw();return}
switch(t.id){case"hbdxf":{const o=[],a=(c,v)=>o.push(c,v),ln=(ly,x1,y1,x2,y2)=>{a(0,"LINE");a(8,ly);a(10,x1);a(20,-y1);a(30,0);a(11,x2);a(21,-~
y2);a(31,0)};a(0,"SECTION");a(2,"HEADER");a(9,"$INSUNITS");a(70,2);a(0,"ENDSEC");a(0,"SECTION");a(2,"ENTITIES");H.levels.forEach((l,i)=>{(l.walls||[]).for~
Each(w=>ln("WALLS-L"+(i+1),w[0],w[1],w[2],w[3]));(l.dims||[]).forEach(w=>ln("DIMS-L"+(i+1),w[0],w[1],w[2],w[3]))});a(0,"ENDSEC");a(0,"EOF");const ~
u=URL.createObjectURL(new Blob([o.join("\n")],{type:"application/dxf"})),k=document.createElement("a");k.href=u;k.download=(H.name||"home")+".dxf";k.click~
();setTimeout(()=>URL.revokeObjectURL(u),2000);break}case"hbroof":roof=!roof;t.textContent="Roof: "+(roof?"on":"off");view="3d";draw();break;case"hbview":~
view=view==="3d"?"2d":"3d";draw();break;case"hbundo":if(undoS.length){H.levels=JSON.parse(undoS.pop());L=Math.min(L,H.levels.length-~
1);store();lvBar();draw()}break;case"hbstart":starter();lvBar();break;case"hbclear":push();H.levels[L]=mk();store();draw();break;case"hbaddlv":if(H.levels~
.length<4){push();H.levels.push(mk());L=H.levels.length-1;store();lvBar();draw()}break;
case"hbsite":{const b=bbox();if(!b){alert("Build some walls first.");break}snapshot();const ~
d=propertyDims(),it=makeItem("home",Math.round(d.w/2-b.w/2),Math.round(d.h/2-b.h/2));it.name=H.name||"Custom home";it.w=b.w;it.h=b.h;it.height=Math.round(~
H.levels.length*LH+4);it.unitCost=H.psf;items().push(it);selectedId=it.id;save();activePage="designer";render()}break}});
document.addEventListener("input",e=>{if(e.target.id==="hbname"){H.name=e.target.value;store()}if(e.target.id==="hbpsf"){H.psf=+e.target.value||0;store();~
draw()}});
document.addEventListener("change",e=>{if(e.target.id==="hbrt")rtype=e.target.value});
document.addEventListener("keydown",e=>{if(activePage!=="homes"||/INPUT|TEXTAREA|SELECT/.test(document.activeElement?.tagName))return;const ~
k=e.key,l=lv();const T={1:"move",2:"room",3:"wall",4:"door",5:"win",6:"erase",7:"dim"};if(T[k]){tool=T[k];prev=null;draw()}else ~
if(k.toLowerCase()==="r"&&sel>=0&&l.furn[sel]){push();l.furn[sel].r=((l.furn[sel].r||0)+90)%360;store();draw()}else ~
if((k==="Delete"||k==="Backspace")&&sel>=0&&l.furn[sel]){push();l.furn.splice(sel,1);sel=-1;store();draw()}else ~
if(k.toLowerCase()==="z"&&(e.ctrlKey||e.metaKey)){e.preventDefault();document.getElementById("hbundo")?.click()}else ~
if((k==="Delete"||k==="Backspace")&&selR>=0&&l.rooms[selR]){push();delRoom(l,selR);selR=-1;store();draw()}},true);
document.addEventListener("click",e=>{const b=e.target.closest&&e.target.closest("#hbinsp [data-act]");if(!b||activePage!=="homes")return;const ~
l=lv();push();
if(sel>=0&&l.furn[sel]){if(b.dataset.act==="dup"){l.furn.push({...l.furn[sel],x:l.furn[sel].x+2,y:l.furn[sel].y+2});sel=l.furn.length-~
1}else{l.furn.splice(sel,1);sel=-1}}
else if(selR>=0&&l.rooms[selR]){if(b.dataset.act==="dup"){const ~
r=l.rooms[selR];addRoom(l,r.x+r.w,r.y,r.w,r.h,r.t);selR=l.rooms.length-1}else{delRoom(l,selR);selR=-1}}
store();document.activeElement&&document.activeElement.blur();draw()});
```

```
document.addEventListener("change",e=>{const el=e.target;if(activePage!=="homes")return;
if(el.id==="hbscale"){H.W=+el.value;store();draw();return}
if(el.id==="hbpreset"){const v=el.value;el.value="";if(v)preset(v);return}
if(el.dataset&&el.dataset.ins!=null){const l=lv(),k=el.dataset.ins;push();
if(sel>=0&&l.furn[sel]){const f=l.furn[sel],n=parseFloat(el.value);if(!isNaN(n)){f[k]=k==="r"?((n%360)+360)%360:n}}
else if(selR>=0&&l.rooms[selR])roomEdit(l,selR,k,el.value);
store();el.blur();draw()}});
const pr=render;render=function(){pr.apply(this,arguments);if(activePage==="homes")mount()};
})();

wireEvents();if(state._recoveryWarning){setTimeout(()=>alert(state._recoveryWarning),200)}render();
})();
</script>
</body>
</html>
```

A: support.html

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>AcreForge – Project status</title>
<style>
body{margin:0;background:#101820;color:#e8eef5;font:16px Segoe UI,Arial,sans-serif}
main{max-width:760px;margin:auto;padding:32px 18px}
h1{margin:0 0 6px}
p{line-height:1.6}
.s,small{color:#9fb3c8}
.c{background:#18242f;border:1px solid #2c4156;border-radius:10px;padding:18px;margin:20px 0}
a{color:#8db8ff}
</style>
</head>
<body>
<main>
<h1>AcreForge</h1>
<p class="s">A U.S.-wide rural-property planning prototype.</p>
<section class="c">
<h2>Current status</h2>
<p>The site designer, Home Builder, map context, and browser-local project saves are available as a prototype. Projects stay in this browser unless you ~
export or deliberately publish a project.</p>
<p><strong>No paid plan, fundraiser, or payment link is active.</strong> Prices, benefits, refunds, and delivery commitments have not been finalized. Do ~
not send payment for access.</p>
<p>AcreForge is a working name only. A comprehensive trademark search has not been completed, so the name is not cleared for commercial launch.</p>
<p><a href="data-notice.html">Read the prototype data &amp;amp; privacy notice</a>. It describes local browser storage, map requests, public exports, and the ~
unverified status of the server-side payment configuration.</p>
<p><a href="./">Open the prototype</a></p>
</section>
<small>This tool is for early planning, not a survey, engineering package, permit, or professional approval. Verify site conditions and applicable ~
requirements with qualified professionals.</small>
</main>
</body>
</html>
```

A: data-notice.html

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<meta name="description" content="Prototype data-use and privacy information for the AcreForge rural-property planning demo.">
<title>Data &amp;amp; privacy notice – AcreForge prototype</title>
<style>
body{margin:0;background:#101820;color:#e8eef5;font:16px/1.65 Segoe UI,Arial,sans-serif}
main{max-width:800px;margin:auto;padding:32px 18px 56px}
h1,h2{line-height:1.2}
h1{margin-bottom:8px}
h2{margin-top:30px}
.muted{color:#9fb3c8}
.notice{background:#18242f;border:1px solid #6b5730;border-radius:10px;padding:16px}
a{color:#9bc2ff}
li{margin:8px 0}
</style>
</head>
<body>
<main>
<p><a href="./">← Back to the prototype</a></p>
<h1>Data &amp;amp; privacy notice</h1>
<p class="muted">AcreForge is an uncleared working name. This page describes the current prototype, not a complete commercial privacy policy or legal ~
advice. Last updated October 24, 2026.</p>

<section class="notice">
<strong>Do not enter sensitive information.</strong> This prototype is not designed for confidential property, financial, contact, or account data. Review~
 any project before exporting or publishing it.
</section>

<h2>Project information saved in your browser</h2>
<p>The planner and Home Builder save project content in local storage in the browser profile you are using. This can include project names, planning ~
assumptions, objects, dimensions, notes, map coordinates, parcel geometry you import, drawn map overlays, and home layouts. Theme and edit-mode ~
preferences are also stored locally. The app does not provide cloud sync or user accounts in its interface. Anyone with access to this browser profile or ~
device may be able to access its saved data.</p>
<p>You can use the app's export and import tools to make or transfer project files. Exported files are not protected by this site; store and share them ~
carefully. Clearing browser storage or using a different browser/profile may make local projects unavailable.</p>

<h2>Map and network requests</h2>
<p>The site is hosted by Netlify. When your browser requests the site, Netlify and the network providers involved may process technical request ~
information such as your IP address, browser, and requested files under their own policies. See <a href="https://www.netlify.com/privacy/" target="_blank"~
 rel="noopener">Netlify's privacy statement</a>.</p>
<p>When you open the terrain map, the browser requests map imagery from the U.S. Geological Survey and elevation tiles from AWS Terrain Tiles. Those ~
services receive the network requests needed to return the map tiles, which can reveal the geographic area being viewed along with ordinary request ~
metadata. See the <a href="https://www.usgs.gov/privacy" target="_blank" rel="noopener">USGS privacy policy</a> and <a ~
href="https://aws.amazon.com/privacy/" target="_blank" rel="noopener">AWS privacy notice</a>.</p>
<p>The optional <strong>Mapped features</strong> control is off until you enable it. If enabled, the app sends the visible map bounds—not your full saved ~
project—to the public OpenStreetMap Overpass API to request nearby mapped features. The bounds identify the small area currently displayed. See the <a ~
href="https://osmfoundation.org/wiki/Privacy_Policy" target="_blank" rel="noopener">OpenStreetMap Foundation privacy policy</a>.</p>
<p>The app does not request device geolocation. It uses map coordinates that you select or provide through an import or map link. Map and public-data ~
coverage can be incomplete or inaccurate.</p>

<h2>Public project exports</h2>
<p>The separate public-export feature creates a project JSON file intended to be placed beside the site files on its host. If you do that, the file and ~
its contents become available to people who can reach the published site. A view-only link is not access control. Check the file for names, coordinates, ~
parcel details, notes, and other information you do not want to disclose before publishing.</p>

<h2>Accounts, payments, and related services</h2>
<p>The current site interface does not offer account sign-in, paid plans, or checkout. No payment link or fundraiser is presented here. However, Stripe ~
webhook code remains in the deployment source and may be active only if separately configured with server-side settings. That configuration and any ~
related provider records or logs have not been audited for this notice. The operator should verify the hosting configuration before making claims about ~
whether Stripe or Supabase process any information.</p>

<h2>Analytics, advertising, and contact</h2>
<p>The application source does not include a configured analytics or advertising script. This statement does not cover technical logs maintained by the ~
site host or services listed above. This prototype currently does not publish a verified business identity or dedicated privacy-contact address. Do not ~
submit sensitive information or expect this prototype to provide a formal data-request process.</p>

<h2>Questions and future changes</h2>
<p>This notice may need substantial revision before commercial use, especially if accounts, payments, analytics, cloud storage, support forms, or other ~
data collection are added. A commercially launched service should identify its operator and provide an appropriate contact channel and complete, ~
jurisdiction-aware privacy disclosures.</p>
<p><a href="support.html">View prototype and payment status</a></p>
</main>
</body>
</html>
```

A: netlify.toml

```
[build]
  publish = "."
  command = "echo static"

[[headers]]
  for = "/*"
  [headers.values]
    X-Content-Type-Options = "nosniff"
    Referrer-Policy = "strict-origin-when-cross-origin"
    X-Frame-Options = "SAMEORIGIN"

[functions]
  directory = "netlify/functions"
```

A: netlify\functions\stripe-webhook.js

```
// Stripe -> Supabase: marks the buyer's account as Pro (or removes Pro on cancel).
// Env vars: STRIPE_WEBHOOK_SECRET, SUPABASE_URL, SUPABASE_SERVICE_KEY
const crypto = require("crypto");

function verify(raw, header, secret) {
  const parts = Object.fromEntries((header || "").split(",").map(p => p.split("=")));
  if (!parts.t || !parts.v1) return false;
  if (Math.abs(Date.now() / 1000 - Number(parts.t)) > 300) return false;
  const expected = crypto.createHmac("sha256", secret).update(parts.t + "." + raw).digest("hex");
  const a = Buffer.from(expected), b = Buffer.from(parts.v1);
  return a.length === b.length && crypto.timingSafeEqual(a, b);
}

const base = () => process.env.SUPABASE_URL.replace(/\/$/, "") + "/auth/v1/admin/users";
const sbHeaders = () => ({
  apikey: process.env.SUPABASE_SERVICE_KEY,
  Authorization: "Bearer " + process.env.SUPABASE_SERVICE_KEY,
  "Content-Type": "application/json"
});

async function setMeta(userId, meta) {
  const r = await fetch(base() + "/" + userId, { method: "PUT", headers: sbHeaders(), body: JSON.stringify({ app_metadata: meta }) });
  if (!r.ok) throw new Error("Supabase update failed " + r.status);
}

async function findByCustomer(customer) {
  for (let page = 1; page <= 20; page++) {
    const r = await fetch(base() + "?per_page=1000&page=" + page, { headers: sbHeaders() });
    if (!r.ok) return null;
    const { users } = await r.json();
    const hit = (users || []).find(u => u.app_metadata && u.app_metadata.stripe_customer === customer);
    if (hit) return hit;
    if (!users || users.length < 1000) return null;
  }
  return null;
}

exports.handler = async (event) => {
  const secret = process.env.STRIPE_WEBHOOK_SECRET;
  if (!secret || !process.env.SUPABASE_URL || !process.env.SUPABASE_SERVICE_KEY) return { statusCode: 500, body: "Not configured" };
  const raw = event.isBase64Encoded ? Buffer.from(event.body, "base64").toString("utf8") : event.body;
  if (!verify(raw, event.headers["stripe-signature"], secret)) return { statusCode: 400, body: "Bad signature" };
  const evt = JSON.parse(raw), obj = evt.data.object;
  try {
    if (evt.type === "checkout.session.completed" && obj.client_reference_id) {
      await setMeta(obj.client_reference_id, { pro: true, stripe_customer: obj.customer });
    } else if (evt.type === "customer.subscription.deleted") {
      const u = await findByCustomer(obj.customer);
      if (u) await setMeta(u.id, { pro: false });
    }
  } catch (e) {
    return { statusCode: 500, body: e.message };
  }
  return { statusCode: 200, body: "ok" };
};
```

A: README.md

```
# AcreForge

A nationwide, local-first planning workspace for rural land, farms, estates, homes, and buildings. It runs in a modern browser with no build step or paid ~
mapping dependency. The MapLibre viewer library is bundled locally so its controls do not depend on a third-party CDN; loading live USGS imagery, AWS ~
elevation tiles, or optional OpenStreetMap references still requires internet access.

## About and first use

The app opens on **About**, which explains the project vision and links directly to the site planner, Home Builder, and project dashboard. U.S. state maps~
 are available; local rules and permitting information are not built in. Use **Project Settings → Appearance** to change the color theme; the choice is ~
saved in the current browser.

**Commercial status:** AcreForge is a working name, not trademark-cleared. The app is a prototype; paid plans, account sign-in, and fundraising are not ~
active.

## Site designer

The conceptual 2D site designer includes a searchable library of 71 site objects across equestrian, fairgrounds, landscape, building, access/parking, and ~
infrastructure categories. The 1,000-acre starter concept includes an access-road layout, circular round pens and turnout, feed/tack and general stores, ~
distinct post-and-rail, board, woven-wire, electric-tape, and no-climb fences, plus native bushes, tree groups, and a grove. Add event centers, livestock ~
halls, stages, vendor rows, camping, parking, orchards, hedges, gardens, and utilities. Work on an effectively infinite, panning canvas with a feet-based ~
grid; use cursor-centered zoom, adjustable grid spacing and snapping, layers, and labels. Select, move, rotate, resize, duplicate, or delete objects; ~
footprints support rectangles, rounded rectangles, ellipses, triangles, and angled polygons, and roads include straight and curved options. The isometric ~
massing preview supports drag-to-orbit, Shift-drag or right-drag to pan, and wheel zoom; use Reset view to recenter it. Only the selected structure is ~
labeled in the preview to reduce clutter, and secondary options are tucked into a compact menu. The template library, selected-object editor, placed-~
object list, and drawing-layer controls are collapsible; their open/closed state is kept while you work.

Map scale is 1 map unit = 1 foot. The plan automatically fits and centers the full property extent to the map viewport, including narrow screens, without ~
changing scale; use Fit plan to return after panning or zooming. The 1,000-acre starter extent is a planning reference, not a limit on the infinite canvas~
 or a property boundary. For parcel-based acreage and siting checks, import a single-boundary WGS84 KML or GeoJSON Polygon/MultiPolygon from a county ~
assessor/GIS or survey source. The imported outline sets acreage and checks new or moved object footprints; the planning grid remains visible across the ~
canvas. Acreage is calculated geodesically from the supplied outline, including holes. Paste a Google Maps or Zillow URL as a reference link; the app does~
 not scrape those services or derive parcel lines from their pages. A reference URL is not a substitute for an authoritative boundary file.

Switch between the editable flat site plan, a U.S. state map, and a state terrain viewer. State outlines come from the U.S. Census Bureau 2024 TIGER/Line ~
cartographic boundary dataset (1:500,000), projected in Albers equal-area, with Alaska and Hawaii shown separately. The state viewer uses MapLibre GL JS, ~
USGS topographic or aerial imagery, and public AWS Terrain Tiles elevation data. These services and libraries do not require a paid API key; an internet ~
connection is needed to load them.

State terrain opens in USGS aerial imagery by default; use **Topo map** to switch to USGS topographic maps. At neighborhood zoom, enable **Mapped ~
features** to request OpenStreetMap home/building footprints and other context. Building footprints render as raised 3D blocks at city scale. OSM heights ~
are used when available; otherwise the illustration estimates a height from floor count or a generic building type. These are visual context, not actual ~
building dimensions.

On a state terrain map, select a drawing mode to add house/building footprints, land areas, landmarks, or proposed clearing polygons. The app saves these ~
overlays with the project and reports horizontal WGS84 perimeter, area, and polygon-side measurements. Imported parcel geometry can be displayed as a ~
reference, but parcel lines are not fetched from Zillow or inferred from imagery. Use an assessor/GIS or survey KML/GeoJSON file for parcel geometry. ~
Census state boundaries, imagery, and elevation are generalized planning references, not navigation, survey, or legal property data. Satellite/topographic~
 imagery is not a complete tree or building inventory; add planning overlays manually and verify locations from authoritative sources.

The **Mapped features** control can optionally request nearby OpenStreetMap homes, other buildings, roads, fields/orchards, woods/brush, and individually ~
mapped trees. It sends the visible map bounds to the public Overpass API only after you enable it, and only at neighborhood zoom. Use **Reference layers**~
 to hide categories, click a mapped feature to see its available name/category, or **Clear reference context** to remove temporary downloaded references. ~
Clearing does not change map imagery, the saved parcel boundary, or project overlays. Reference data is temporary and is not saved with the project; ~
coverage varies and may be missing or out of date. OpenStreetMap data is © OpenStreetMap contributors (ODbL); an internet connection is required. ~
**Proposed clearing** is only a saved planning annotation; it does not alter satellite imagery or real-world vegetation.

The map can show coordinates parsed from a Google Maps link or the center of an imported parcel, along with an inset of the parcel and its site objects. ~
Switching views does not change saved project data. New starter projects default to 80 acres (1,000 is the maximum); existing saved project acreage is ~
preserved and can be edited in Project Settings until an imported parcel boundary sets it from the source geometry.

The designer is for early planning only: its rectangular planning extent is acreage-derived, not surveyed parcel geometry. Object dimensions and cost ~
assumptions are illustrative and do not establish feasibility, permits, or approvals.

## Building & Lot Builder
Open **Building & Lot Builder** in the sidebar. Pick a starter (3-bed home, horse barn, shop/garage, retail, full lot) and a drawing area (72–600 ft). ~
Browse 12 library folders (rooms, kitchen, bath, garage/shop, barn, commercial, lot items). With the Move tool, click a room or item to edit its size, ~
position, rotation, type and height, or duplicate/delete it. Keys: 1 Move, 2 Room, 3 Wall, 4 Door, 5 Window, 6 Erase, R rotate, Delete remove, Ctrl+Z ~
undo. Toggle 3D to spin the model; **Place on site plan** drops it on the property.

## Run locally

For the most reliable local run, extract the ZIP and serve the complete folder with any static file server. In PowerShell, from that folder:

```powershell
npx serve .
```

Deploy to Netlify

1. Extract `acreforge.zip`.
2. Drag the extracted folder into Netlify Drop, or create a Netlify site from a Git repository containing these files.
3. No build command is needed. Set the publish directory to the folder containing `index.html`.

The included `netlify.toml` configures the static publish directory, function directory, and response headers. A Stripe webhook source file remains in ~
`netlify/functions`; it is not a checkout flow, and its server-side configuration has not been audited.

Contents

- `index.html` — application.
- `vendor/` — pinned MapLibre GL JS/CSS files and upstream license notice.
- `us-states.js` — bundled U.S. Census state-boundary paths required by the location map.
- `CALCULATIONS.md` — formulas, model boundaries, and planning assumptions.
- `CHANGELOG.md` — implementation notes.
- `support.html` — prototype and payment status.
- `data-notice.html` — factual prototype data-use and privacy notice.
- `CAMPAIGN.md` — unpublished fundraising planning draft.
- `netlify.toml` — static-host defaults.
- `acreforge.zip` — deployable archive of the above files (excluding itself); regenerate it after changing the app or documentation.

Data and privacy
```

Project data autosaves to this browser's local storage and is not cloud-synced by the app. The terrain map requests USGS imagery and AWS elevation tiles; ~
the optional OpenStreetMap feature lookup sends the visible map bounds only after the user enables it. Public project export is intended to publish its ~
JSON file alongside the site, so review it for sensitive information first. The [prototype data and privacy notice](data-notice.html) describes these ~
flows and the unverified status of any server-side Stripe webhook configuration. It is not a complete commercial privacy policy.

The app can read the legacy `st3` site-planner local-storage data and migrate its site objects and financial assumptions on first launch.

Limits

The site map is an acreage-derived rectangular planning boundary, not a survey, GIS parcel, or legal property boundary. Terrain overlays are measured from~
map clicks and their accuracy is limited by map resolution and source data; they are not survey-grade. Terrain relief does not convert horizontal ~
footprint measurements into slope-adjusted surface area. The imagery may not show every building or tree. The isometric view is a WebGL-free conceptual ~
massing illustration. Cost, revenue, staffing, and regulatory prompts are illustrative only. No output establishes property-specific feasibility, ~
compliance, approval, financing, or professional design.

See `CALCULATIONS.md` and the in-app Project Settings for the full disclaimer and formulas.

Sharing with family

Each browser keeps its own copy. To share: click **Export**, send the dated .json file, and the other person clicks **Import**. Import shows both save ~
times and warns if the file is older. The newest exported file wins, so agree on who edits at a time.

Terrain and themes

Use **Terrain** in the Site Designer to sculpt elevation (shown in the isometric view). Change the color theme in **Project Settings → Appearance**.

Publishing publicly
1. Click **Public export** (edit mode) and save published-project.json next to index.html.
2. Drag the folder (not the ZIP) onto Netlify Deploys.
3. Visitors see a view-only copy; open your-site/?edit=1 once on your own browser to edit. This is a convenience link, not security. Anyone who knows it ~
can edit their own copy, but nobody can change the published one.

Accounts and payments

Paid plans, account sign-in, and payment processing are currently disabled. Do not add payment URLs, account credentials, customer email allowlists, or ~
secret keys to `index.html`: browser code is public, and local license flags are not secure access control. The included Stripe webhook is not a complete ~
production billing system by itself. Before accepting payments, implement and test server-side checkout/entitlements, cancellation and refund handling, ~
customer support, and truthful pricing disclosures; have the resulting flow and legal terms reviewed for the jurisdictions where you sell.

A: CALCULATIONS.md

```
# Calculation reference

All values are user-editable planning assumptions where controls are provided. Outputs are illustrative and are not professional advice or project-~
specific determinations.

## Property and land allocation

- Without an imported parcel, site area is the manually entered acreage (`acreage × 43,560` square feet), and the displayed rectangular boundary is ~
generated from the entered aspect ratio.
- The rectangular boundary is a planning extent, not a placement fence: without an imported parcel, the site canvas can be panned, zoomed, and extended ~
beyond it. One map unit equals one foot; zooming and panning do not alter object coordinates or measurements.
- With an imported KML or GeoJSON Polygon/MultiPolygon, the displayed outline uses its WGS84 longitude/latitude coordinates, projected into local feet for~
 the site plan. Parcel acreage is calculated from the geodesic polygon area on an authalic-sphere approximation; interior rings are subtracted as holes.
- Imported geometry defines parcel acreage and the placement-check boundary. The grid remains across the viewport; new or moved objects are checked ~
against the imported outline. Keep source survey/assessor data as the authority: input accuracy depends on the imported boundary and its datum/quality, ~
and this lightweight planner is not a survey/GIS or legal parcel determination.
- The U.S. location view uses generalized 2024 U.S. Census TIGER/Line cartographic state boundaries at 1:500,000. The contiguous states use an Albers ~
equal-area projection; Alaska and Hawaii are shown separately. Location markers are plotted only when supported coordinates are available. This overview ~
is not a navigation map or survey. The parcel/site-plan inset retains local feet coordinates and the imported outline.
- The state terrain viewer uses MapLibre GL JS, USGS topographic or aerial imagery, and public AWS Terrain Tiles elevation data without a paid API key. ~
Online map services and the JavaScript library require an internet connection.
- Terrain-map polygon side lengths and perimeter sum haversine distances between the clicked WGS84 vertices, using a mean Earth radius of 6,371,007.1809 m~
 and converting meters to feet. Polygon area uses the app's spherical geodesic area calculation and is converted with 43,560 square feet per acre. ~
Landmark coordinates are shown directly in latitude and longitude.
- Terrain footprint areas are horizontal plan-view areas: elevation and visible relief do not slope-correct the footprint. Results depend on click ~
placement, map scale, imagery/elevation resolution, and the source data; they are planning measurements, not survey-grade dimensions. A displayed parcel ~
overlay comes only from user-imported geometry; the viewer does not derive or verify parcel lines, zoning, restrictions, houses, or trees from imagery or ~
listing pages.
- The optional mapped-context layer queries OpenStreetMap through the public Overpass API for the current small map viewport. It displays mapped building ~
ways, selected road classes, forest/orchard/vineyard land-use ways, wooded ways, and individual tree nodes. The viewport is sent only after the user ~
enables the layer; the data is not saved into the project. OSM coverage is incomplete and can be stale, and mapped features are not proof that a building,~
 road, or tree currently exists. Attribution: © OpenStreetMap contributors, ODbL.
- Rectangles use `width × length`; ellipses use `width × length × π ÷ 4`; triangles use `width × length ÷ 2`.
- Rounded rectangles use `width × length − (4 − π) × radius²`, with `radius = 0.2 × min(width, length)`.
- Angled polygons bevel all four corners by `cut = 0.18 × min(width, length)`; area is `width × length − 2 × cut²`.
- Curved-road cost uses an approximate centerline length: the rendered cubic Bézier curve is sampled at 40 equal parameter intervals and the segment ~
lengths are summed. Its width and length fields describe the road's bounding footprint; the centerline and stroke stay within that footprint.
- Object acreage is object area divided by 43,560.
- Open acreage is total acreage less conceptual building/hardscape area and pasture area, floored at zero. Linear infrastructure is not subtracted as a ~
polygon footprint.

## Capital allowance

- Fixed-price object: entered unit cost per placed object.
- Area-priced object: square feet × entered unit cost per square foot.
- Acre-priced object: object acreage × entered unit cost per acre.
- Linear object: the longer object dimension × entered unit cost per linear foot.
- Site capital allowance: sum of object costs × `(1 + contingency percentage ÷ 100)`.
- Phase estimates use each object's assigned phase and apply contingency proportionally.

The unit costs are starter placeholders. They exclude site-specific construction scope, land, soft costs, fees, escalation, financing, taxes, and other ~
project costs unless a user includes them in an editable allowance.

## Annual operations

- Boarding revenue: occupied stalls × monthly board rate × 12.
- Event revenue: annual events × average event site fee.
- Linked ancillary revenue: $42,000 if a farm stand is placed, $48,000 if an indoor arena is placed, and $30,000 if placed pasture reaches 10 acres. These~
 are illustrative editable-model defaults, not forecasts.
- Gross revenue: boarding + event + ancillary revenue.
- Operating expense: fixed operating expense less the optional $90,000 family-labor planning offset, plus occupied stalls × variable cost per stall, plus ~
events × direct cost per event. Expense is floored at zero.
- NOI: gross revenue less operating expense.
- Stall breakeven: `ceil((fixed expense − offset + event costs − event revenue − ancillary revenue) ÷ (monthly rate × 12 − variable cost per stall))`, ~
bounded by the current model's inputs. If the needed contribution is non-positive, breakeven is zero; if per-stall margin is non-positive, it is not ~
reached.
- Simple payback: site capital allowance ÷ positive annual NOI. It is not shown as meaningful when NOI is zero or negative.
- Financing: standard fixed-rate fully amortizing monthly-payment estimate using entered principal, annual rate, and term. Estimated annual debt service ~
is 12 monthly payments; NOI after debt service subtracts that estimate.
- Five-year illustration: gross revenue and operating expenses each grow at a fixed 3% per year. The illustration has no ramp-up, variable growth ~
controls, taxes, or other forecast adjustments.

## Important model boundaries

The model does not calculate taxes, depreciation, working capital, insurance, replacement reserves, debt covenants, loan eligibility, development charges,~
 property acquisition, or legal/engineering/site constraints. Linked revenue rules are simplified scenario toggles and should not be interpreted as ~
confirmation that an activity is permitted.

## Full disclaimer

This application provides illustrative planning estimates, conceptual layouts, and organizational prompts only. It is not legal, architectural, ~
engineering, environmental, land-use, zoning, building-code, fire-code, employment, tax, lending, construction, insurance, accessibility, or financial ~
advice. Property-specific conditions, costs, designs, approvals, and requirements must be verified with the appropriate county departments, state ~
agencies, licensed professionals, contractors, lenders, insurers, and legal counsel.
```

Appendix B - Changelog

```
# Changelog

## Unreleased
- Live on GitHub Pages: https://jacobbaltins.github.io/acreforge/ (editing on by default; the view-
only mode is local/legacy).
- Added a Free / Pro (planned) / Early Access pricing preview and a Pro waitlist link; payments are
not active.
- Decluttered the sidebar (shorter labels, hidden footer on short screens, scrollable mobile nav).
- Site Designer objects now have a minimum ~18px invisible hit area so they stay draggable at large
acreage.
- Default parcel size is now 80 acres with 1,000 acres as the maximum (selectable in settings).
- Home Builder became the Building & Lot Builder: 12 library folders (~50 new items), editable rooms
and items (size, position, rotation, type, height, duplicate, delete), starter presets (home, barn,
shop, retail, full lot) and 72/150/300/600 ft drawing scales.
- Made USGS aerial imagery the default state terrain basemap, clarified the aerial/topographic layer
switch, and added raised 3D building context at neighborhood zoom using reported OSM heights or
explicitly estimated defaults.
- Expanded the state terrain map's optional, neighborhood-scale OpenStreetMap context to distinguish
homes, other buildings, fields/orchards, woods/brush, roads, and mapped trees; added category
visibility controls, safe click-to-inspect labels, and a clear action that preserves parcel geometry
and user overlays.
- Clarified that mapped context is incomplete reference data, parcel boundaries must come from an
assessor/GIS or survey file, and proposed-clearing polygons do not alter imagery or real-world land
cover.
- Rebranded the working product name to AcreForge and updated positioning for U.S.-wide rural-
property planning; the name still needs trademark clearance before commercial launch.
- Disabled the incomplete browser-only Pro/account/trial controls, removed a client-side email
allowlist, and clarified that paid plans and fundraising are not active.
- Updated the support page and campaign draft to remove unconfirmed prices, rewards, and delivery
promises.
- Added a prototype data/privacy notice covering browser-local data, map requests, public exports,
and the unverified server-side webhook configuration; linked it from the app and status page.
- Added an unpublished, support-only GoFundMe campaign draft with suggested title, short
description, and story, plus official platform/IRS references; it avoids inventing a goal, budget,
organizer identity, reward, or launch date.
- Added a private, evidence-based campaign budget worksheet and organizer-only GoFundMe setup
checklist; no personal identity or payout details are requested or stored.
- Added an About landing page as the app's first screen, with a summary of the project vision and
shortcuts into planning and design.
- Moved the persistent color-theme control from the obstructive top-right overlay into Project
Settings.

## 1.18.0
- Home Builder: Roof toggle (gable roof) in the 3D view.
- Home Builder: Dimension tool (key 7) draws ft/in measurement lines; Erase removes them.
- Home Builder: DXF export of all levels (walls and dimensions, feet) for AutoCAD/Inventor.

## 1.16.0
- CAD workspace: full-screen AutoCAD-style mode (menus, quick-access bar, tool strip, navigation
bar, crosshair, command line, coordinate/status bar). Drag & drop or File menu to load project JSON
or GeoJSON/KML parcels.
- Print sheet with title block; DXF export for AutoCAD.

## 1.15.0
- New default 'CAD' theme: dark drafting workspace, blue accents, square controls, crosshair cursor,
model-space status bar.

## 1.14.1
- Live drag preview: a snapped ghost follows the cursor and turns red outside the parcel.

## 1.14.0
- Map view: measurement grid (50-1000 ft auto-spacing) and flat-plan overlay placed on the terrain
map. Objects outside the parcel get red outlines; Zoom to plan button.

## 1.13.0
- Slope & earthwork panel: land by slope, per-object slope warnings, rough cut/fill yardage.
- City tools: generate a street grid and house lots, remove generated objects.

## 1.12.0
- 7-day free trial, free-access emails, Supabase sign-in, Stripe webhook for automatic Pro.
- Favicon and share-preview tags.

## 1.11.0
- Palette grouped by category with counts.
- Pro tier groundwork: Go Pro dialog, license-key unlock, watermark on free printed reports.
- Fixed Farmland and Town layers missing from new projects.

## 1.10.0
```

- Public/private mode: on a public web address visitors get a view-only plan; add ?edit=1 once to unlock editing in your browser (?edit=0 to lock). 'Public export' downloads published-project.json with boundary, location, finances and notes removed. Added map-data credits. 3D preview zoom now reaches 80x.

1.9.0
- Added foliage (oak, Douglas-fir, fruit trees, forest, riparian buffer, meadow, lawn, flower beds), a Farmland group (row crops, hay, vineyard, berries, greenhouse, pond, silo, sheds) and a Town & city group (houses, townhomes, apartments, shops, school, park, plaza, streets). New Farmland and Town layer toggles.

Change log

1.8.1

- Added Forest, Ranch and Daylight color themes (picker at top right, remembered per browser).
- Family sharing: exported files now carry a date/time in the name, and Import shows both save times, warns if the file is older than your current project, and keeps a recovery copy.

1.8.0

- Added editable terrain: a **■ Terrain** tool in the Site Designer with raise, lower, smooth and flatten brushes, adjustable size/strength, undo/redo, "Gentle hills" and "Flatten all".
- Elevations are saved in `project.terrain` (feet), shown as a blue-to-tan overlay on the plan, and lift the ground and objects in the isometric preview (adjustable vertical exaggeration).
- Terrain travels with **Export JSON / Import**, so family members can pass the project file around. Edits are not live-synced; the latest exported file wins.

1.7.3

- Bundled MapLibre GL JS/CSS locally so the map viewer does not depend on loading its library from a third-party CDN.
- Moved the isometric preview into a full-width panel below the plan and editing sidebar.
- Made the template library, selected-object editor, placed-object list, and drawing layers collapsible, preserving panel state while working.

1.7.2

- Added panning to the interactive isometric preview using Shift-drag or right-drag, alongside drag-to-orbit and wheel zoom.
- Added a reset-view control and tucked cutaway/auto-rotate switches into a compact preview-options menu.
- Reduced visual clutter in the isometric view by labeling only the selected structure.

1.7.1

- Fit and center the complete site plan to the actual map viewport on entry, including narrow panels, without distorting the feet-based scale.
- Keep fit-to-plan and cursor-centered zoom consistent with the responsive viewBox; recalculate the view on window resize.

1.7.0

- Set new starter projects to a 1,000-acre conceptual extent while preserving acreage in existing saved projects and imported parcel boundaries.
- Expanded the starter layout with access roads, circular round pens and turnout, feed/tack and general stores, multiple horse-safe fence concepts, native bushes, and a grove.
- Added 14 site-object templates for road types, circular pens, retail/support buildings, fence variants, and shrub planting; refined their map symbols for easier visual distinction.
- Kept dimensions, costs, road alignments, and siting illustrative and editable; they do not represent surveyed or engineered plans.

1.6.0

- Added an opt-in OpenStreetMap context overlay for mapped buildings, local roads, wooded/land-use areas, and individually mapped trees.
- Limited lookups to a small viewport at neighborhood zoom; map bounds are sent to the public Overpass API only while the user enables the layer.
- Kept external map references temporary and separate from saved project geometry, with visible OpenStreetMap attribution and coverage caveats.

1.5.0

- Added a selectable state terrain viewer using MapLibre GL JS, free USGS topographic/aerial basemaps, and public AWS Terrain Tiles elevation data; no paid API key is required.
- Added persistent georeferenced overlays for house/building footprints, land areas, landmarks, and proposed clearing, with WGS84 side, perimeter, and area measurements.
- Clarified that terrain imagery, elevation, and click-derived overlay measurements are planning references, not a complete real-world inventory, survey, or legal parcel determination.

1.4.0

- Replaced illustrative state shapes with bundled U.S. Census Bureau 2024 TIGER/Line cartographic state boundaries, including Albers equal-area contiguous-state geometry and Alaska/Hawaii insets.
- Added an effectively infinite, feet-based planning canvas with visible grid beyond the acreage-derived planning extent, cursor-centered zoom, panning, fit-to-plan, and scale/zoom HUD.
- Added rounded-rectangle and beveled-polygon area formulas, unrestricted-degree footprint rotation, and a curved gravel road whose centerline length informs linear cost.
- Clarified that Census boundaries and conceptual layouts are not survey or navigation sources; parcel-specific acreage and siting checks use imported authoritative geometry.

1.3.0

- Added a flat-plan / U.S. location-map toggle with coordinate markers and an imported parcel/site-plan inset.
- Set new starter projects to 300 acres; imported parcel boundaries still determine the acreage.
- Kept the U.S. map illustrative rather than a live basemap; switching views does not modify saved project data.

1.2.0

- Added KML and GeoJSON Polygon/MultiPolygon parcel-boundary import, coordinate validation, geodesic acreage calculation, non-rectangular outline rendering, and map-grid clipping.
- Added Google Maps/Zillow reference links and clear source labeling; acreage is tied to imported boundary geometry, and object placement checks against the outline.
- The input file must come from an authoritative assessor/GIS or survey source; pasted listing/map links are references only and are not scraped for parcel data.

1.1.0

- Expanded the 2D site designer into a searchable, category-filtered library of 56 site-planning objects, including event centers, exposition and livestock halls, show venues, vendor and food areas, camping, parking, fences, individual trees, groves, orchards, hedges, gardens, and access features.
- Refined the conceptual map with parcel texture, perimeter landscaping, object-specific map details, labels, a compass, and a scale bar.
- Added adjustable 10/20/40/100-foot grid spacing, optional placement snapping, map zoom and pan, grid and label visibility controls, and responsive palette styling.
- Kept new objects additive and optional so existing saved scenarios are preserved; the starter layout remains unchanged.

1.0.0

- Built a self-contained, local-first Oregon Estate Planner with 11 planning sections.
- Added an acreage-derived 2D conceptual site map with structure placement, object editing, selection, movement, rotation, undo/redo, scenario-specific layouts, layer visibility, and object-cost controls.
- Added editable conceptual building diagrams and an interactive isometric canvas preview that does not depend on WebGL.
- Connected site quantities and financial assumptions to capital estimates, operating revenue/expenses, breakeven, financing illustration, phase estimates, and a five-year illustrative projection.
- Added infrastructure notes, editable staffing ratios, a categorized due-diligence checklist, scenario comparison, print report, JSON backup/restore, CSV schedule export, and searchable reference manual.
- Added versioned local storage, recovery snapshot, legacy `st3` migration, accessibility labels/focus handling, responsive layout, reduced-motion behavior, and the full planning disclaimer.